Title Page

SBT-DF205 Mobile and IoT Forensics Case Study

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Course Name: Mobile and IoT Forensics Case Study

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Date of Submission: 27/10/2026

Introduction

This case study involved the forensic examination of an Android 10 file-system extraction from a Google Pixel 3. The examination was carried out under case reference **SBT-DF205-C1** to identify the device configuration and user profiles, reconstruct relevant communications, review application activity, and examine available location and media artefacts.

I worked only with the supplied extraction. The original Pixel 3.zip archive was preserved separately, and a working copy was created before examination. SHA-256 hashing was used to verify that the working copy remained identical to the original evidence before extraction. The analysis was performed from Kali Linux using command-line forensic tools and SQLite queries rather than altering the original evidence.

The investigation focused on what could be supported directly by the recovered records. Where artefacts were missing, empty, duplicated, or inconclusive, I recorded those limitations instead of making assumptions.

Objectives

The objectives of my examination were to:

preserve and verify the integrity of the supplied Android evidence;

identify the Pixel 3 hardware, Android build, security patch level and user profiles;

identify account associations and determine whether they could be corroborated from other artefacts;

examine system settings, Wi-Fi and Bluetooth configuration;

reconstruct SMS and MMS communications and correlate numbers with contacts and call records;

compare AOSP communication records with Google Messages and Google Dialer;

examine available WhatsApp and TextNow artefacts;

examine Google Maps and photograph evidence;

create a UTC-normalised timeline from multiple artefact families; and

assess whether the recovered evidence supports, contradicts or leaves unresolved the allegation of unauthorised communication or information sharing.

Laboratory / Tools Used

The investigation was performed in Kali Linux 2025.3 running in VMware Workstation.

The tools and commands used during the examination included:

sha256sum for SHA-256 integrity verification;

unzip for testing and extracting the working evidence archive;

find, grep, ls, stat and related Linux commands for locating and documenting artefacts;

sqlite3 in read-only mode for examination of Android application databases;

xmllint and text-based examination for Android XML configuration records;

ExifTool for media metadata and GPS examination; and

standard Linux date conversion methods for validating Unix epoch timestamps and converting them to UTC

Method

I first created a structured case directory and separated the original archive from the working copy. The original archive had a SHA-256 value of:

ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a

The working archive produced the same SHA-256 value, confirming integrity before analysis. The working archive was then tested for errors and extracted to the dedicated examination directory.

Device, user, communication, application, location and media artefacts were then examined using their original source paths. Database queries were performed read-only. Raw timestamps were retained and converted to UTC only after identifying their units or stored time-zone information.

Device Identity and User Profiles

The recovered build properties identified the device as a Google Pixel 3 running Android 10. The build identifier was QQ1A.200105.003, and the recorded Android security patch level was 2020-01-01.

Two Android profiles were identified:

User 0, profile name “This Is”

User 11, profile name “User 2”

User 0 was associated with the Google account thisisdfir@gmail.com, while User 11 was associated with thisisdfirtwo@gmail.com. The first account was also present in Android sync configuration and Google Photos preferences, providing corroboration across independent sources.

These records support account-to-profile association, but they do not on their own prove which physical person was operating the phone at any particular time.

Settings and Connectivity Findings

The stored Android settings showed several relevant values. In particular:

adb_enabled = 1

development_settings_enabled = 1

wifi_on = 1

bluetooth_on = 0

airplane_mode_on = 0

location_mode = 3

location_providers_allowed = gps,network

device_name = Pixel 3

The two Wi-Fi configuration files were present, but both contained zero saved network entries. I therefore could not identify a stored SSID or prove a Wi-Fi connection at a particular time.

Bluetooth artefacts were present, including bt_config.conf, bluetooth_db, bluetooth_db-wal and bluetooth_db-shm. A stored Bluetooth identifier was recovered, but the available configuration did not contain sufficient timestamp information to prove a connection at a specific time.

Lab Folder Structure and Evidence Preparation

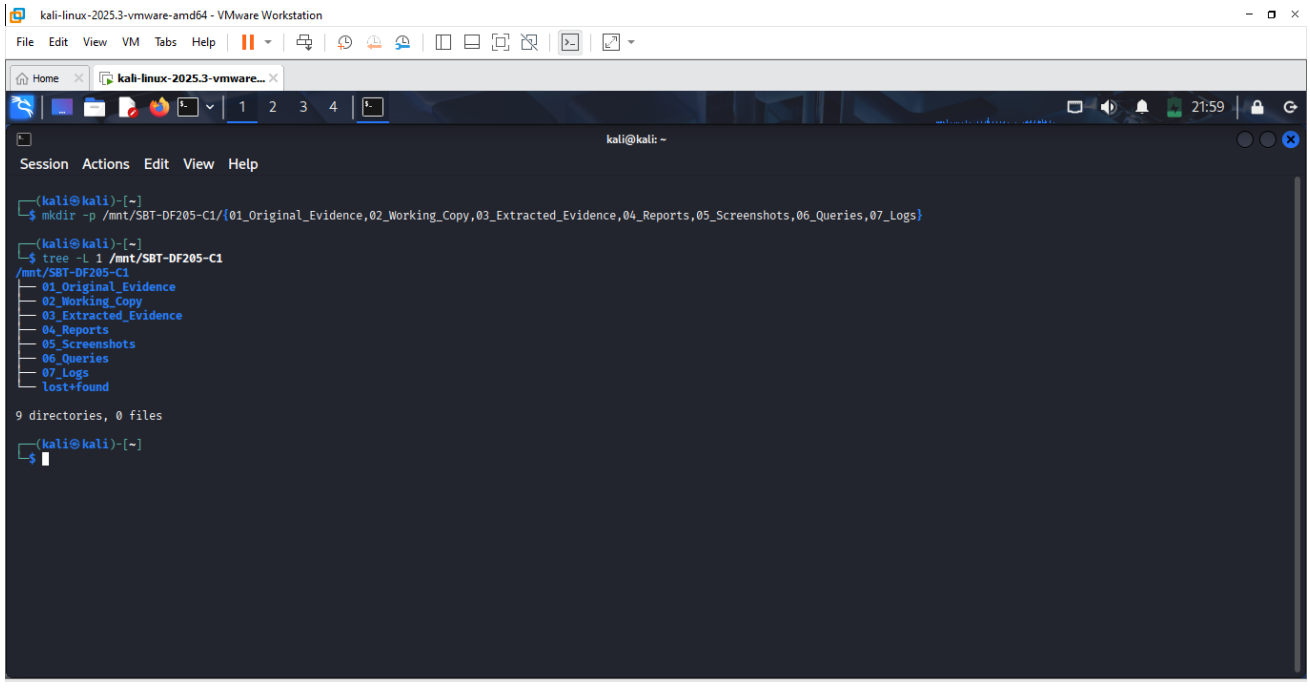


Fig 1.0 Forensic Investigation Directory Structure.

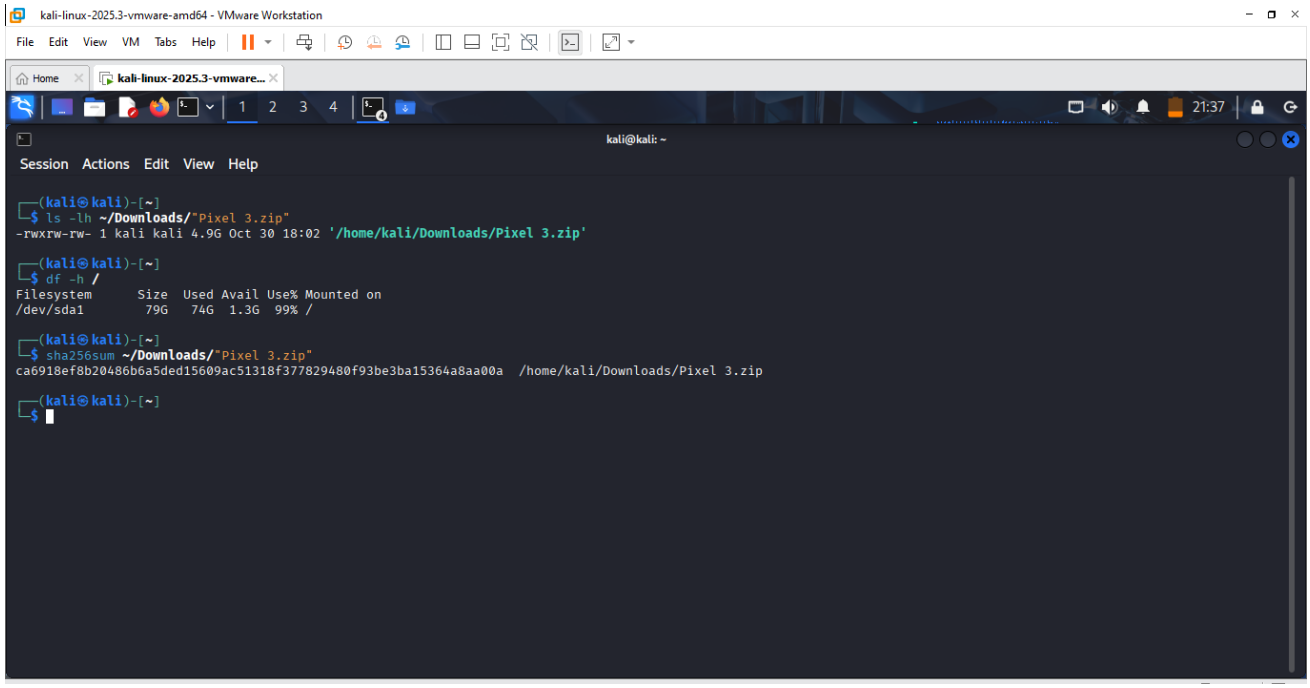


Fig 1.1 Original Pixel 3 Evidence Archive SHA-256.

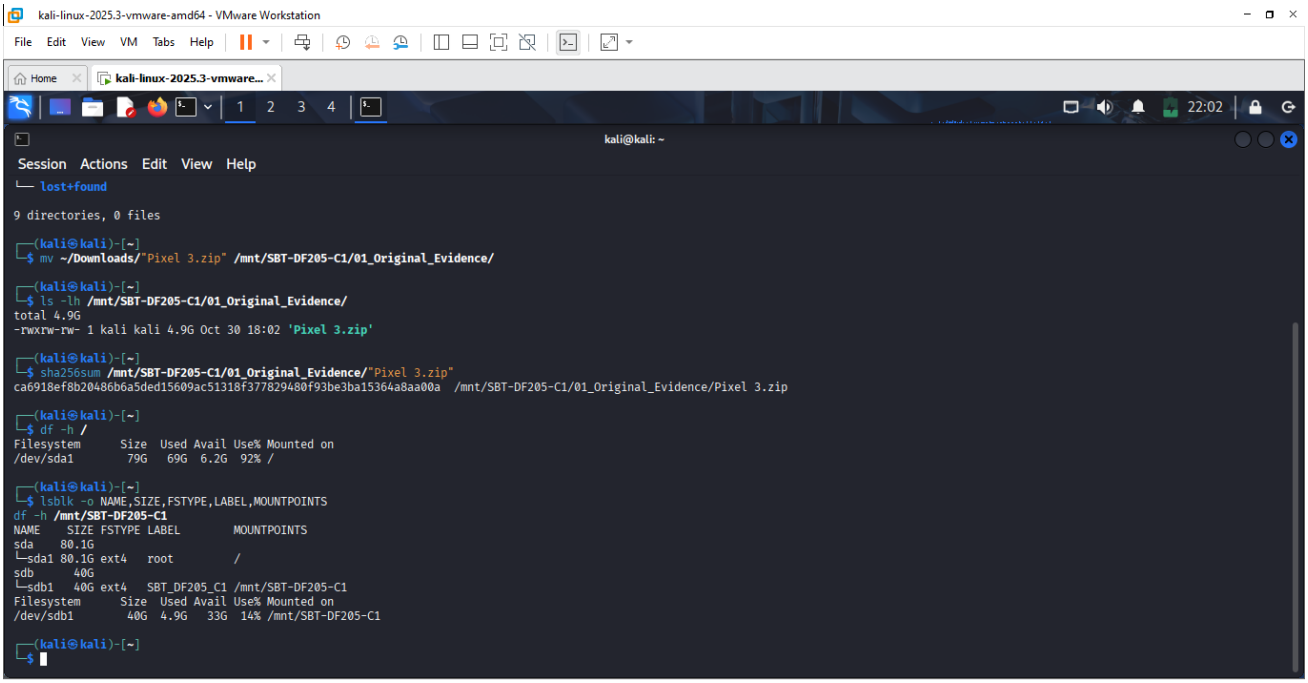


Fig 1.2 Dedicated Forensic Evidence Disk Configuration.

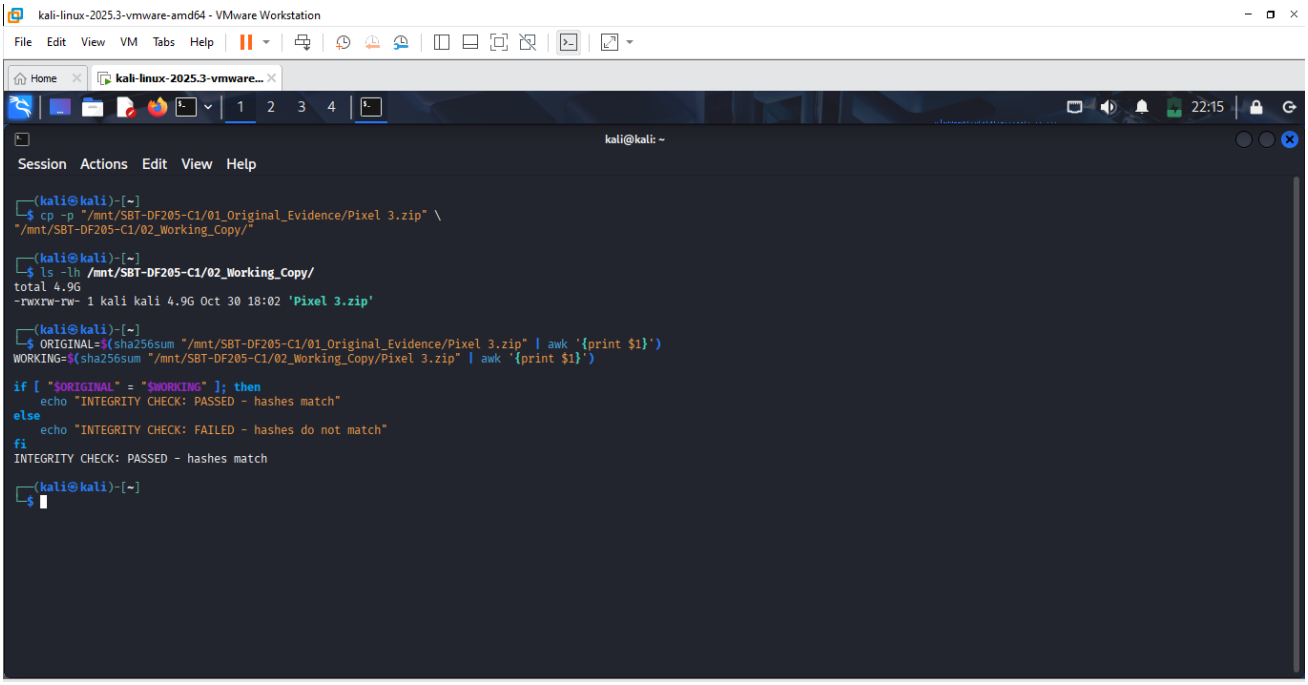


Fig 1.3 Original and Working Copy SHA-256 Integrity Verification.

Explanation

I compared the SHA-256 value of the preserved Pixel 3 archive with the working copy before carrying out the examination. Both hashes matched, confirming that the working evidence was an exact copy and that evidence integrity had been maintained.

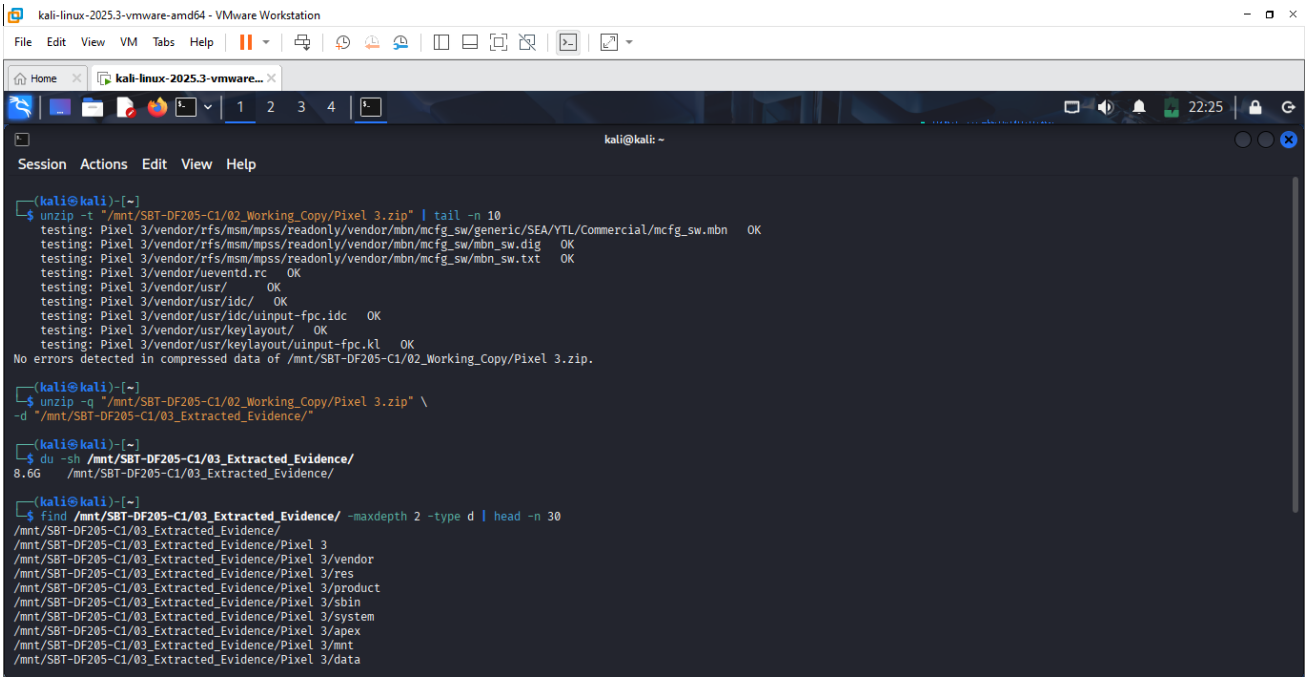


Fig 1.4 Pixel 3 Working Evidence Extraction.

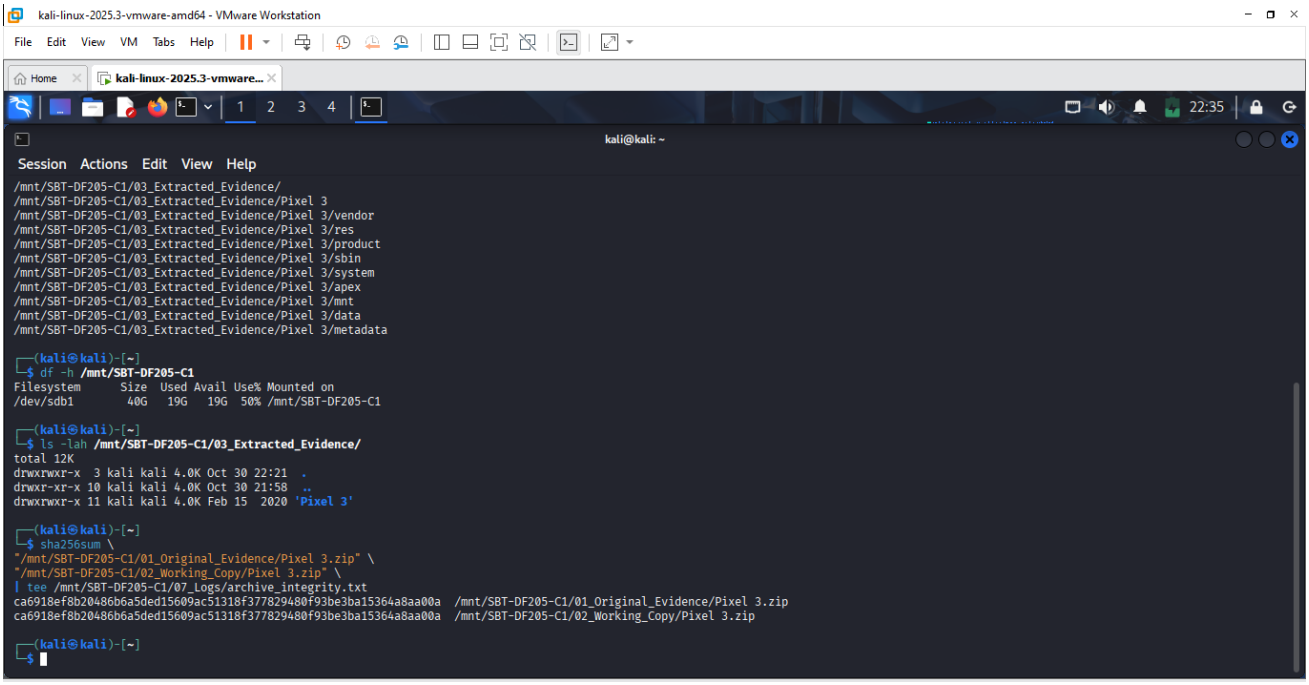


Fig 1.5 Pixel 3 Working Evidence Extraction page 1.

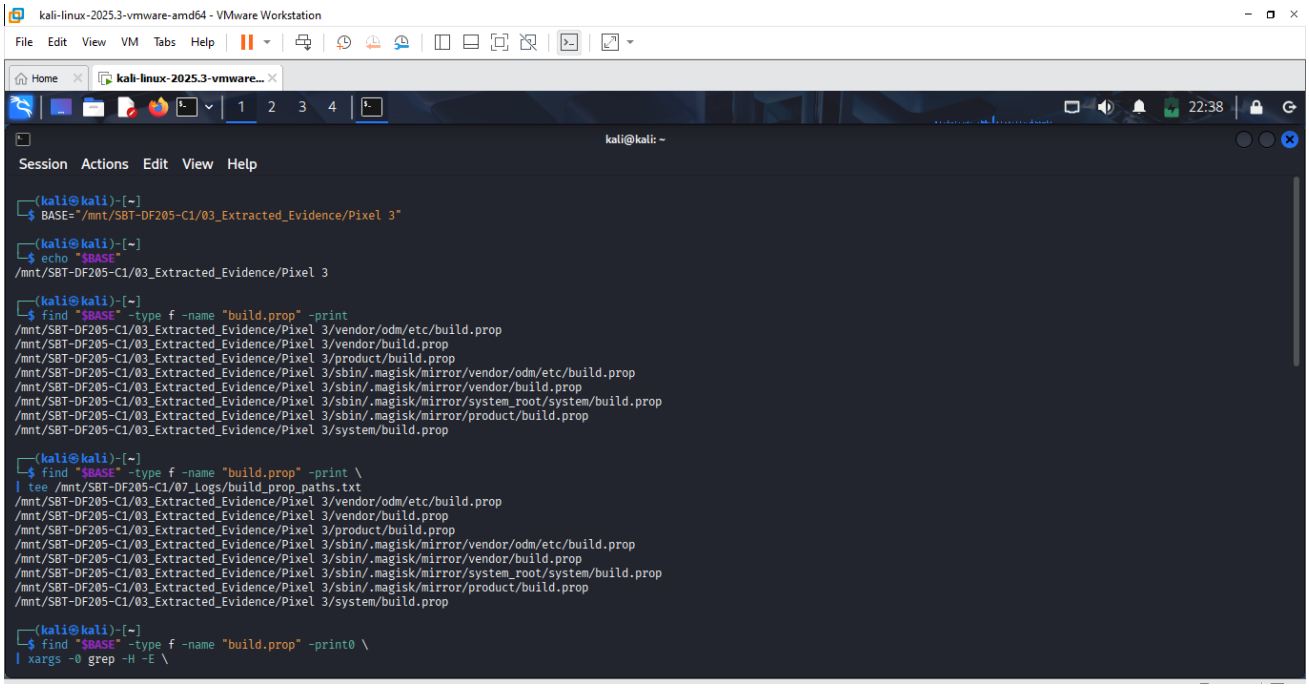


Fig 1.6 Android Device Identity from Build Properties.

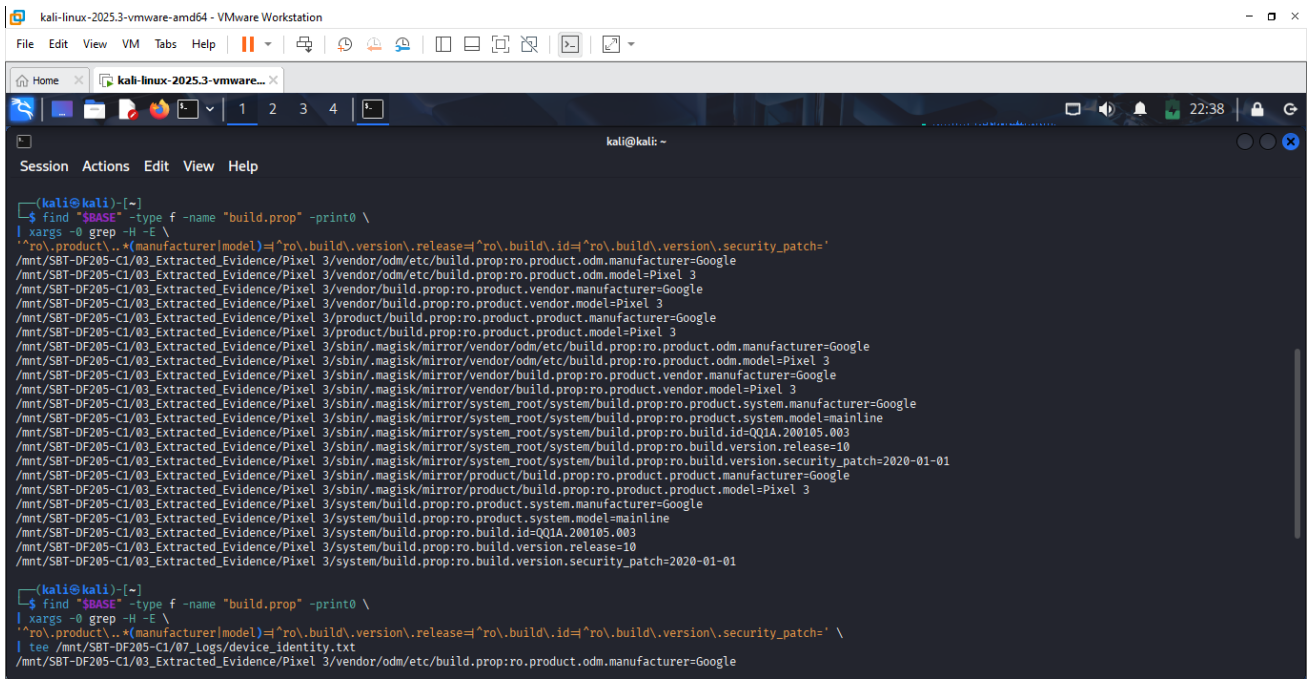


Fig 1.7 Android Device Identity from Build Properties page 1

Explanation

The build-property records identify the extracted device as a Google Pixel 3 running Android 10. They also provide the build identifier and January 2020 security patch level, giving me a reliable technical description of the device under examination.

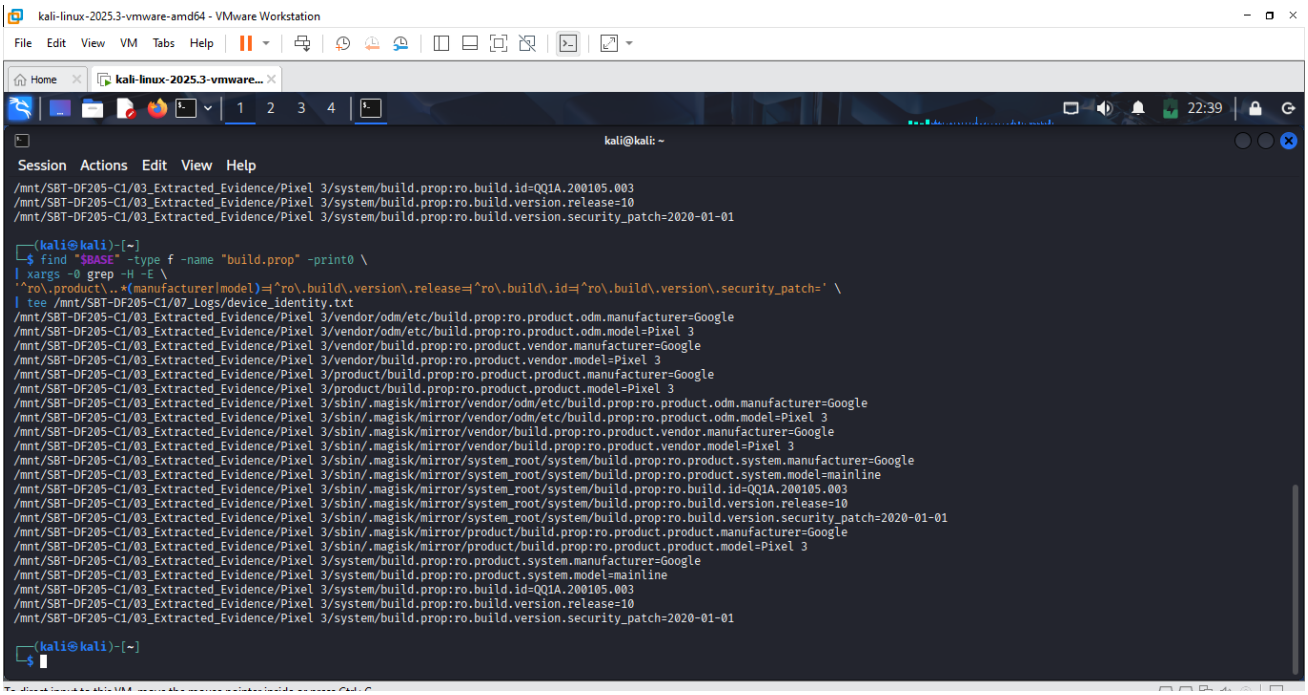


Fig 1.8 Android Device Identity from Build Properties page 2.

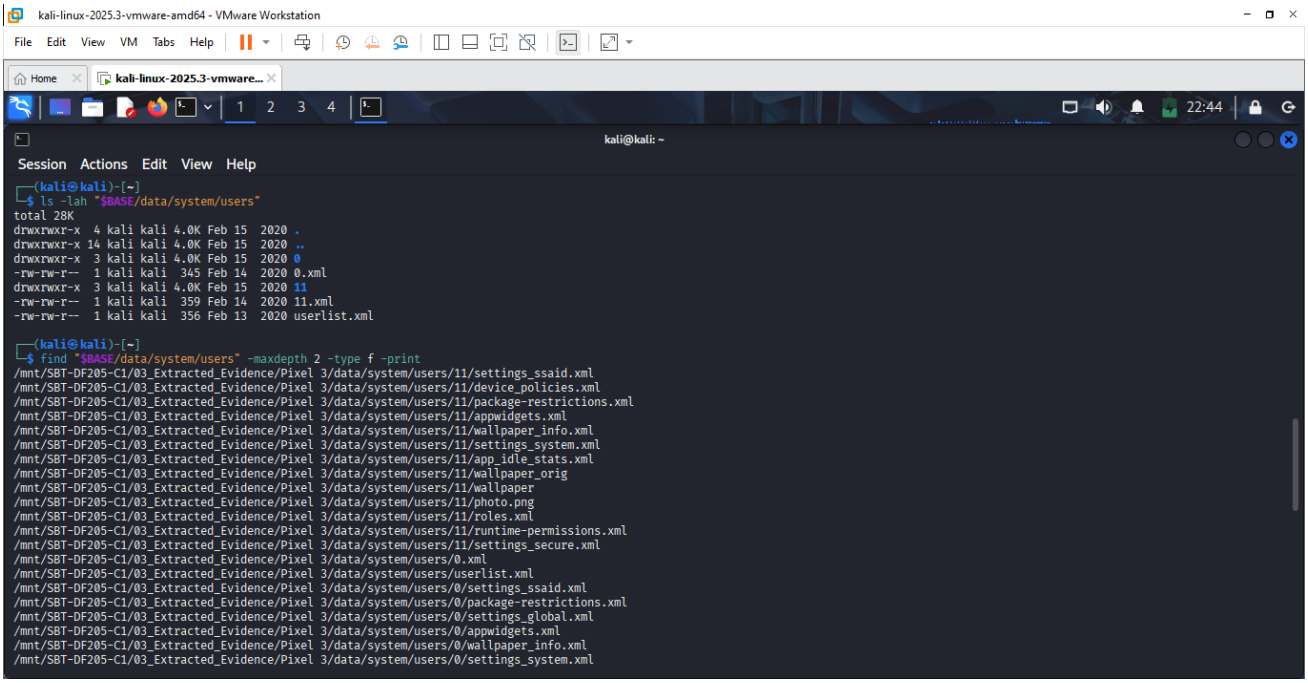


Fig 1.9 Android User Profile List.

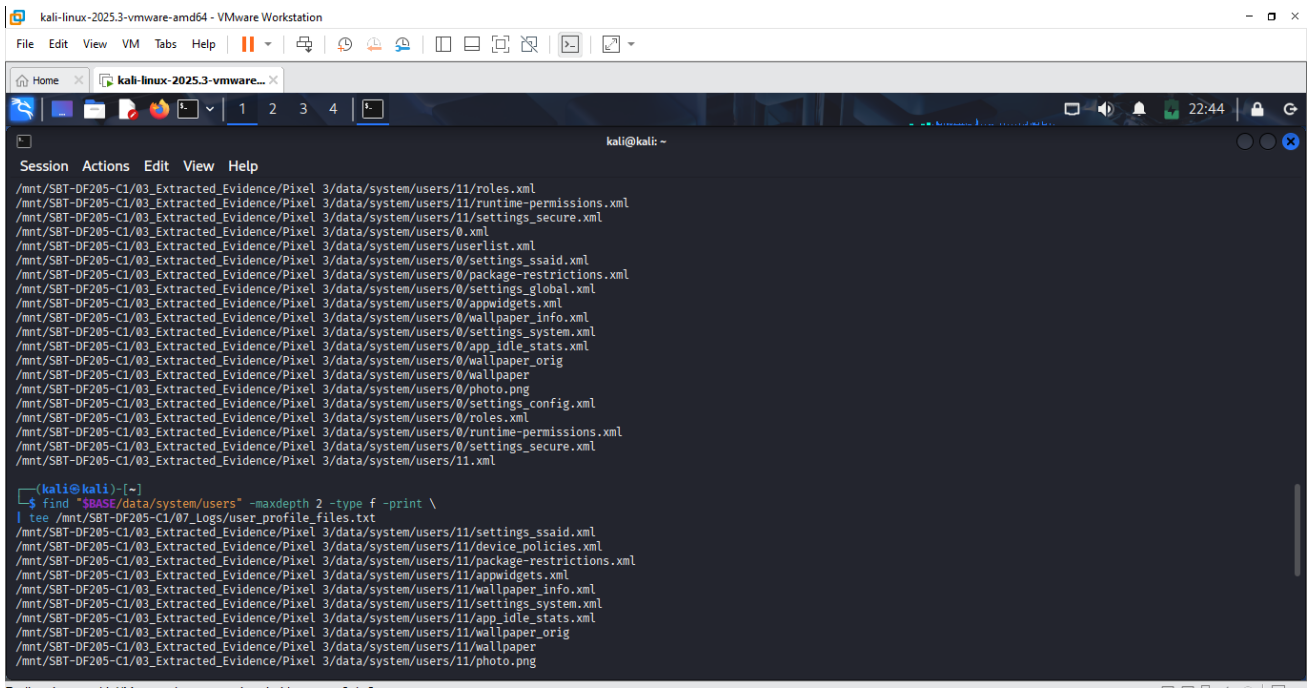


Fig 2.0 Android User Profile List page 1.

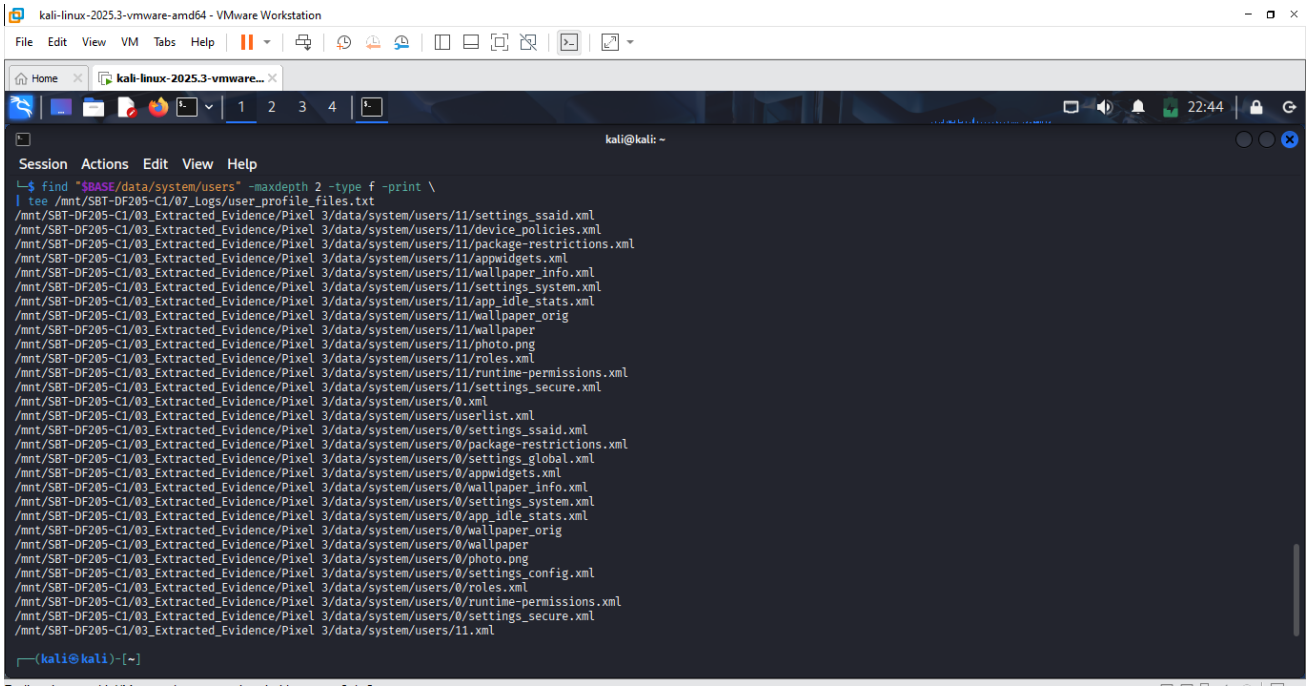


Fig 2.1 Android User Profile List page 2.

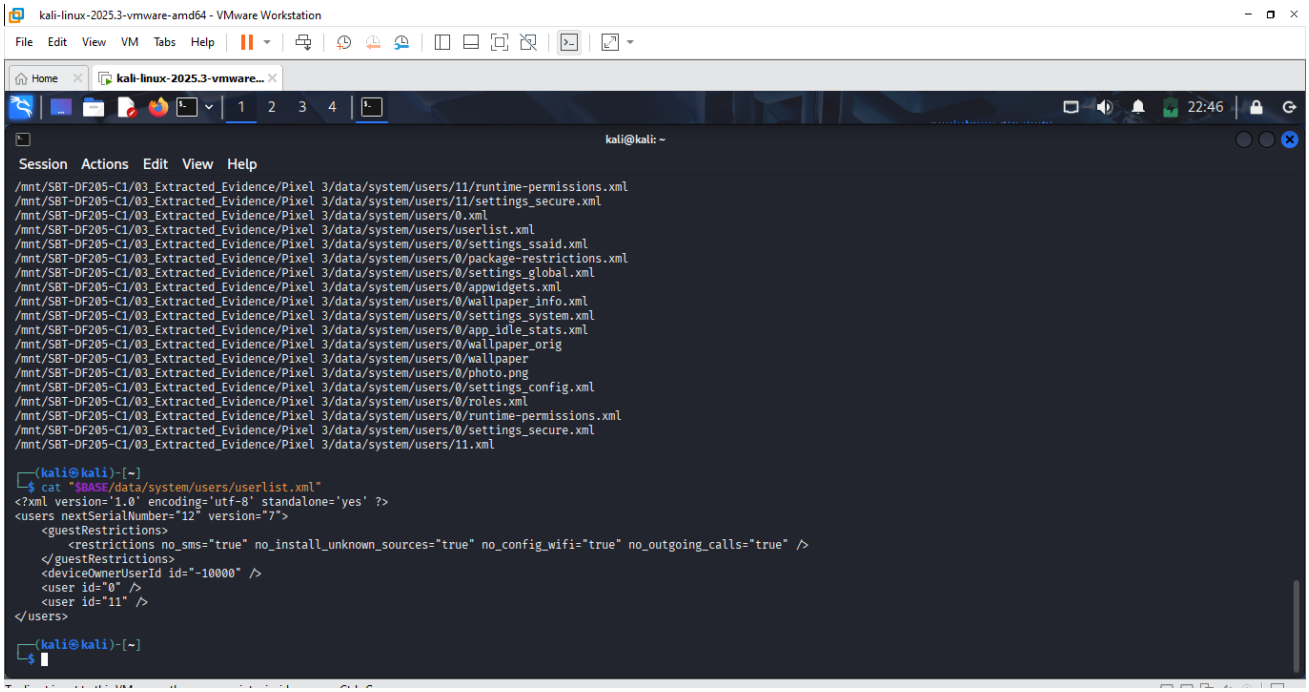


Fig 2.2 Android User Profile List page 3.

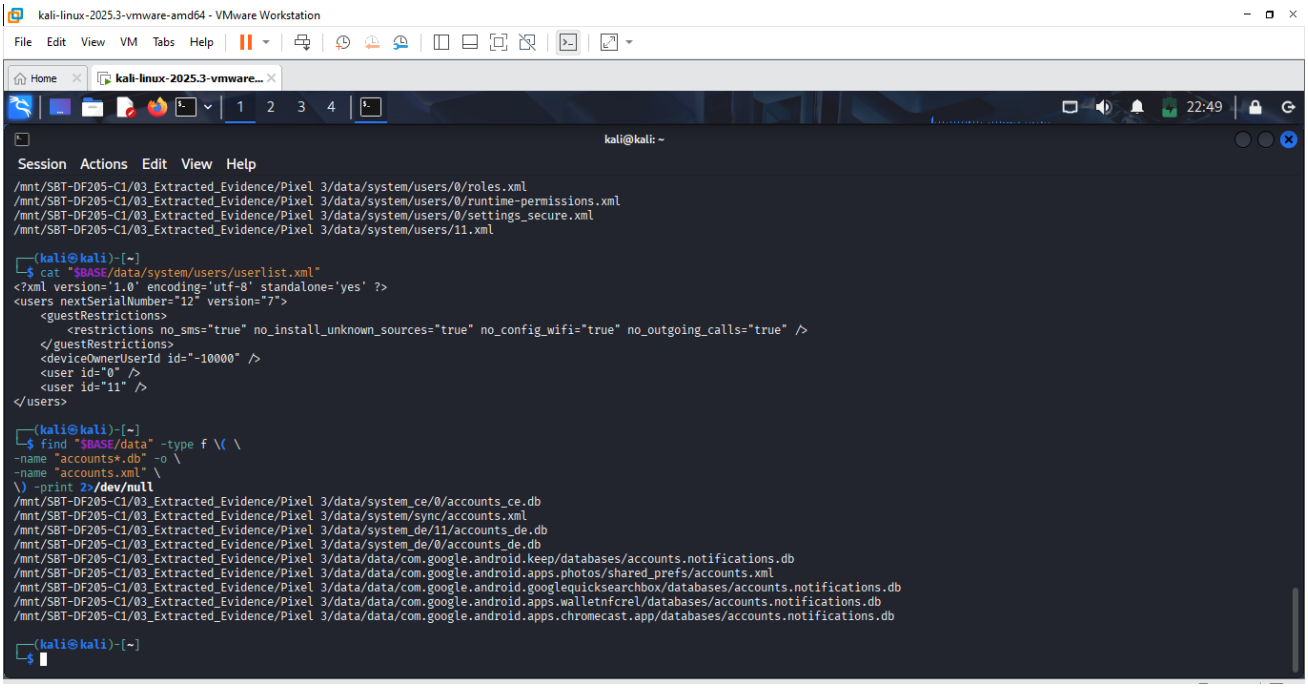
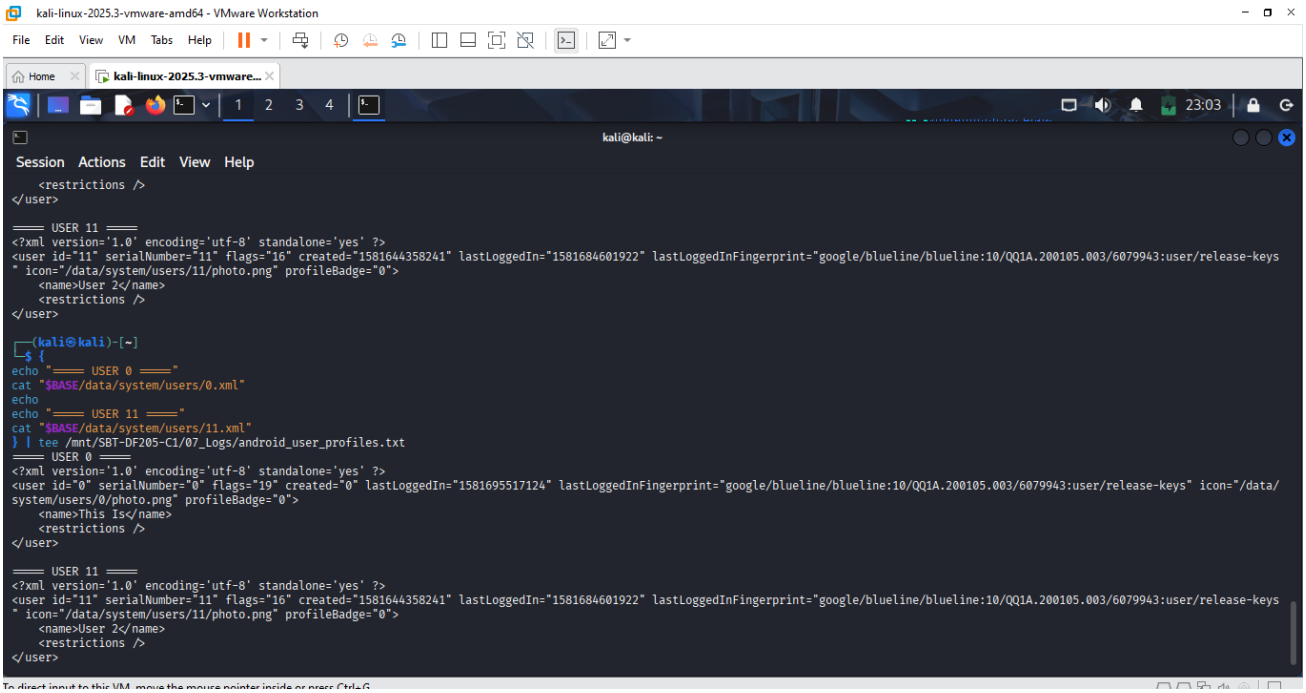


Fig 2.3 Finding account databases.

Findings

Property	Evidence value	Property key / source
Manufacturer	**Google**	ro.product.vendor.manufacturer=Google / ro.product.product.manufacturer=Google
Device model	**Pixel 3**	ro.product.vendor.model=Pixel 3 / ro.product.product.model=Pixel 3
Android version	**10**	ro.build.version.release=10
Build identifier	**QQ1A.200105.003**	ro.build.id=QQ1A.200105.003
Security patch	**2020-01-01**	ro.build.version.security_patch=2020-01-01
Android profiles	**User 0 and User 11**	userlist.xml contains <user id="0" /> and <user id="11" />



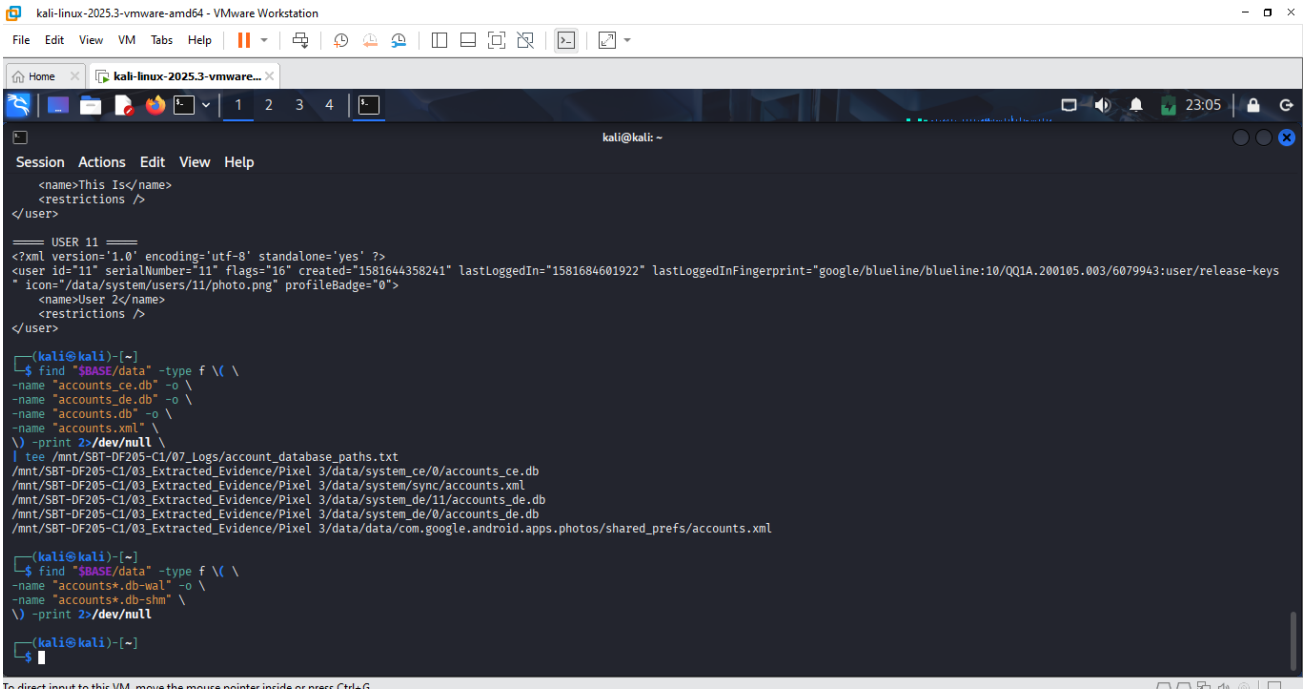

```
    <restrictions />
</user>

═══ USER 11 ═══
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
" icon="/data/system/users/11/photo.png" profileBadge="0">
    <name>User 2</name>
    <restrictions />
</user>

┌──(kali㉿kali)-[~]
└─$ {
echo "═══ USER 0 ═══"
cat "$BASE/data/system/users/0.xml"
echo
echo "═══ USER 11 ═══"
cat "$BASE/data/system/users/11.xml"
} | tee /mnt/SBT-DF205-C1/07_Logs/android_user_profiles.txt
═══ USER 0 ═══
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="0" serialNumber="0" flags="19" created="0" lastLoggedIn="1581695517124" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys" icon="/data/
system/users/0/photo.png" profileBadge="0">
    <name>This Is</name>
    <restrictions />
</user>

═══ USER 11 ═══
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
" icon="/data/system/users/11/photo.png" profileBadge="0">
    <name>User 2</name>
    <restrictions />
</user>
```

Fig 2.4 Android User 0 and User 11 Profile Details.




```
    <name>This Is</name>
    <restrictions />
</user>

═══ USER 11 ═══
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<user id="11" serialNumber="11" flags="16" created="1581644358241" lastLoggedIn="1581684601922" lastLoggedInFingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys
" icon="/data/system/users/11/photo.png" profileBadge="0">
    <name>User 2</name>
    <restrictions />
</user>

┌──(kali㉿kali)-[~]
└─$ find "$BASE/data" -type f \( \
-name "accounts_ce.db" -o \
-name "accounts_de.db" -o \
-name "accounts.db" -o \
-name "accounts.xml" \
\) -print 2>/dev/null \
| tee /mnt/SBT-DF205-C1/07_Logs/account_database_paths.txt
/mnt/SBT-DF205-C1/03_Extracted_Evidence/Pixel 3/data/system_ce/0/accounts_ce.db
/mnt/SBT-DF205-C1/03_Extracted_Evidence/Pixel 3/data/system/sync/accounts.xml
/mnt/SBT-DF205-C1/03_Extracted_Evidence/Pixel 3/data/system_de/11/accounts_de.db
/mnt/SBT-DF205-C1/03_Extracted_Evidence/Pixel 3/data/system_de/0/accounts_de.db
/mnt/SBT-DF205-C1/03_Extracted_Evidence/Pixel 3/data/data/com.google.android.apps.photos/shared_prefs/accounts.xml

┌──(kali㉿kali)-[~]
└─$ find "$BASE/data" -type f \( \
-name "accounts*.db-wal" -o \
-name "accounts*.db-shm" \
\) -print 2>/dev/null

┌──(kali㉿kali)-[~]
└─$
```

Fig 2.5 Android User 0 and User 11 Profile Details page 1.

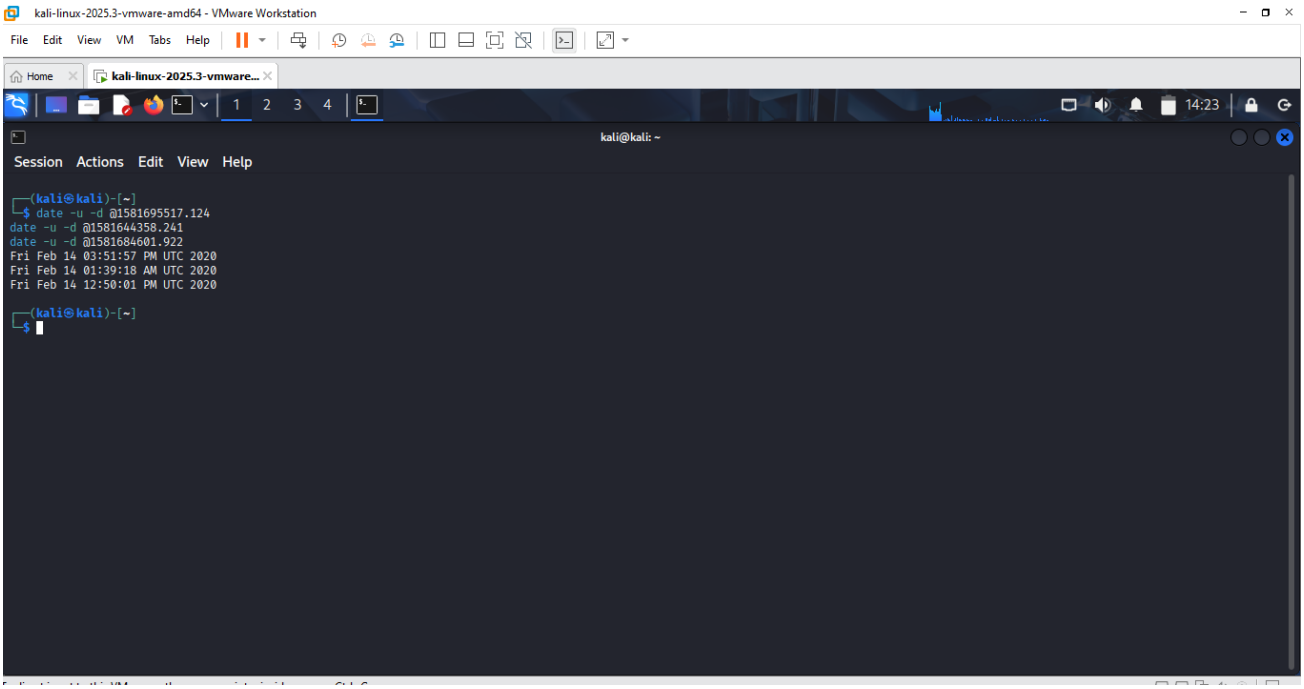


Fig 2.6 The UTC conversions.

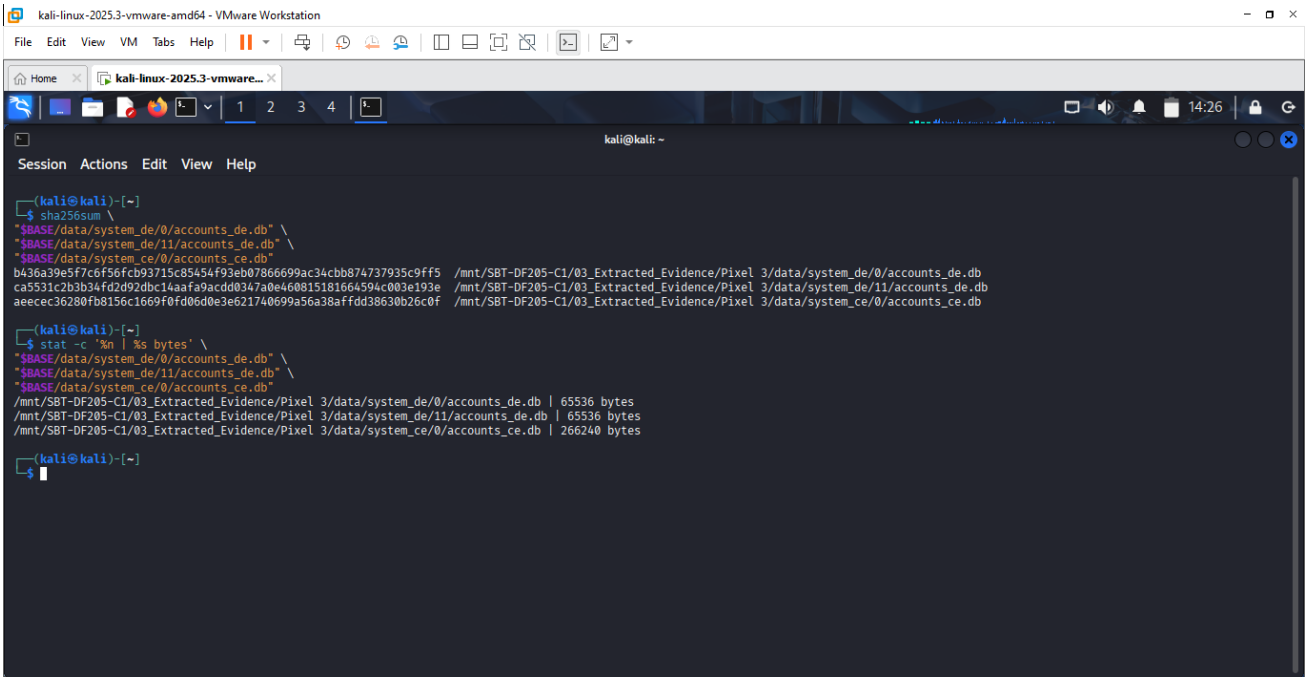


Fig 2.7 Android Account Database Hashes and File Sizes.

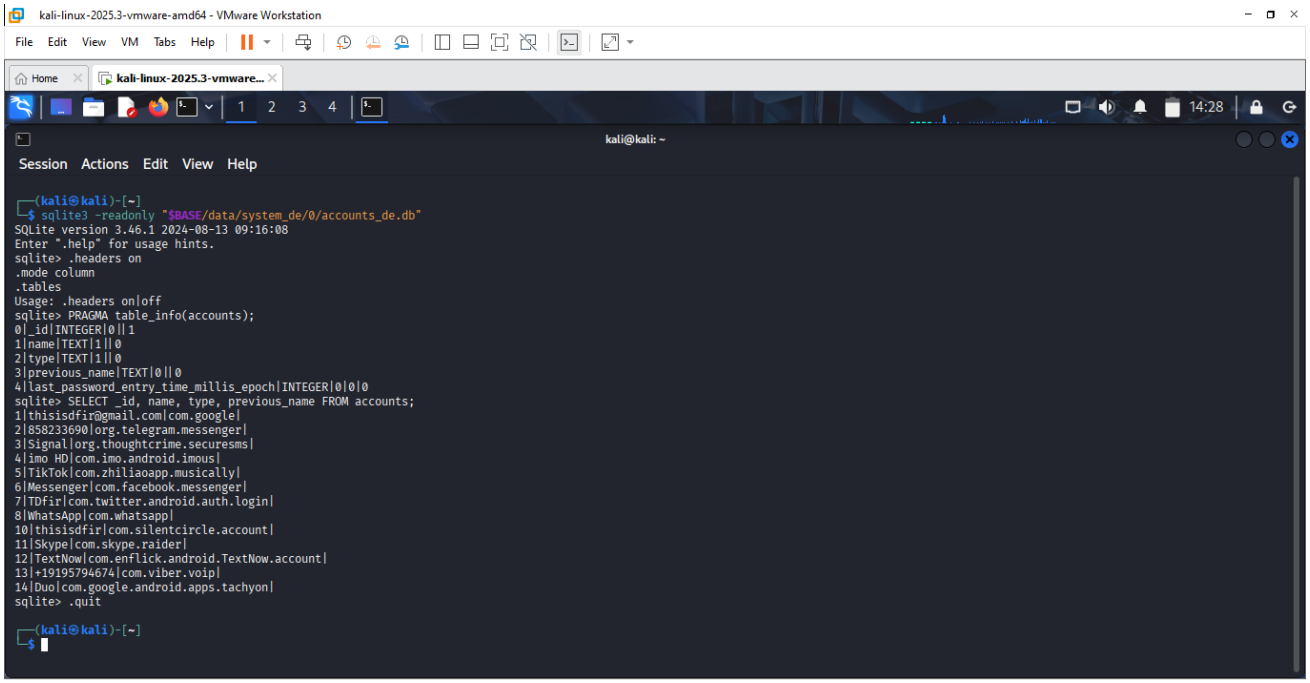


Fig 2.8 User 0 Android Account Association.

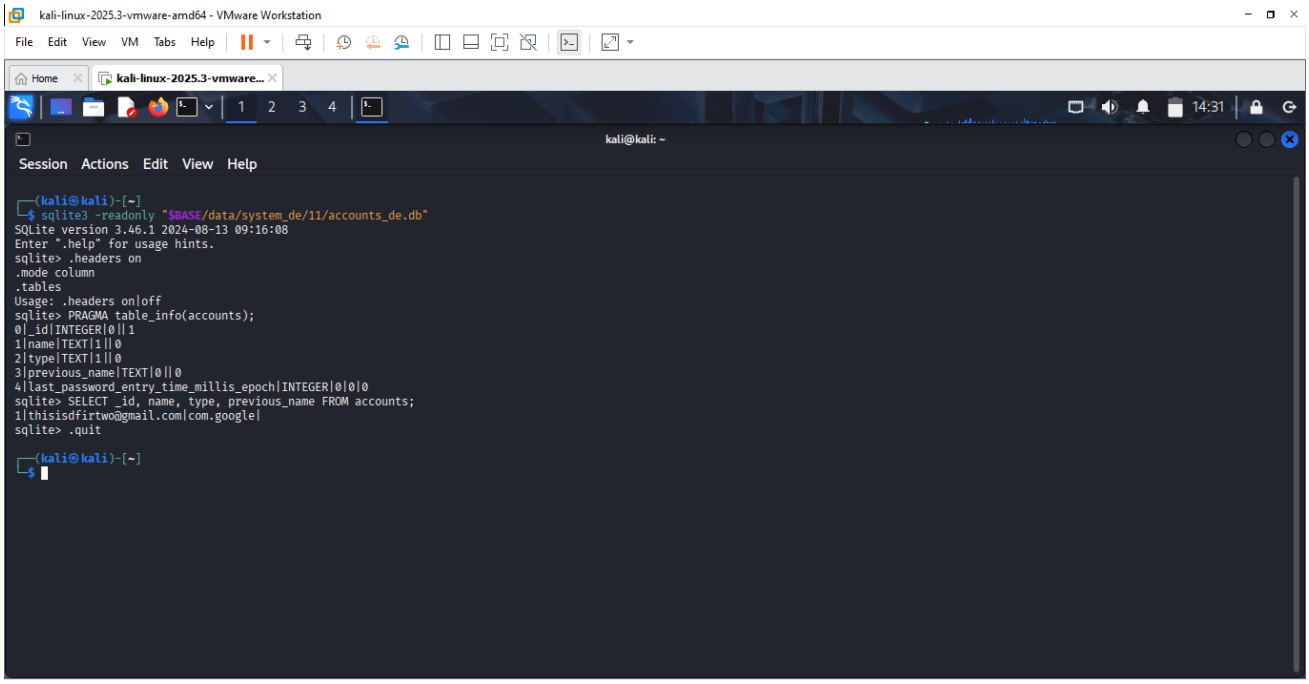


Fig 2.9 User 11 Android Account Association.

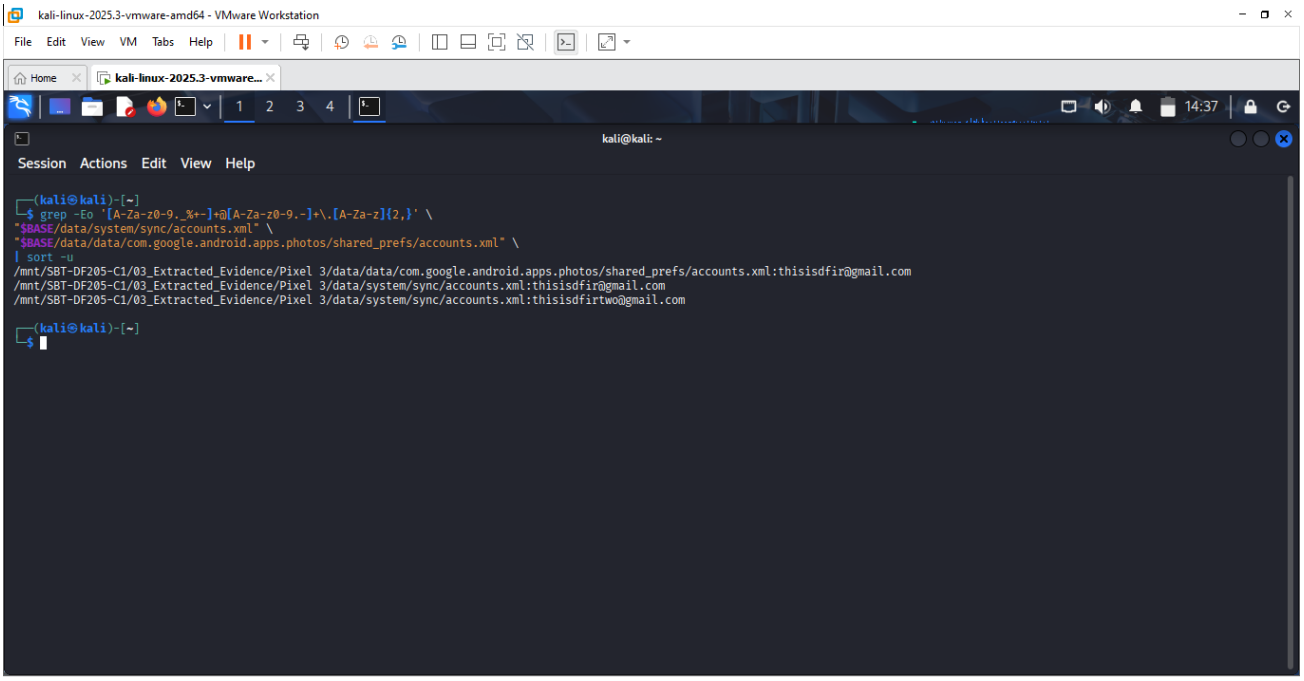


Fig 3.0 Android Account Corroboration Across Source.

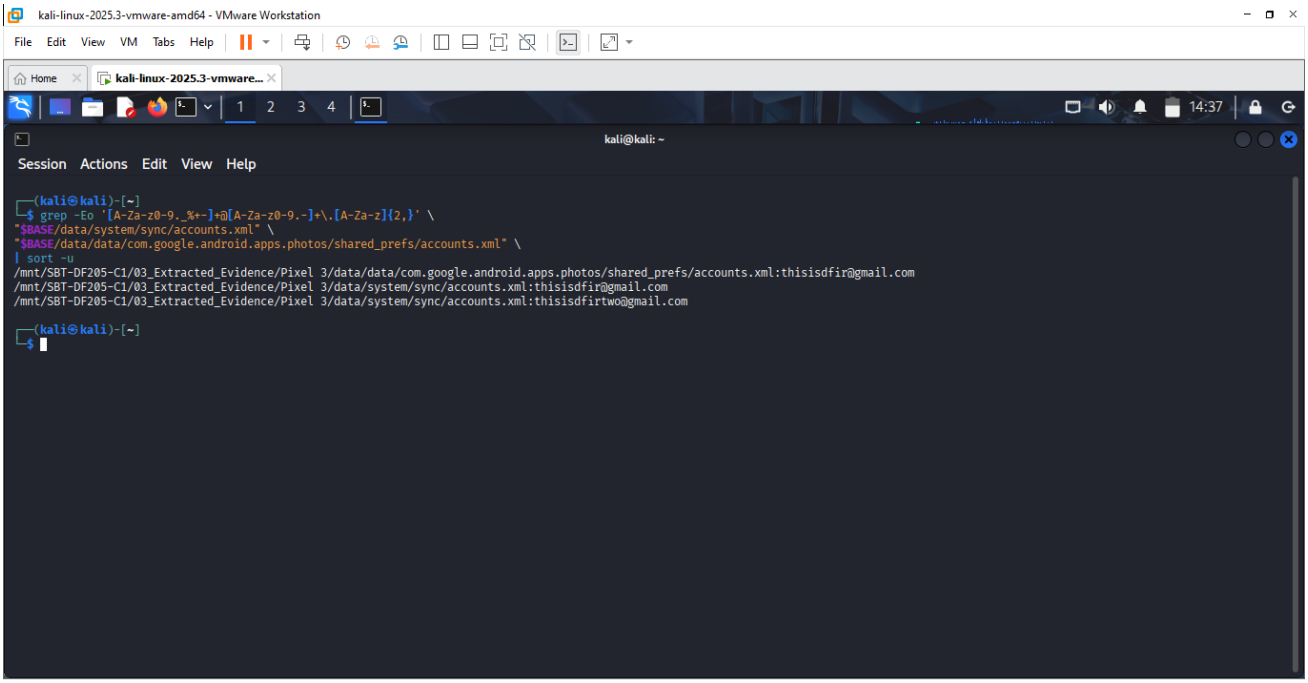


Fig 3.1 Android Account Corroboration Across Sources.

Task 2 findings

Android profile	Profile name	Account record	Account	Type
User 0	**This Is**	1	thisisdfir@gmail.com	com.google

Android profile	Profile name	Account record	Account	Type
User 11	**User 2**	1	thisisdfirtwo@gmail.com	com.google

Explanation

The account identifiers recovered from Android account records were compared with other system and application artefacts. The same Google account appeared across multiple sources, strengthening the association without relying on a single database.

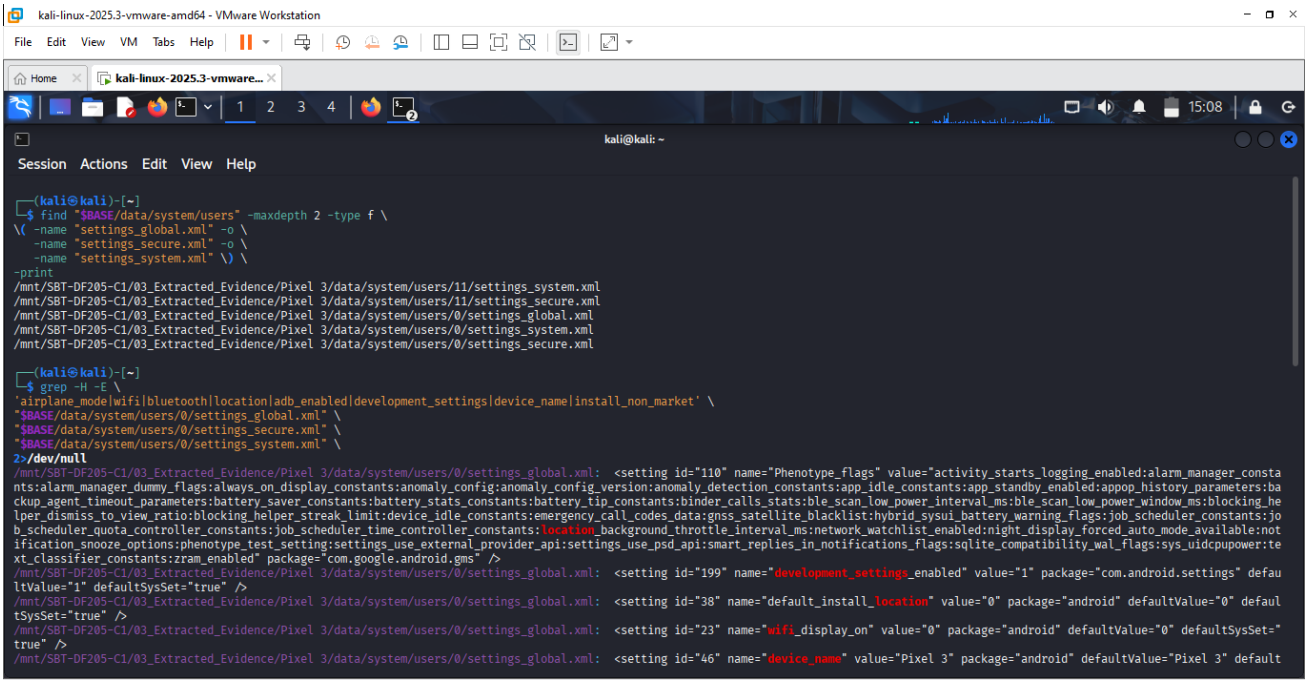


Fig 3.2 Android Relevant System Settings.

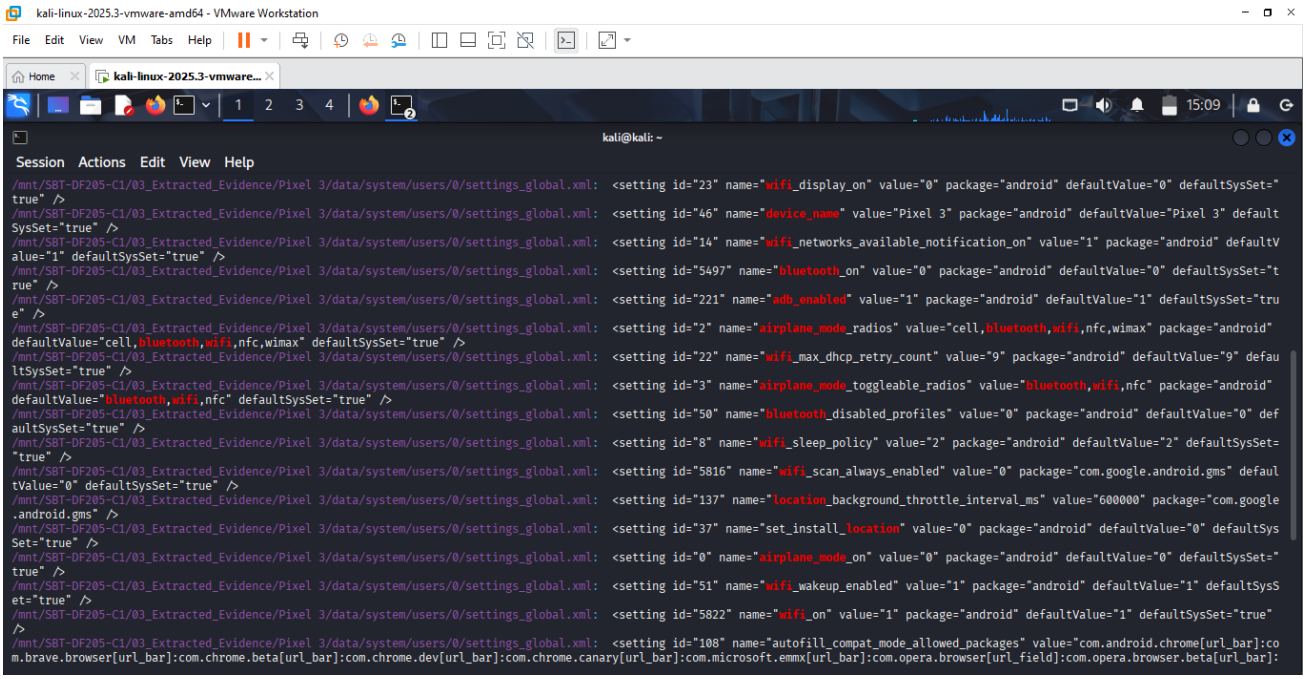


Fig 3.3 Android Relevant System Settings page 1.

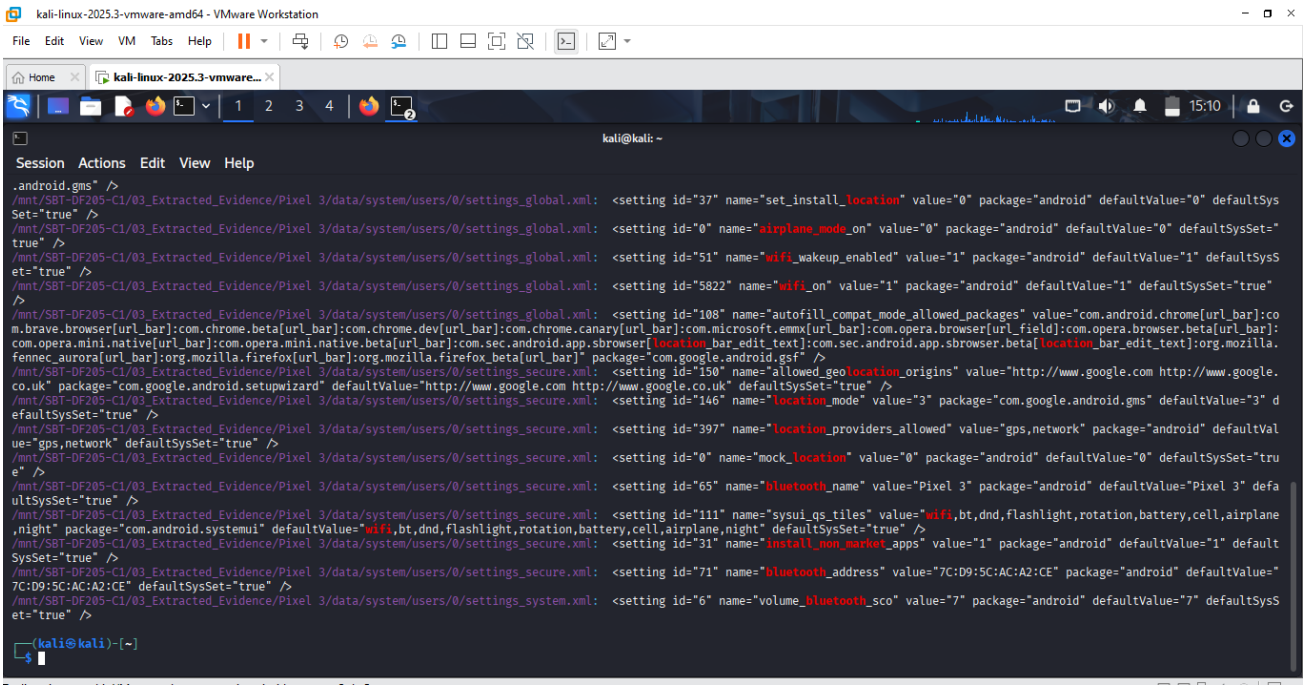


Fig 3.4 Android Relevant System Settings Page 2.

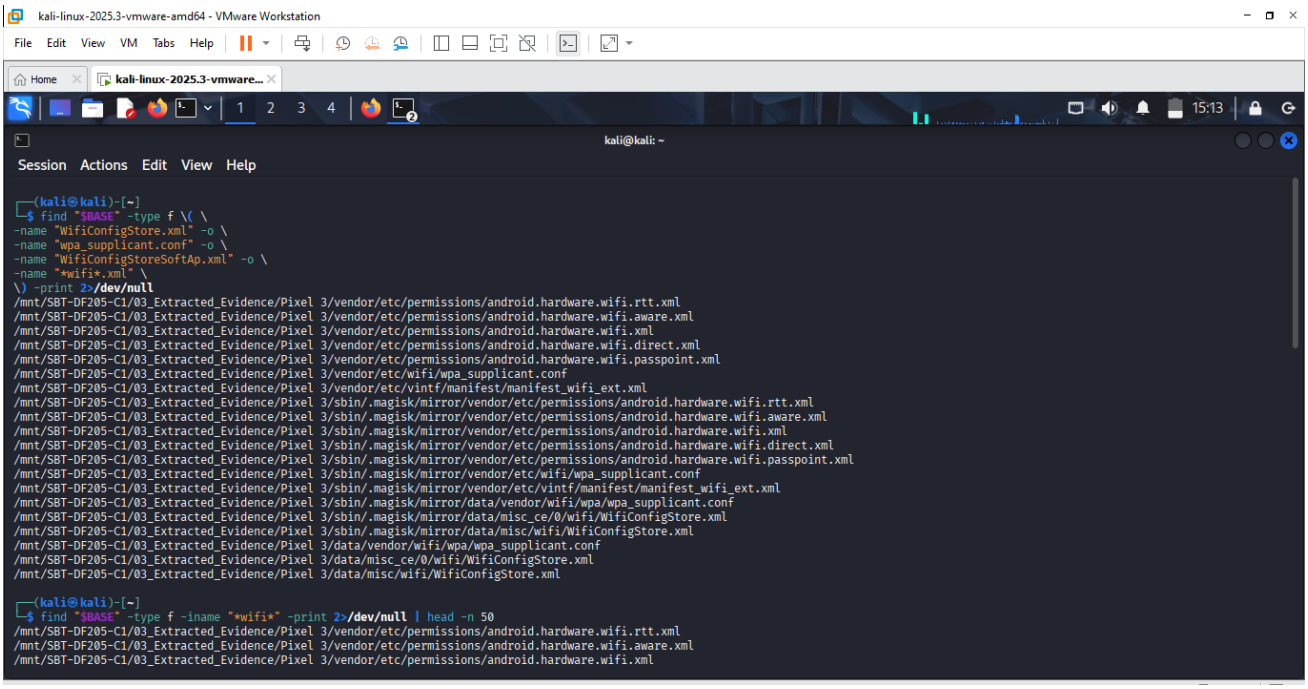


Fig 3.5 Locating Wi-Fi evidence.

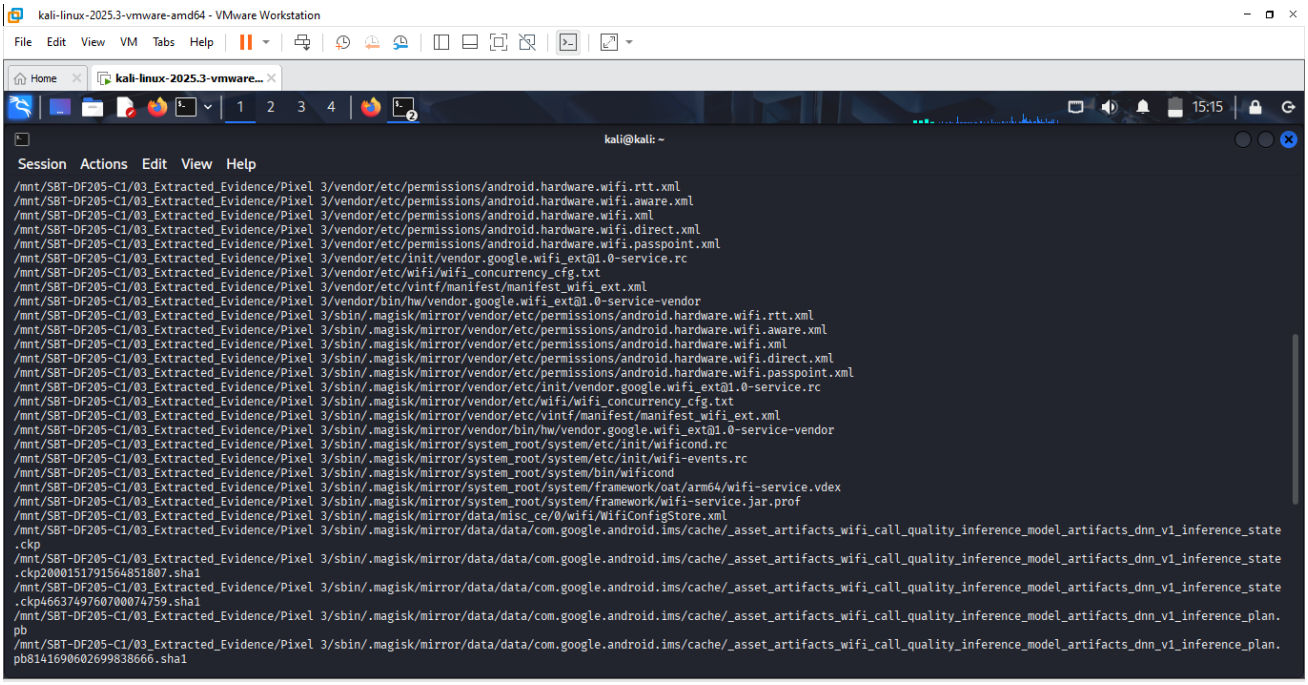


Fig 3.6 Locating Wi-Fi evidence page 1.

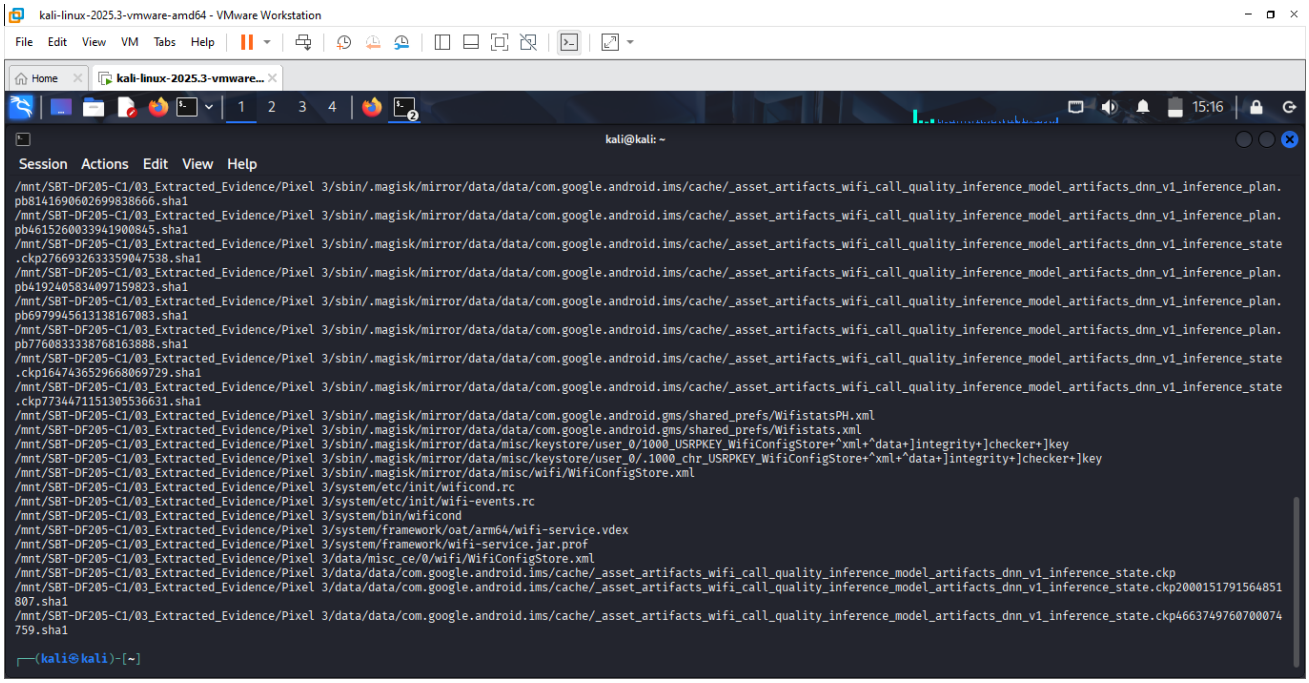


Fig 3.7 Locating Wi-Fi evidence page 2.

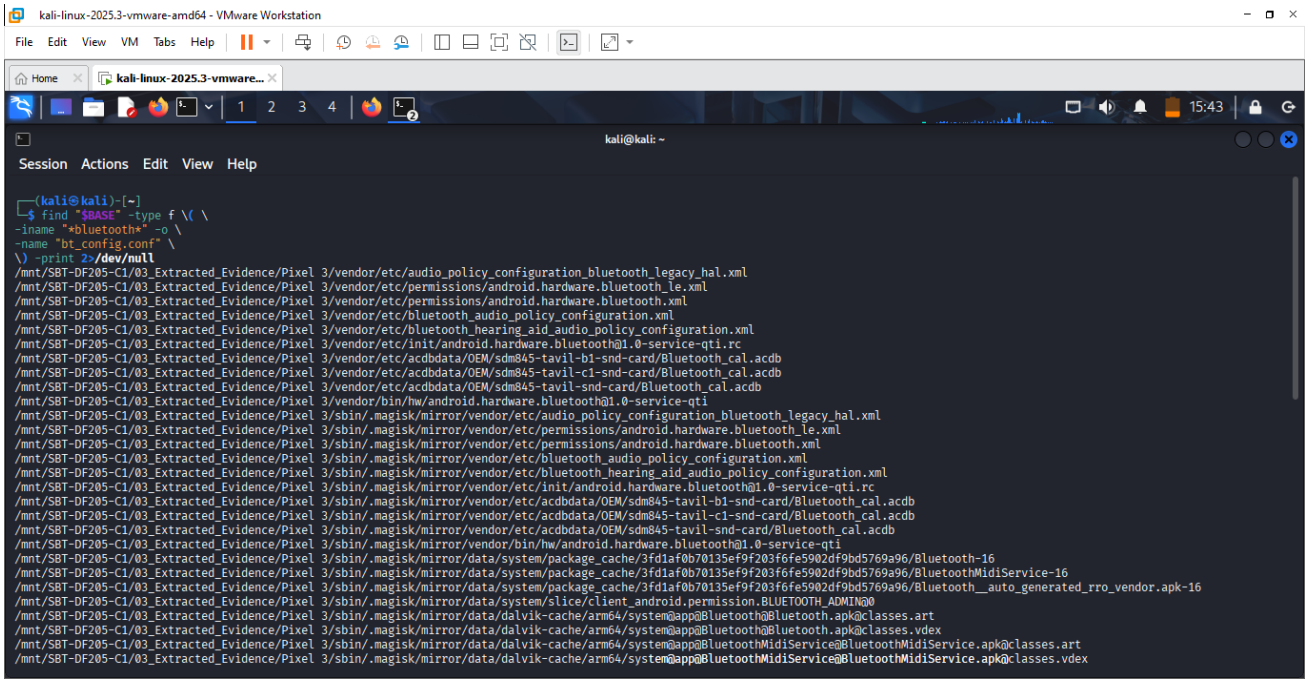


Fig 3.8 Locating Bluetooth evidence.

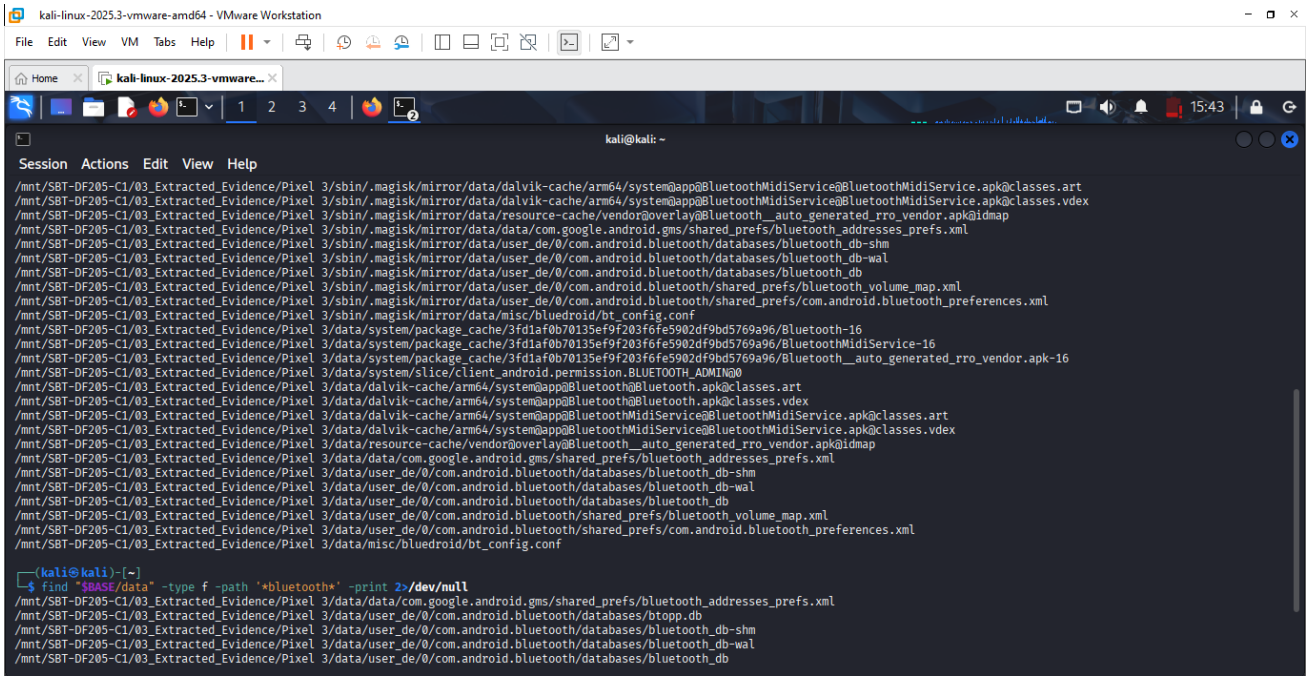


Fig 3.9 Locating Bluetooth evidence page 1.

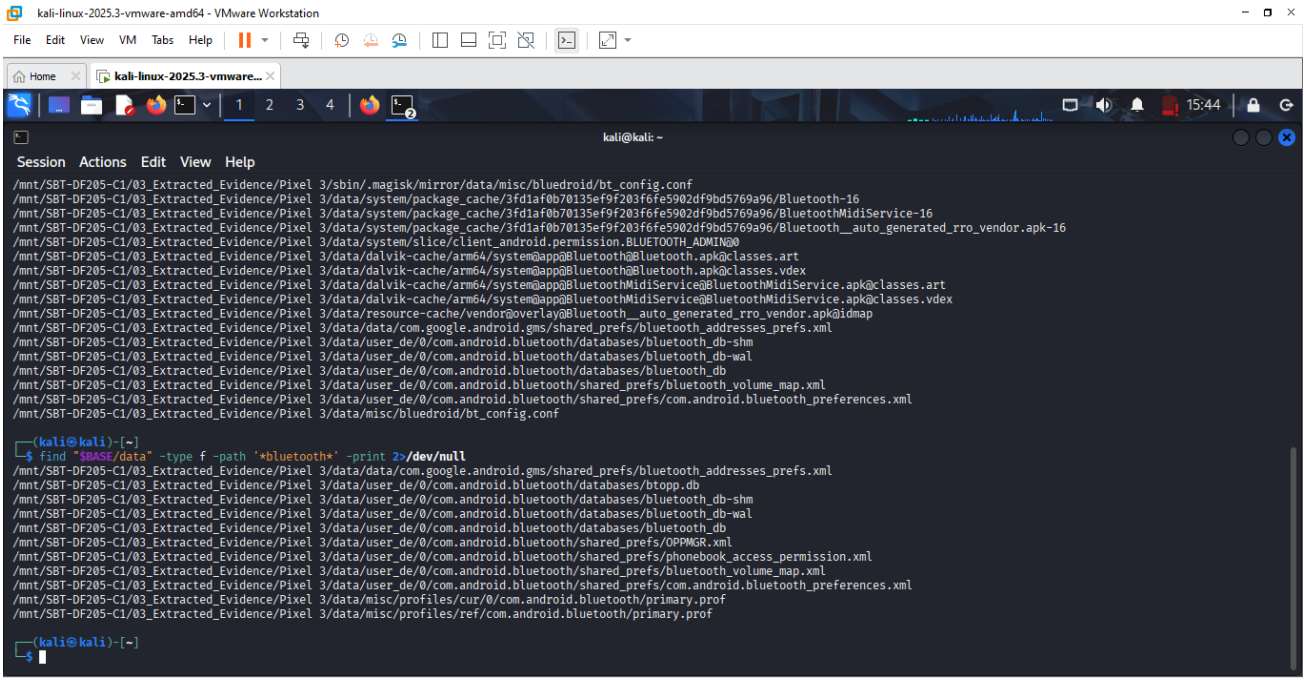


Fig 4.0 Locating Bluetooth evidence page 2.

From User 0's settings files:

Source	Stored key	Value
settings_global.xml	adb_enabled	1
settings_global.xml	wifi_on	1
settings_global.xml	bluetooth_on	0
settings_global.xml	airplane_mode_on	0
settings_global.xml	device_name	Pixel 3
settings_global.xml	development_settings_enabled	1
settings_secure.xml	location_mode	3
settings_secure.xml	location_providers_allowed	gps,network
settings_secure.xml	bluetooth_name	Pixel 3
settings_secure.xml	mock_location	0

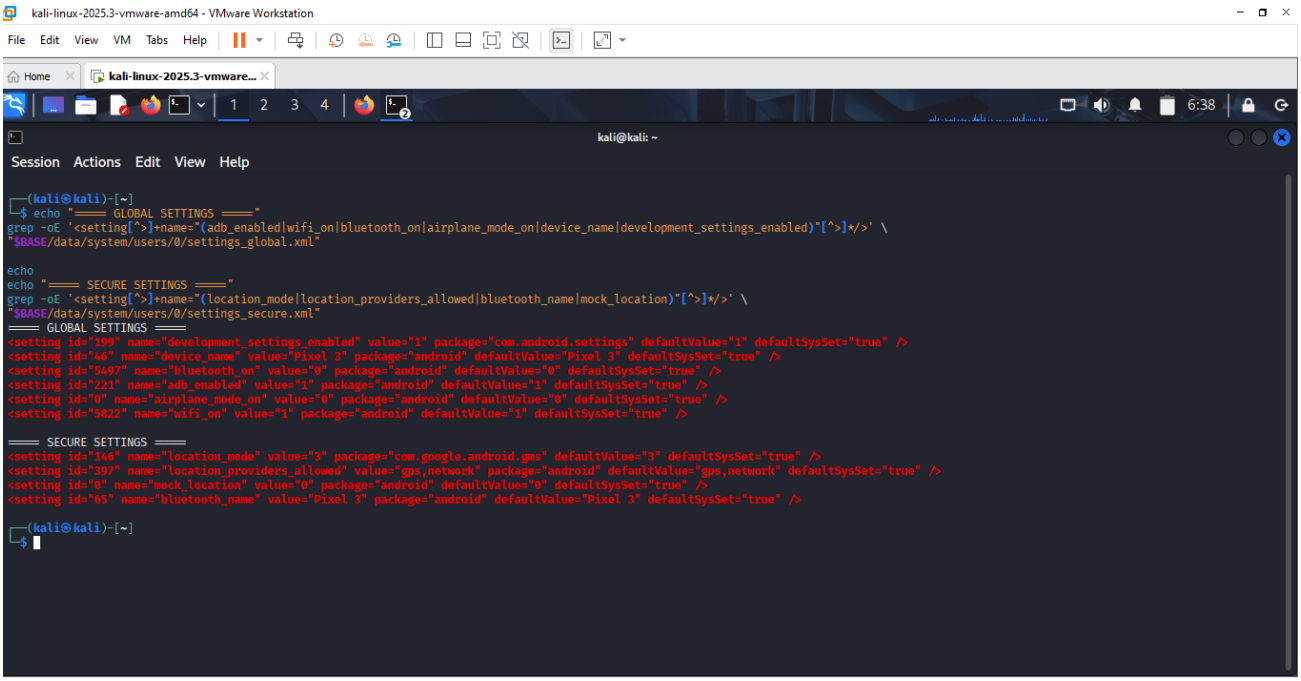


Fig 4.1 Android Relevant Global and Secure Settings.

Explanation

This screenshot shows the Android settings that were relevant to the investigation, including ADB, Wi-Fi, Bluetooth and location configuration. These values describe the stored device state at the time of acquisition but do not by themselves prove when a particular connection occurred.

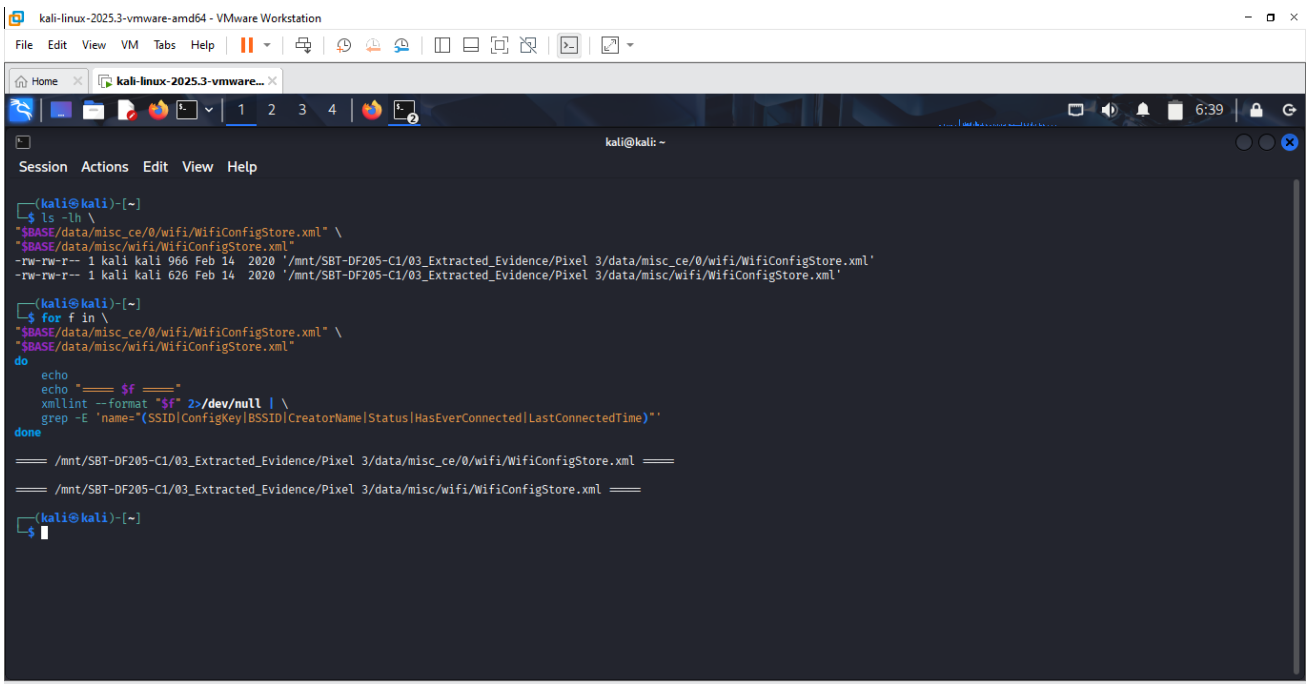


Fig 4.2 Saved Wi-Fi Network Configuration.

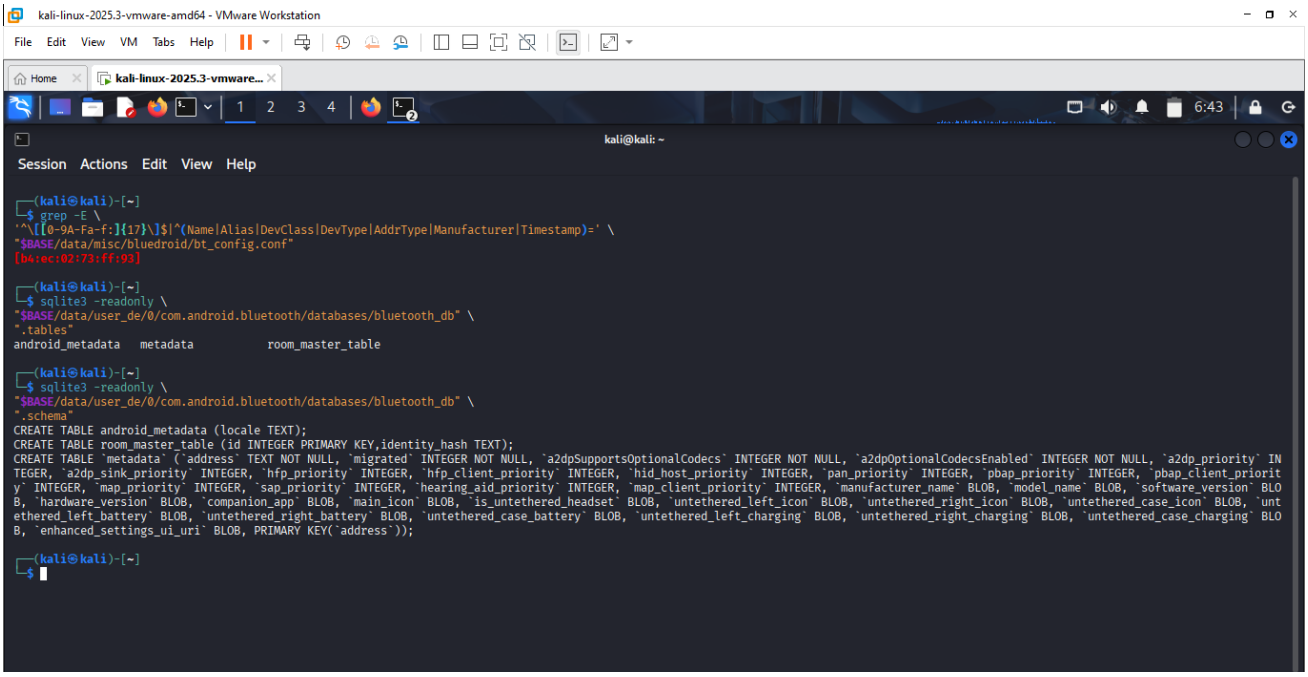


Fig 4.3 Bluetooth Configuration and Database Artefacts.

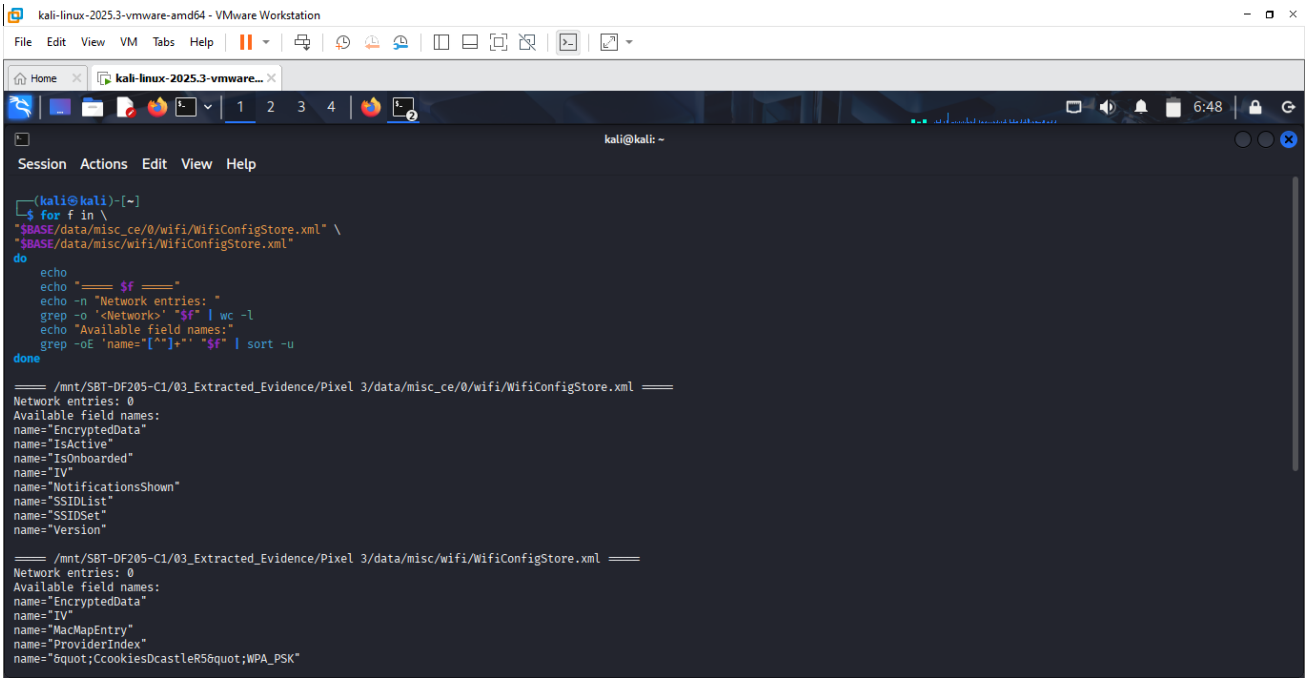


Fig 4.4 Wi-Fi network count/field output.

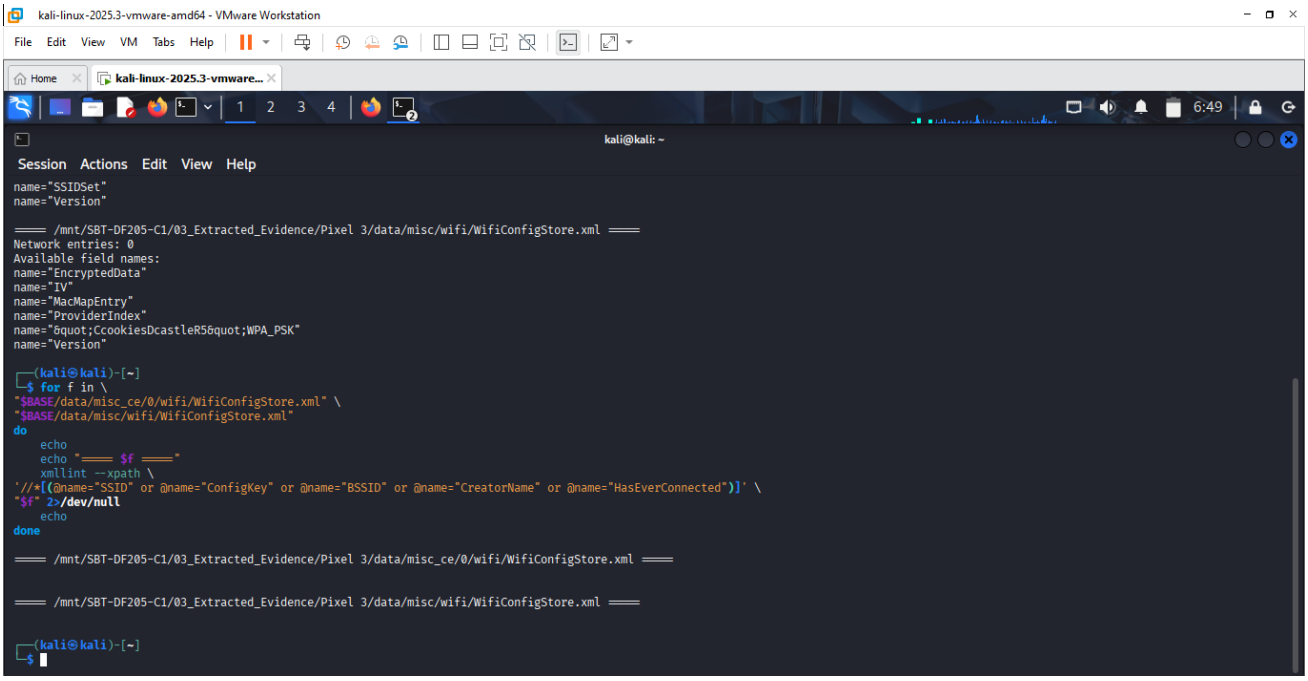


Fig 4.5 Wi-Fi network count/field output Page 1.

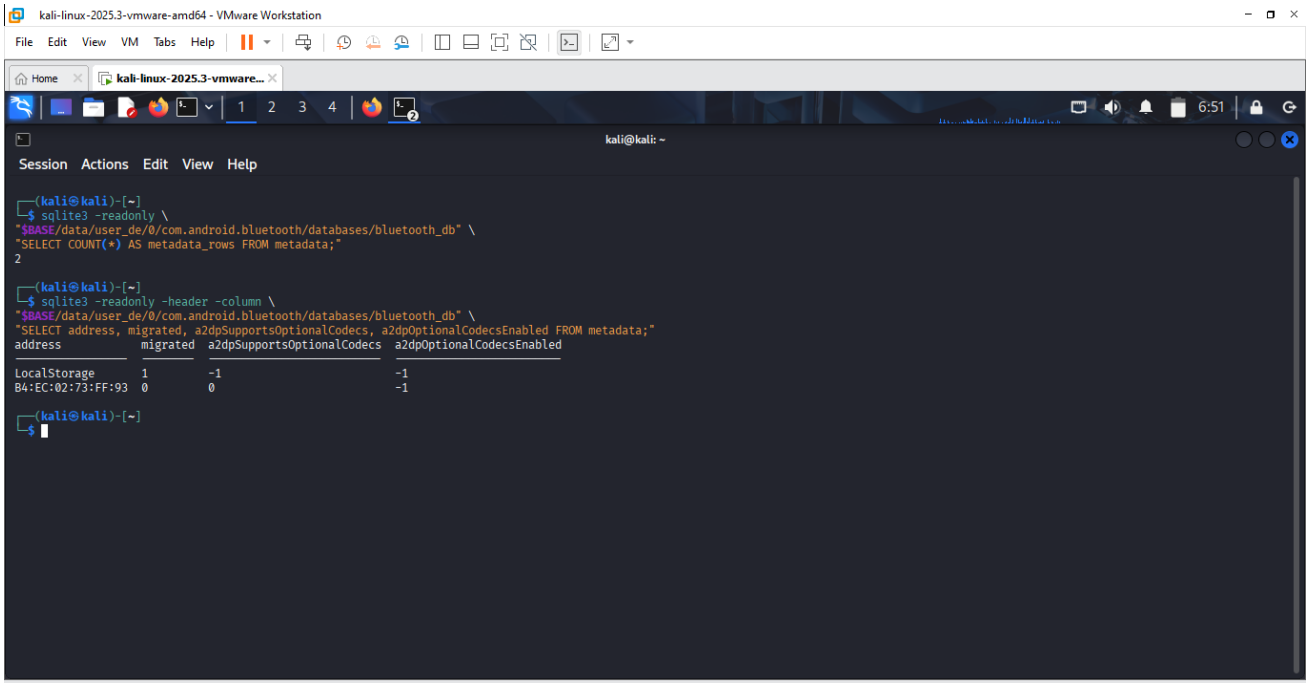


Fig 4.6 Bluetooth Stored Device Identifier and Metadata.

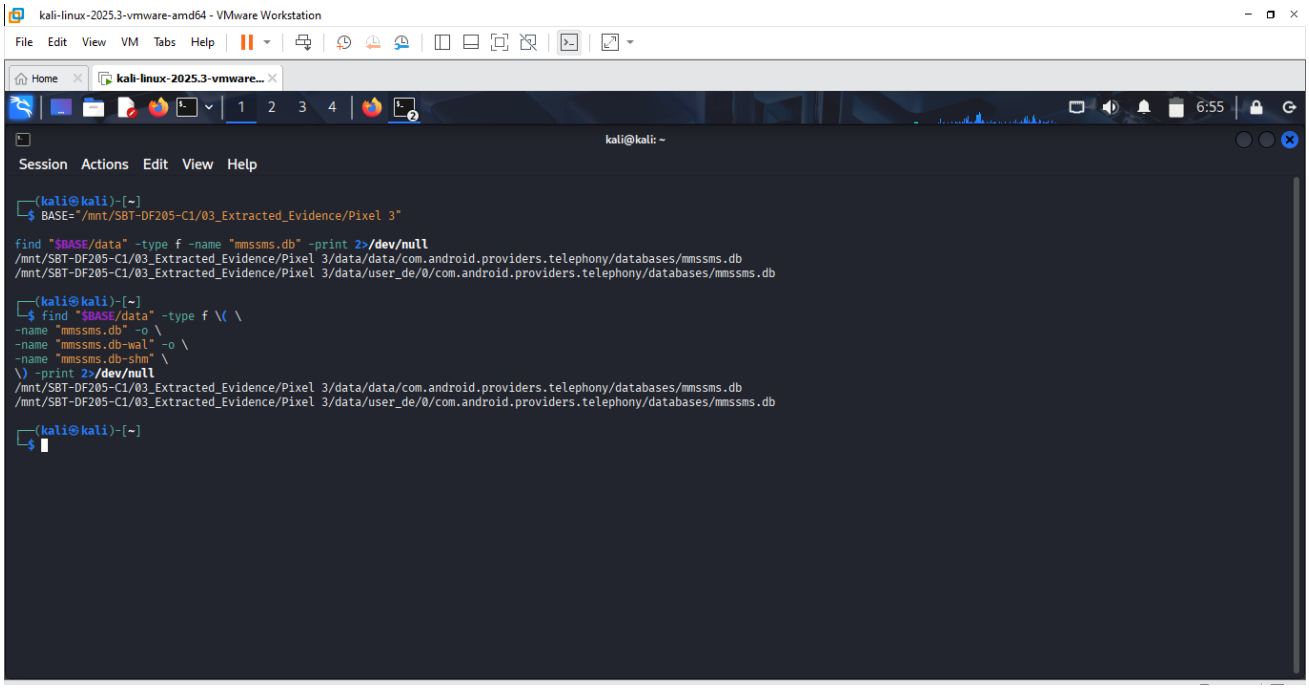


Fig 4.7 Telephony Provider Database Locations and Sidecar.

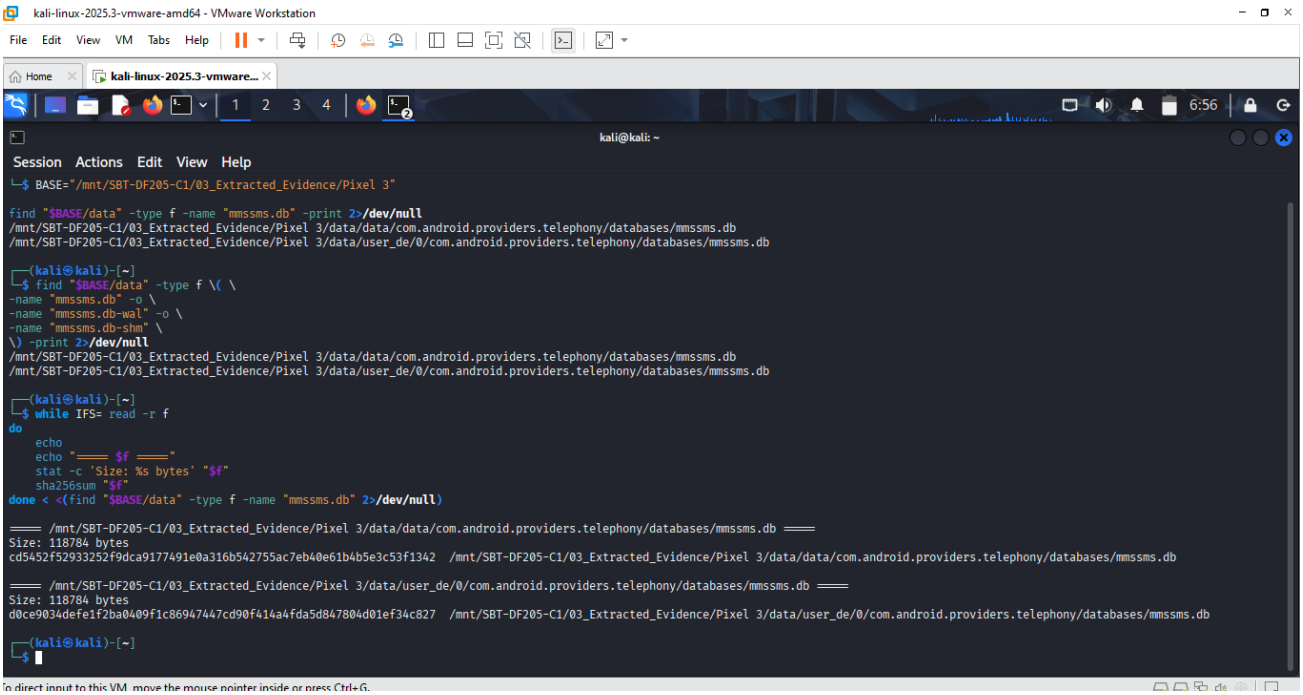


Fig 4.8 Telephony Database SHA-256 and File Size Comparison.

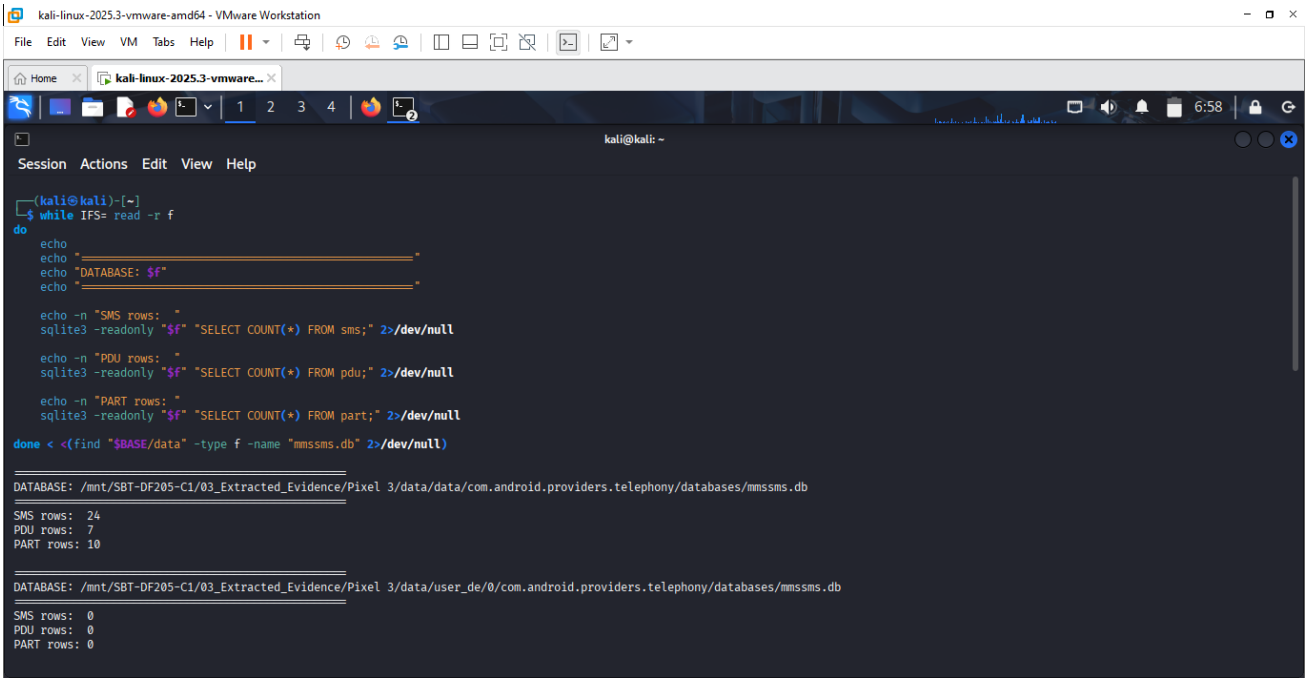


Fig 4.9 Telephony Provider Database Comparison and Row Count.

Explanation

The two mmssms.db files were examined separately because their paths and contents differed. The primary database contained 24 SMS records, 7 PDU records and 10 parts, while the user_de copy contained no communication rows.

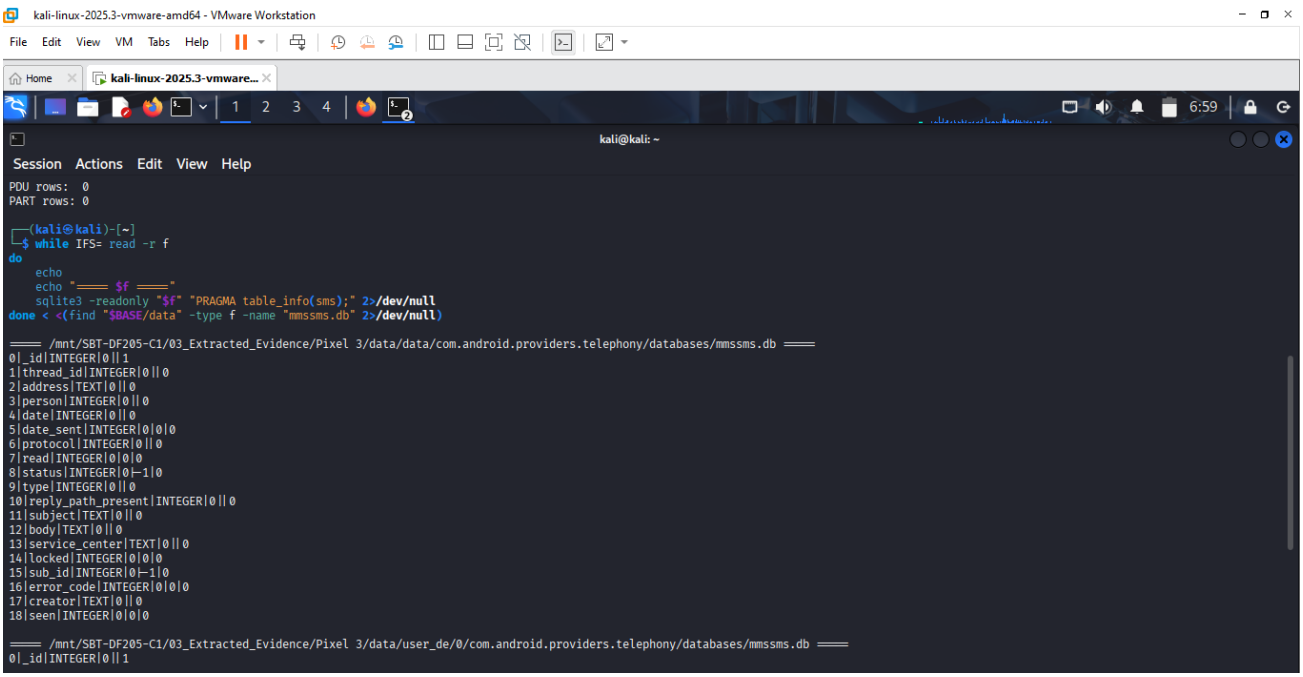


Fig 5.0 Telephony Provider Database Comparison and Row Count page 1.

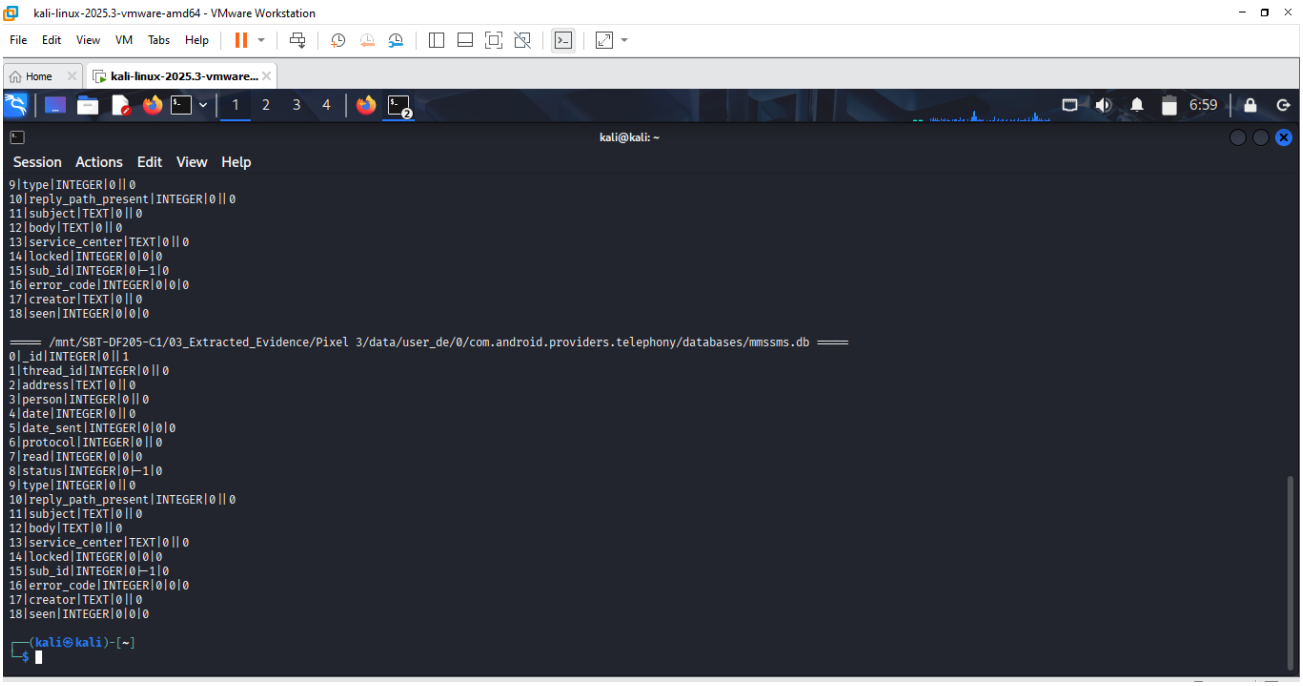


Fig 5.1 Telephony Provider Database Comparison and Row Count page 2.

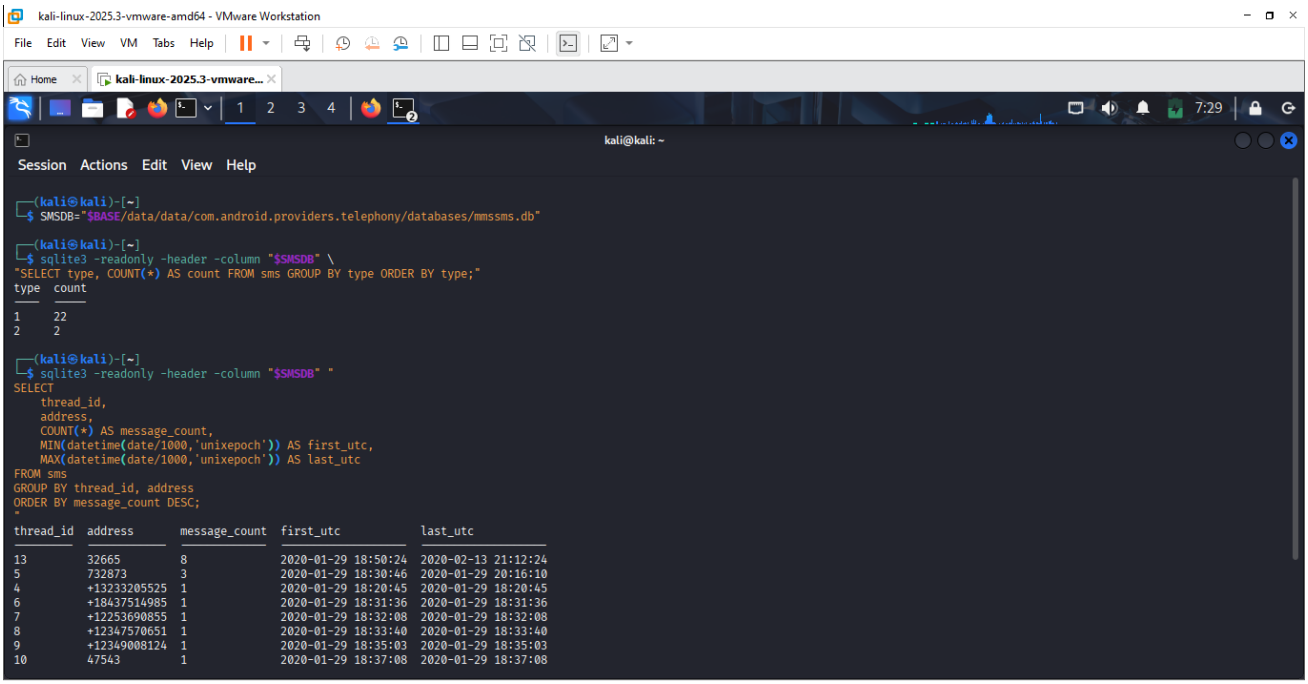


Fig 5.2 SMS Conversation and Participant Summary.

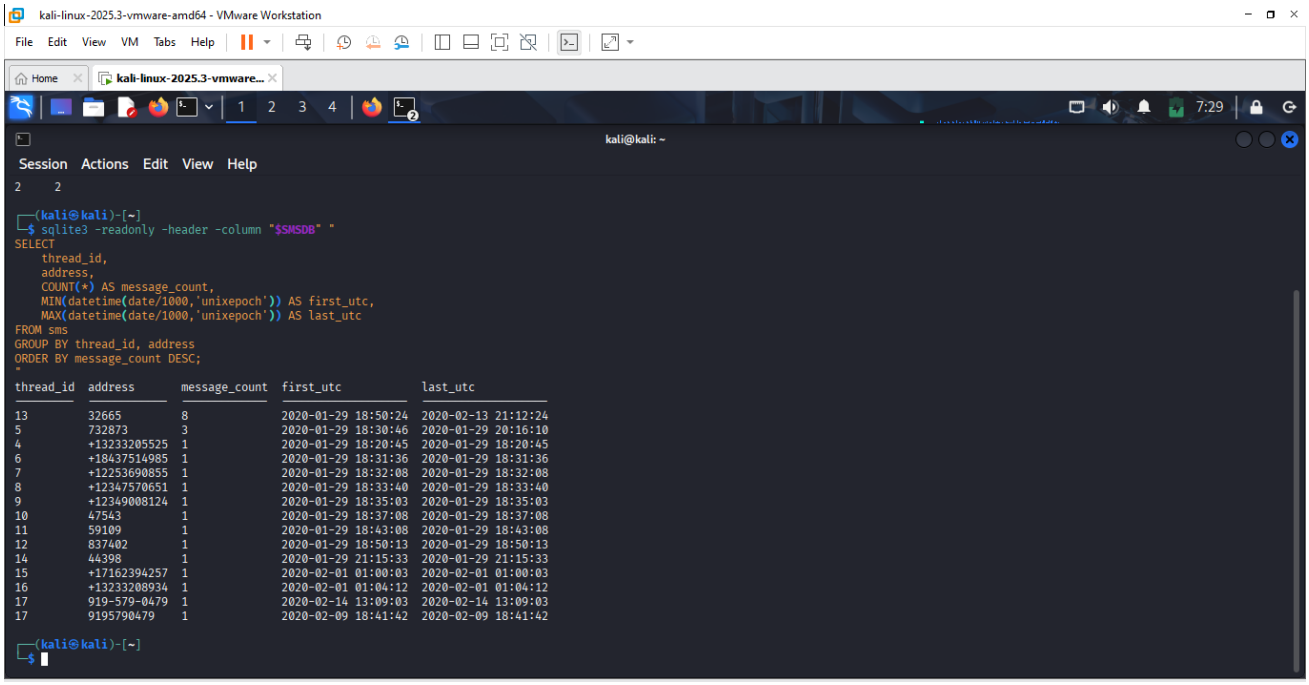


Fig 5.3 SMS Conversation and Participant Summary Page 1.

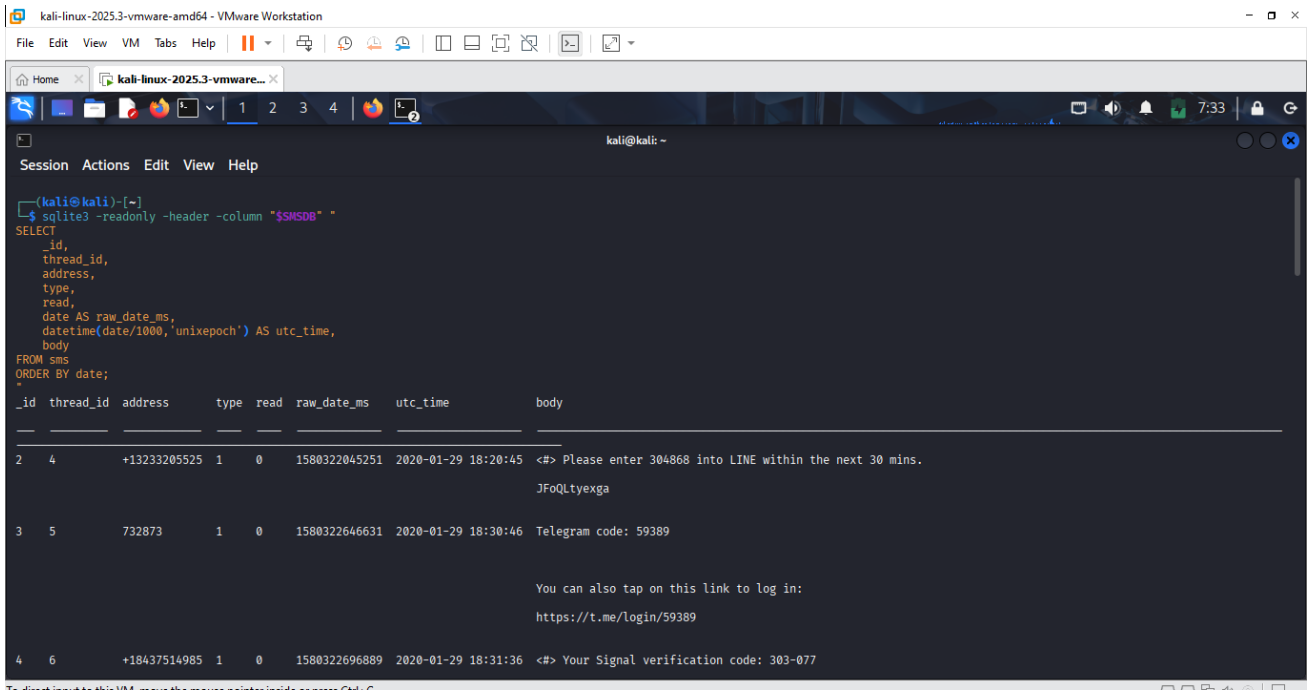


Fig 5.4 Display all 24 SMS records chronologically.

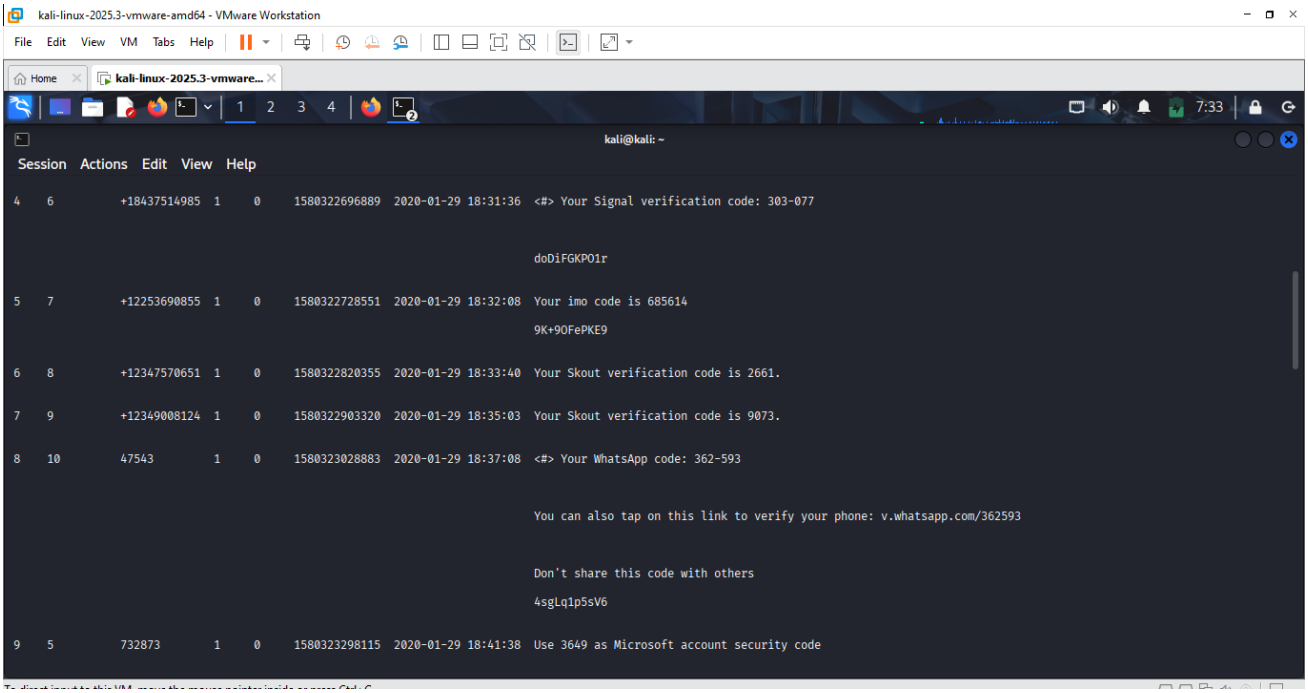


Fig 5.5 Display all 24 SMS records chronologically Page 1.

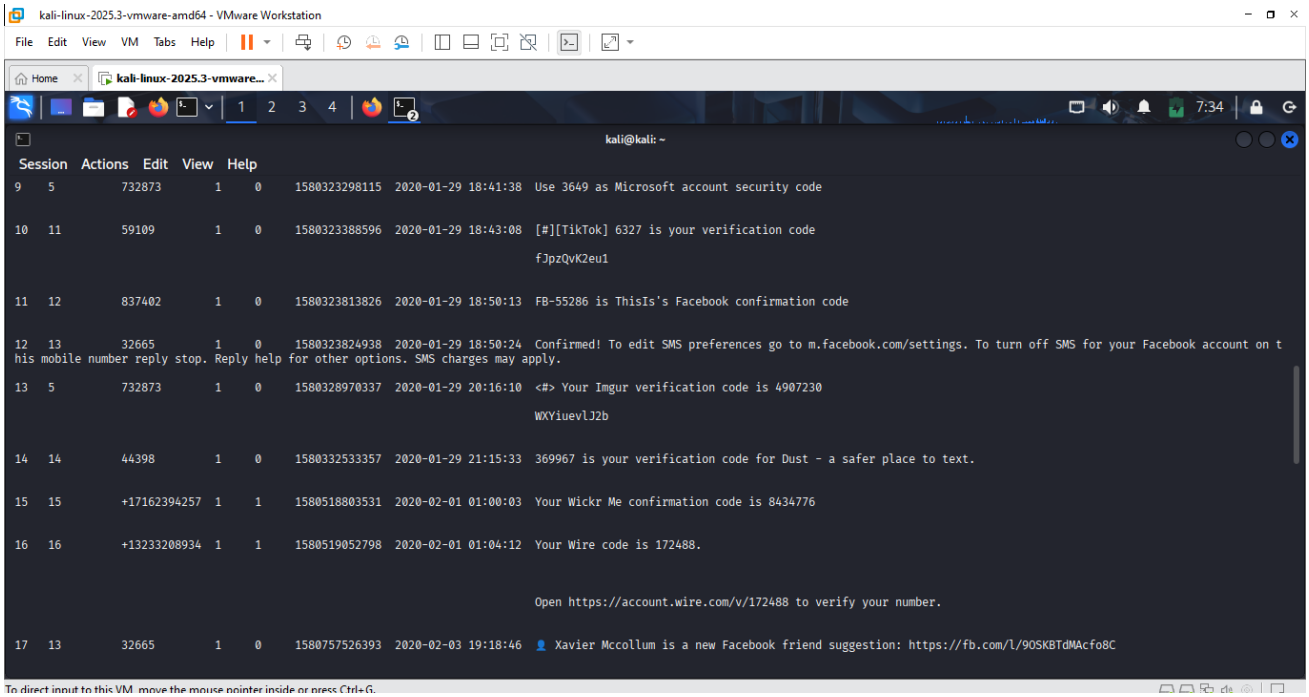


Fig 5.6 Display all 24 SMS records chronologically Page 2.

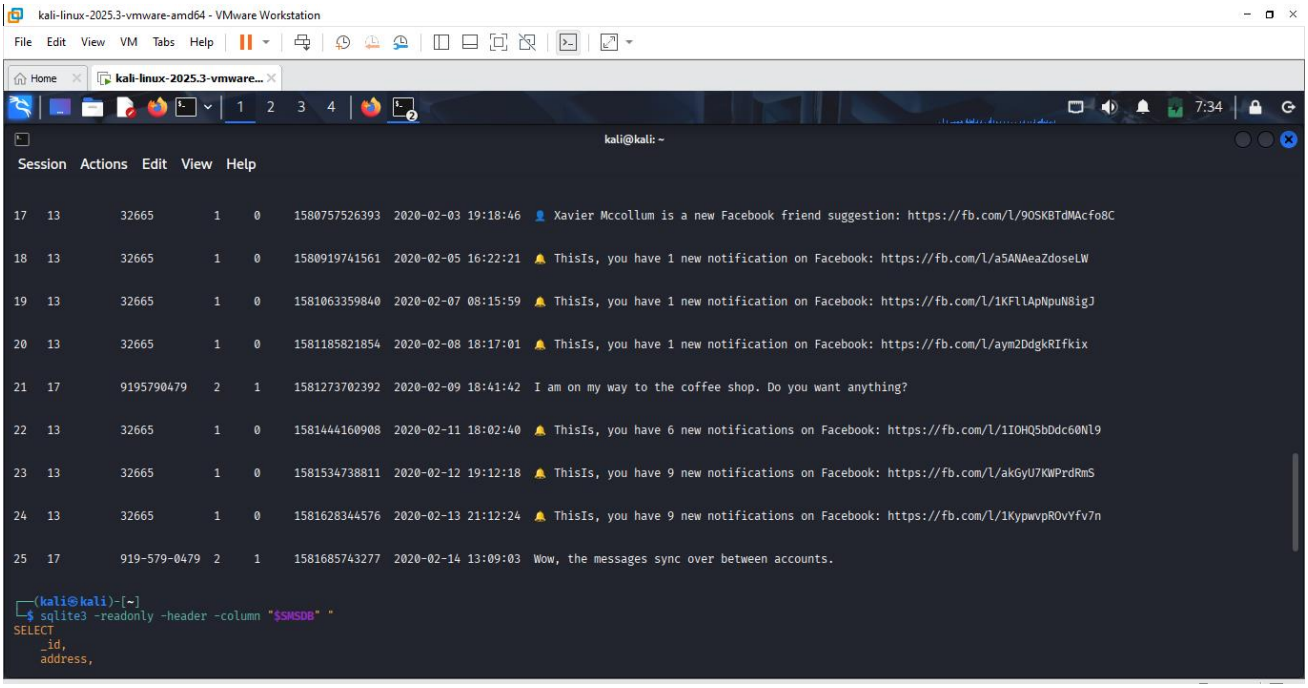


Fig 5.7 Display all 24 SMS records chronologically Page 3.

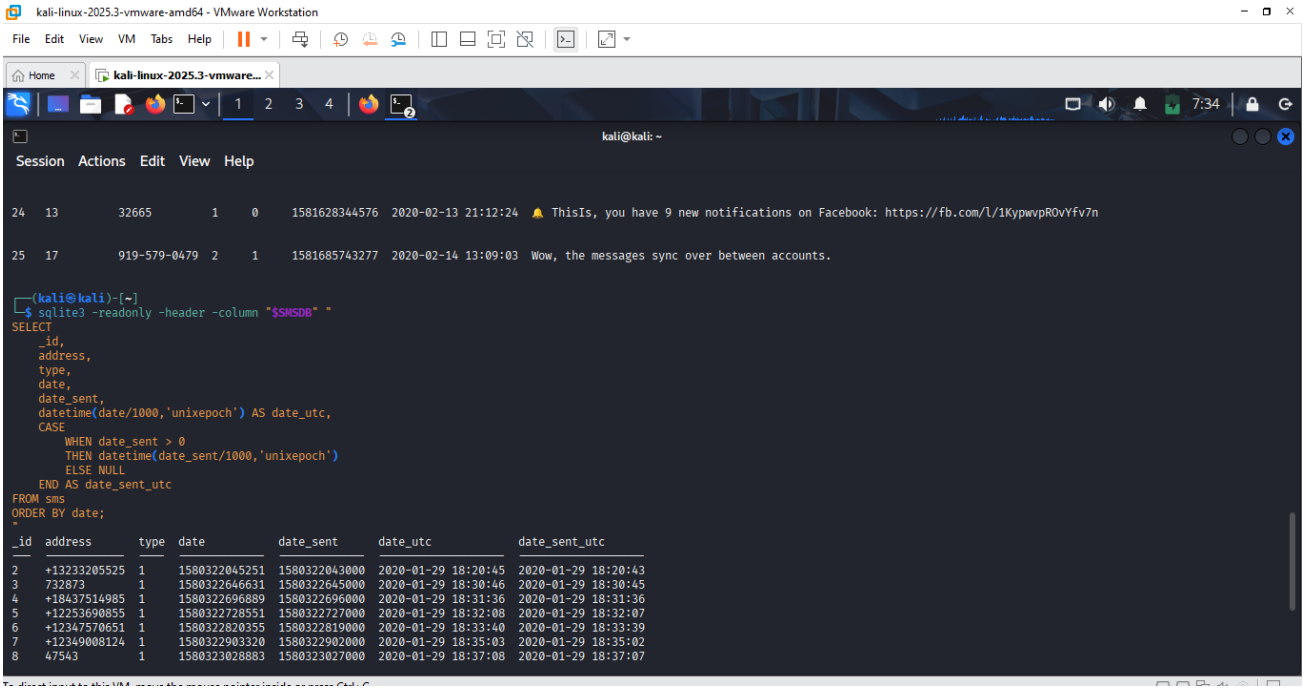


Fig 5.8 Display all 24 SMS records chronologically Page 4.

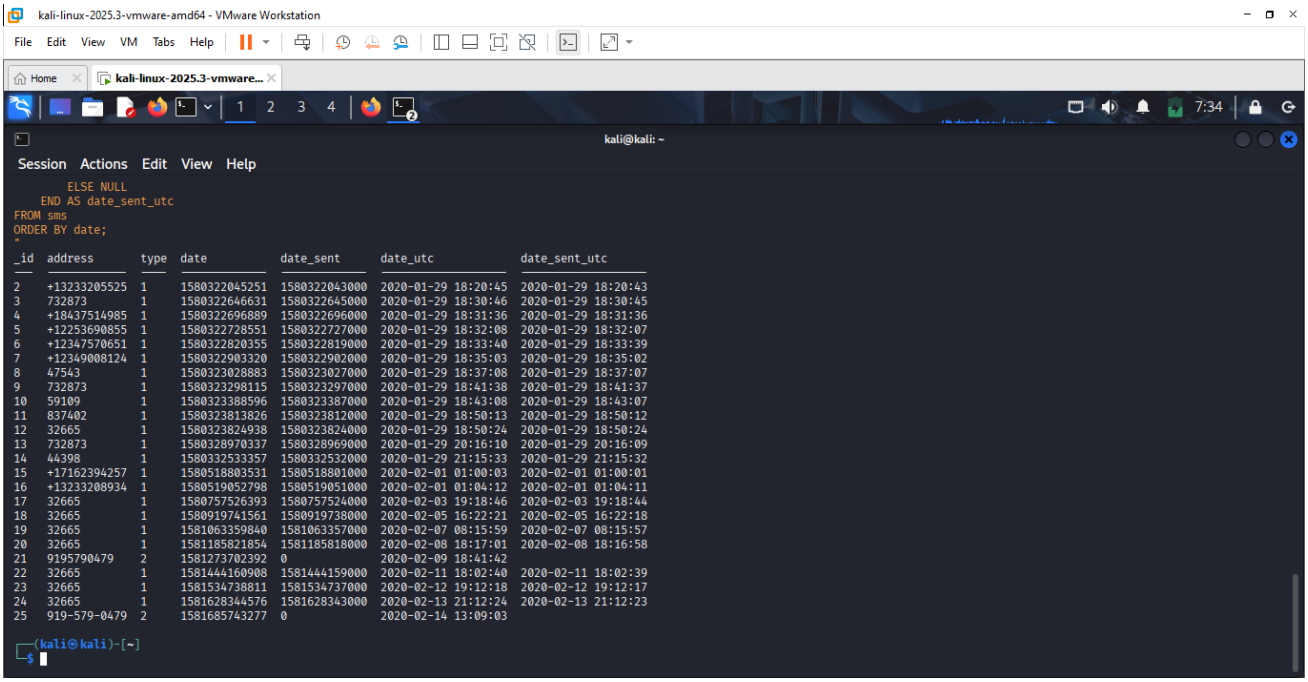


Fig 5.9 Display all 24 SMS records chronologically Page 5.

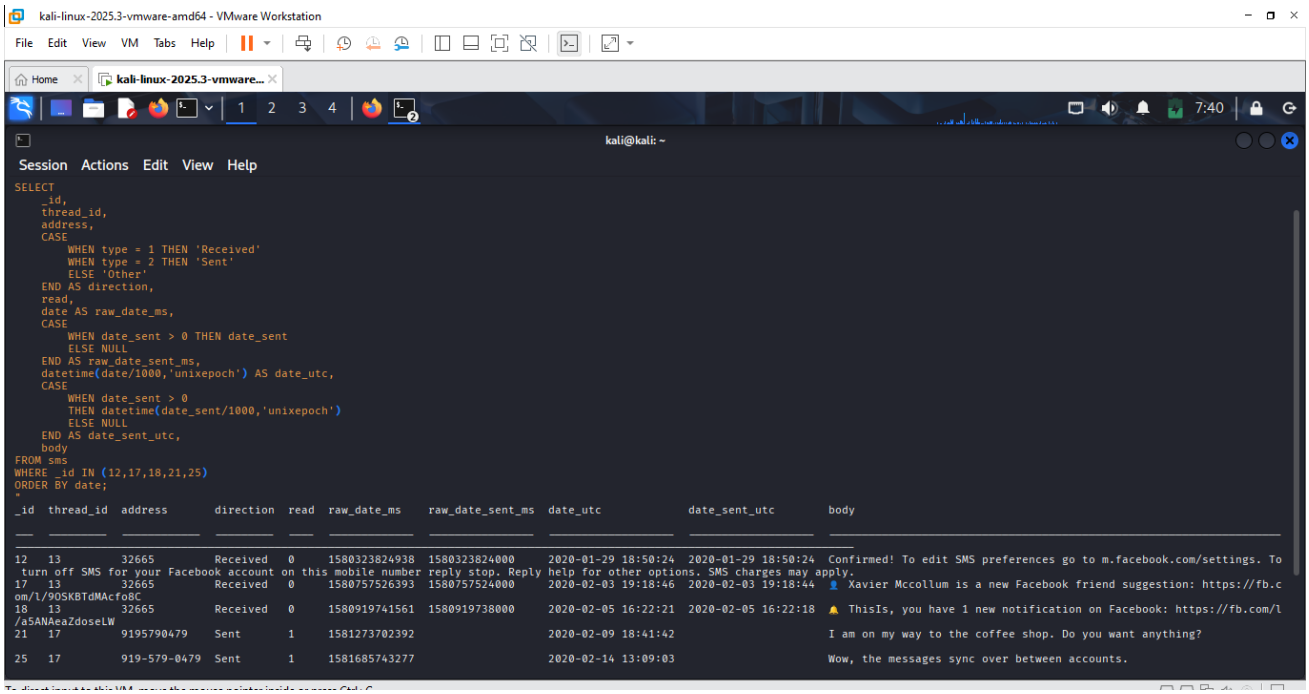


Fig 6.0 Selected Five SMS Records with UTC Conversion.

Explanation

Five SMS records were extracted with their record IDs, addresses, direction, read state, raw timestamps and UTC conversions. This reduced the large SMS dataset to a small set of reproducible records that could be used in the report and timeline without exposing unnecessary verification codes.

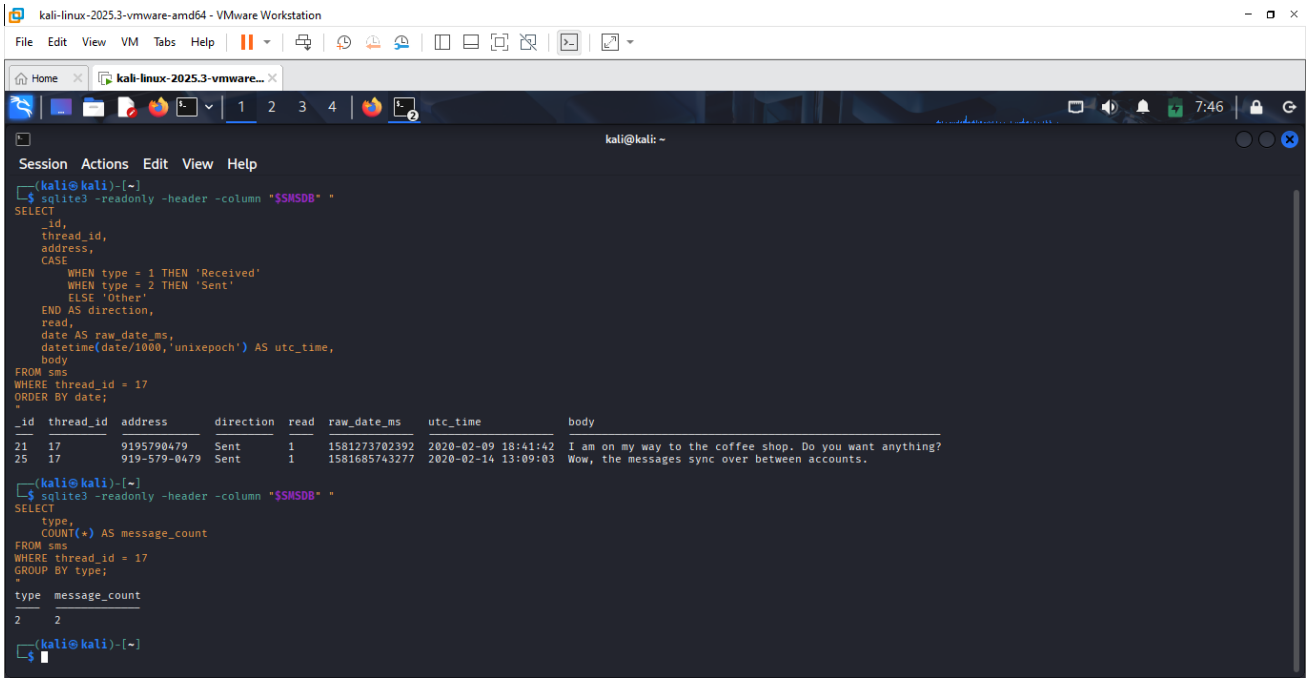


Fig 6.1 SMS Thread 17 Conversation Reconstruction.

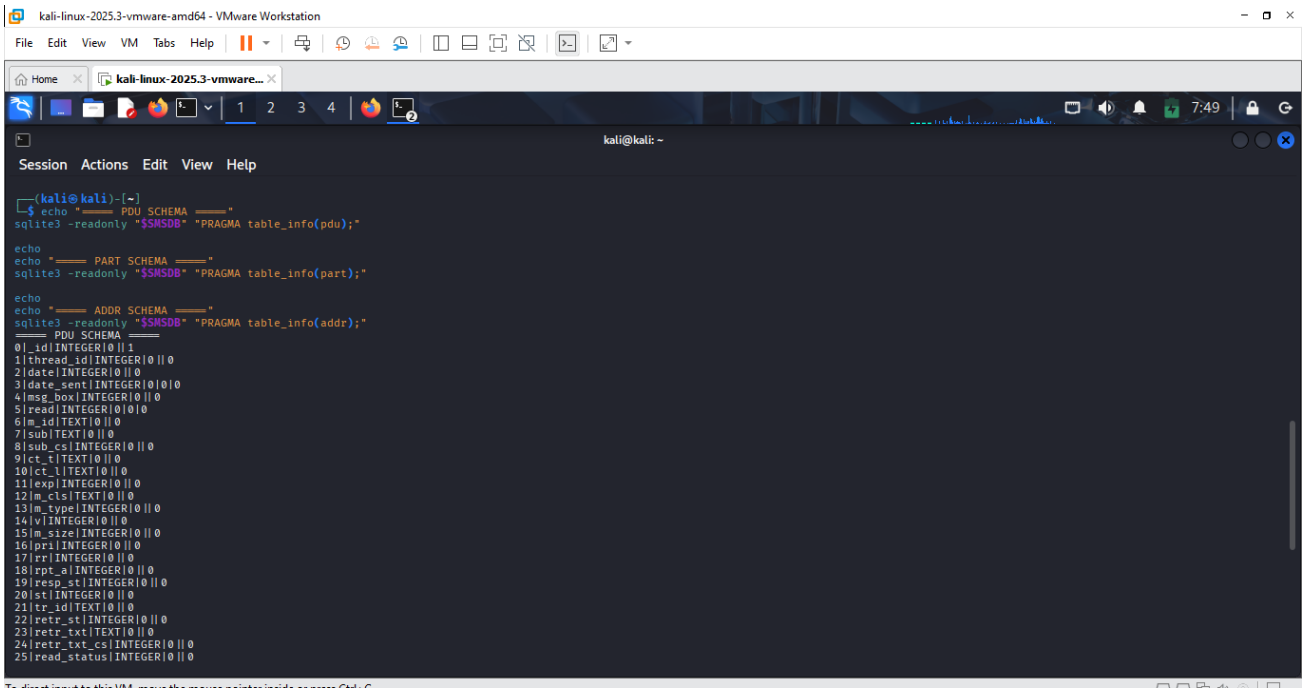


Fig 6.2 MMS PDU Records and Validated Timestamps.

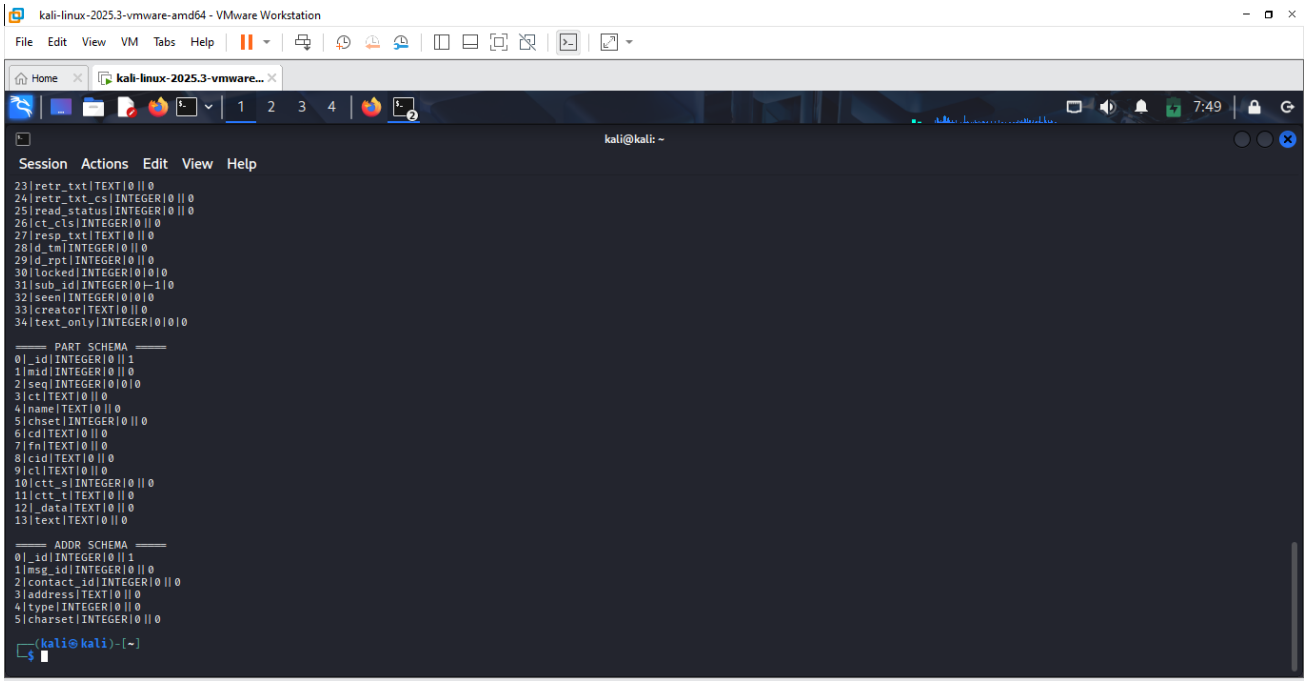


Fig 6.3 MMS PDU Records and Validated Timestamps Page 1.

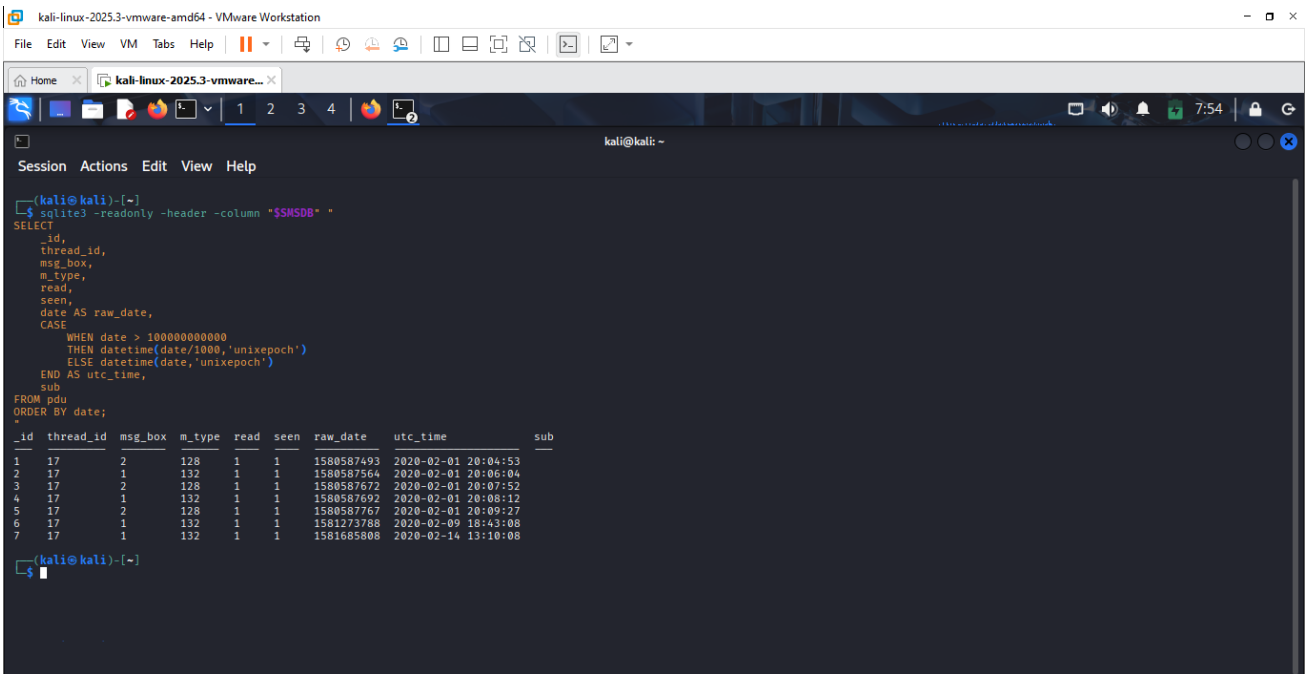


Fig 6.4 MMS PDU Records and Validated Timestamps.

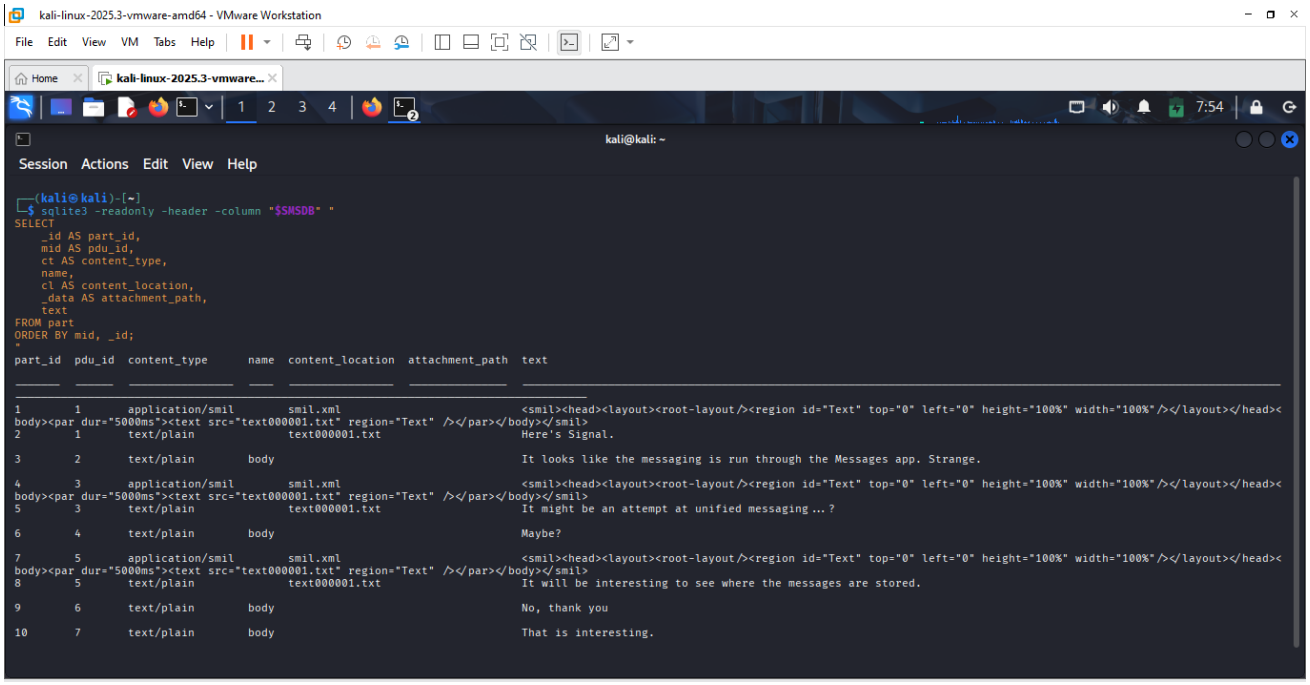


Fig 6.5 MMS Parts Text and Attachment References.

Explanation

The part table was used to connect MMS PDU records to their text components and attachment references. This allowed the MMS communication to be reconstructed from database relationships rather than relying on a single isolated table.

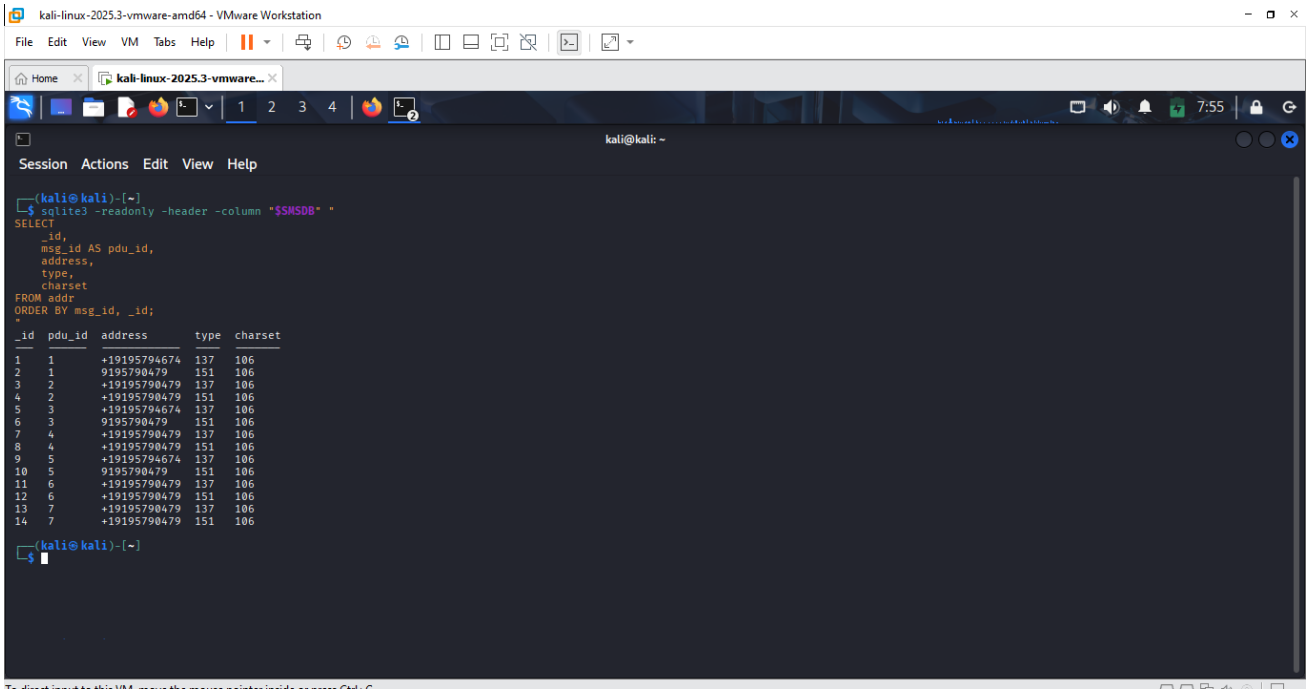


Fig 6.6 MMS Participant Address Records.

MMS — completed

Field	Finding
PDU ID	1
Thread	17
Message box	2 — Sent
MMS type	128
Raw timestamp	1580587493
UTC	2020-02-01 20:04:53 UTC
From	+19195794674
To	9195790479
Part ID	2
Content type	text/plain
Text	“Here's Signal.”

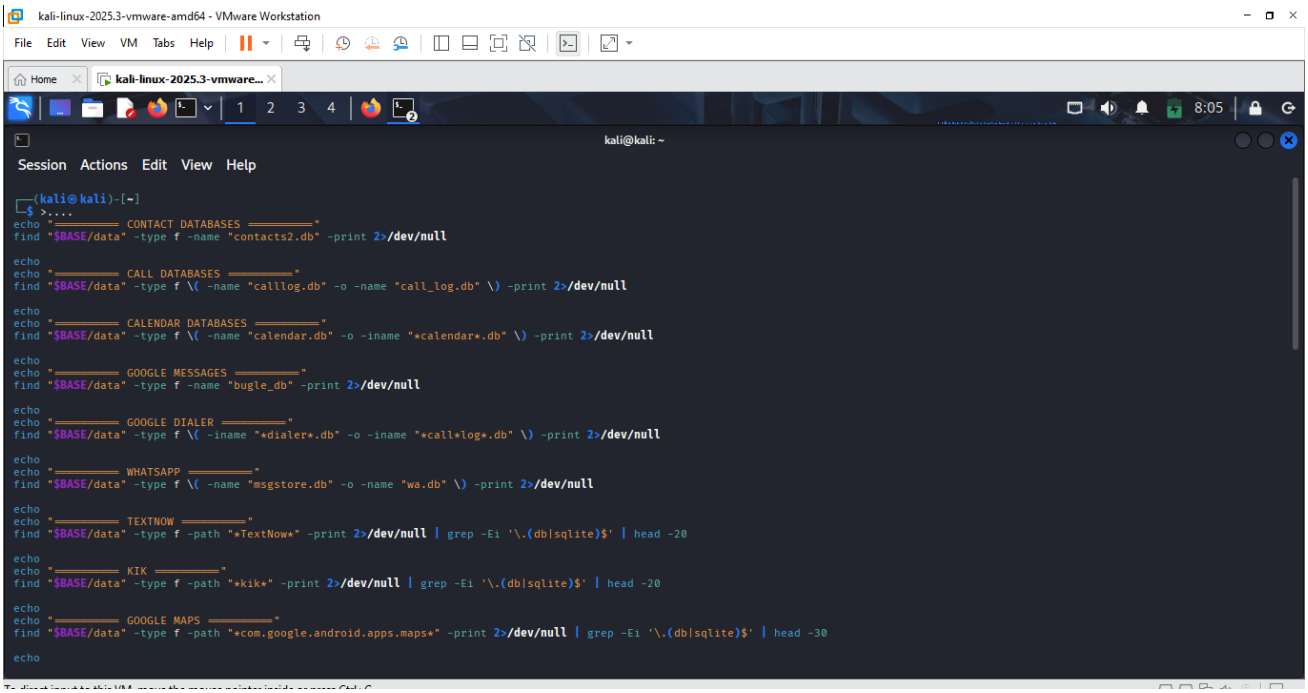


Fig 6.7 Remaining Android Application Database Locations.

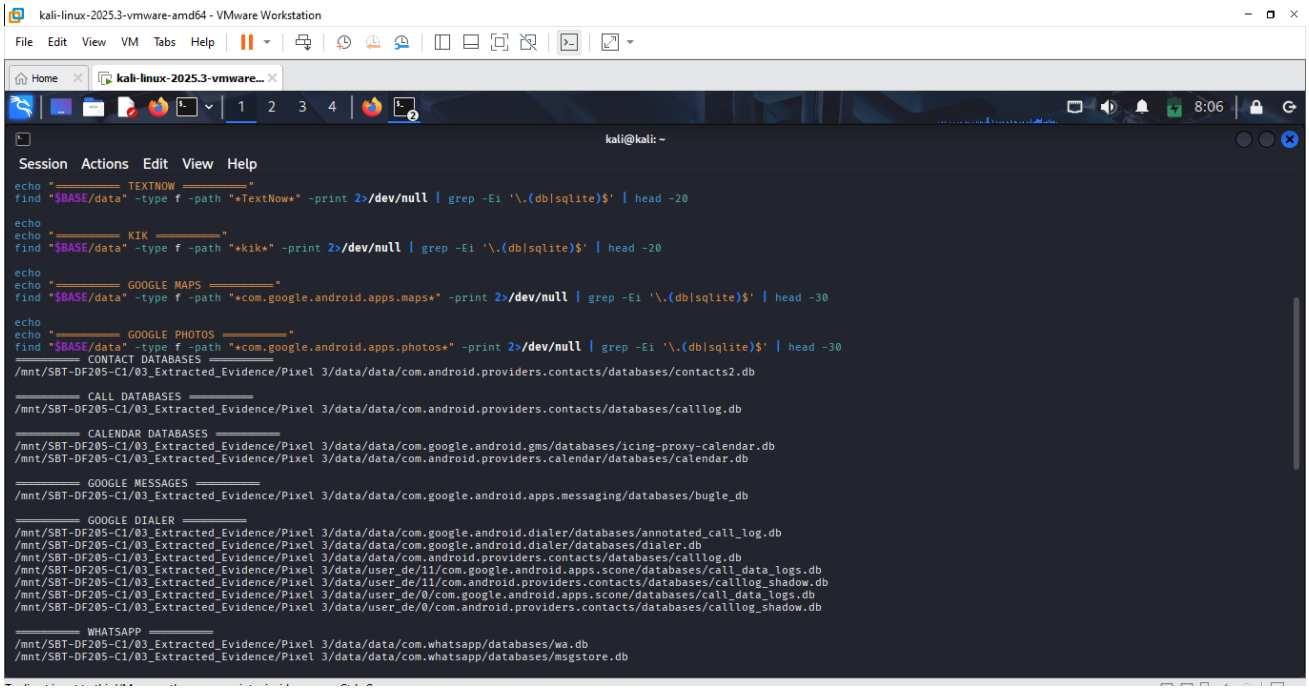


Fig 6.8 Remaining Android Application Database Locations page 1.

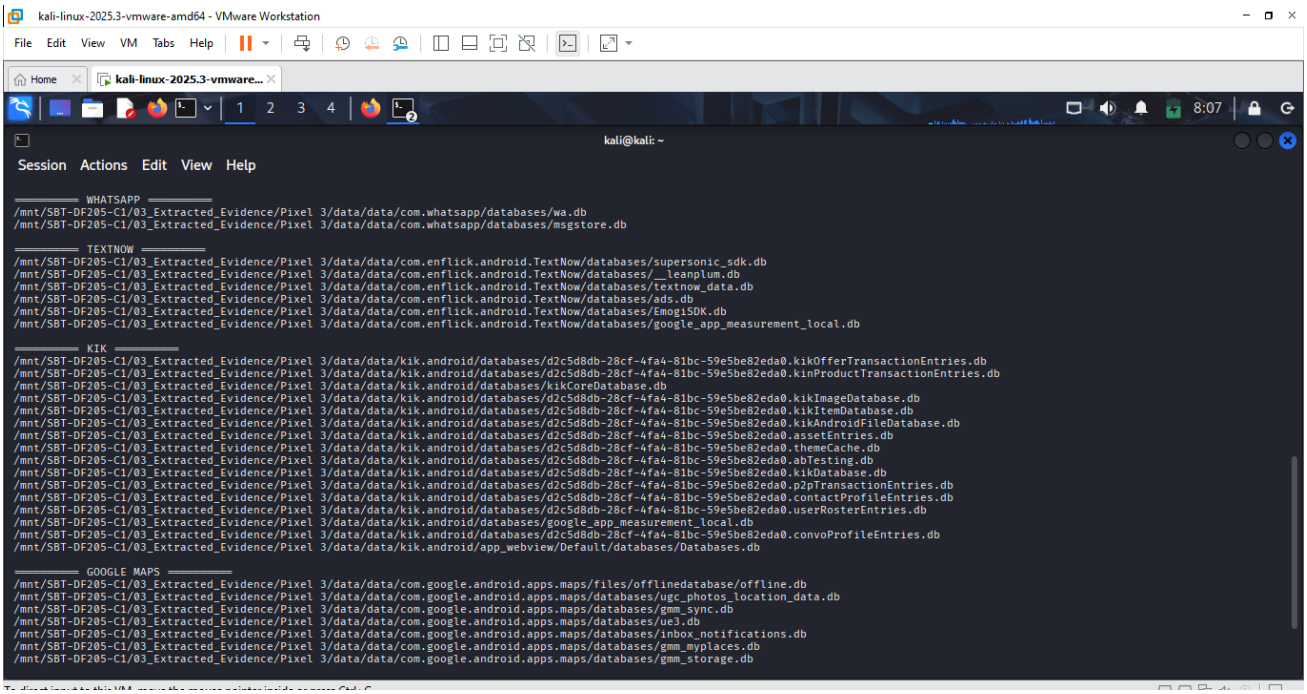


Fig 6.9 Remaining Android Application Database Locations page 2.

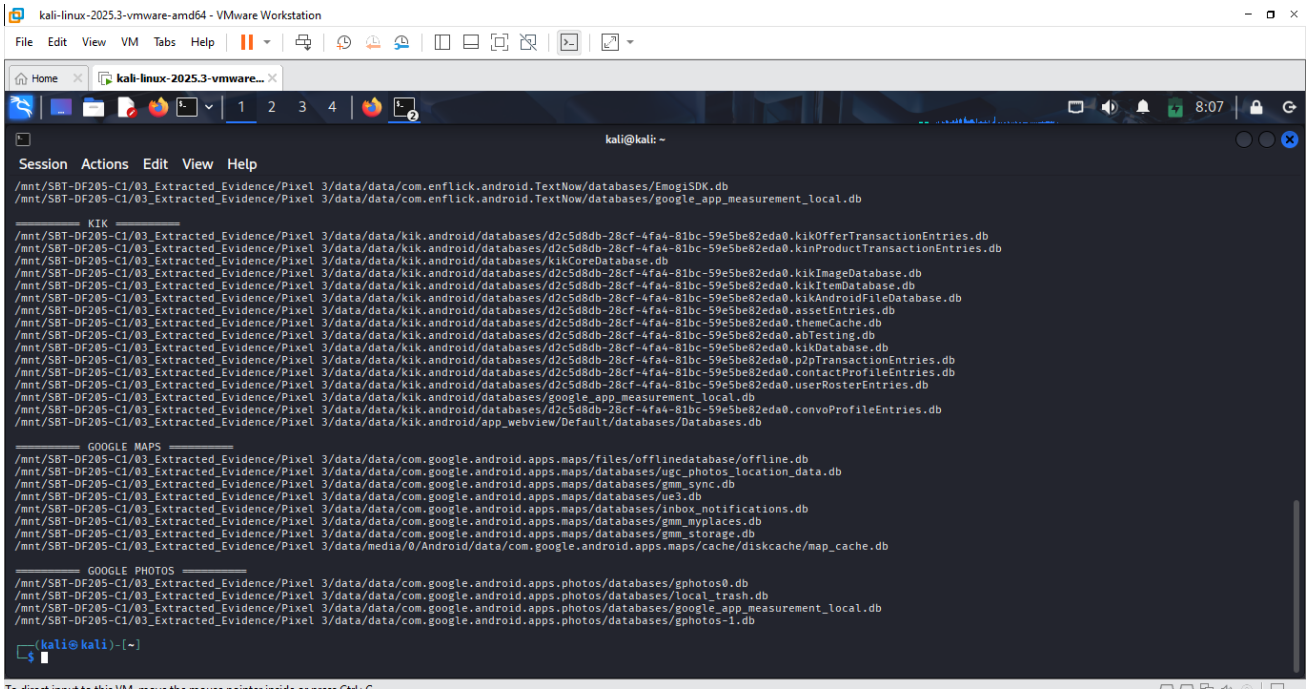


Fig 7.0 Remaining Android Application Database Locations page 3.

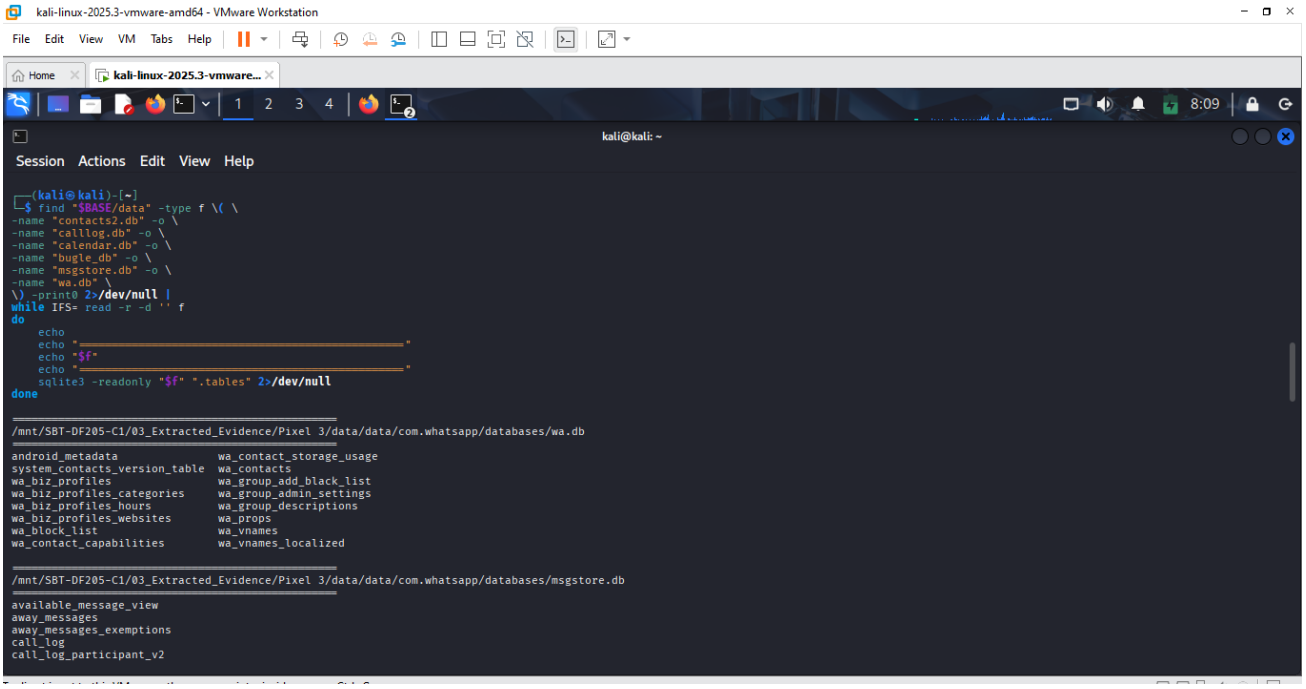


Fig 7.1 Inspecting all SQLite table names.

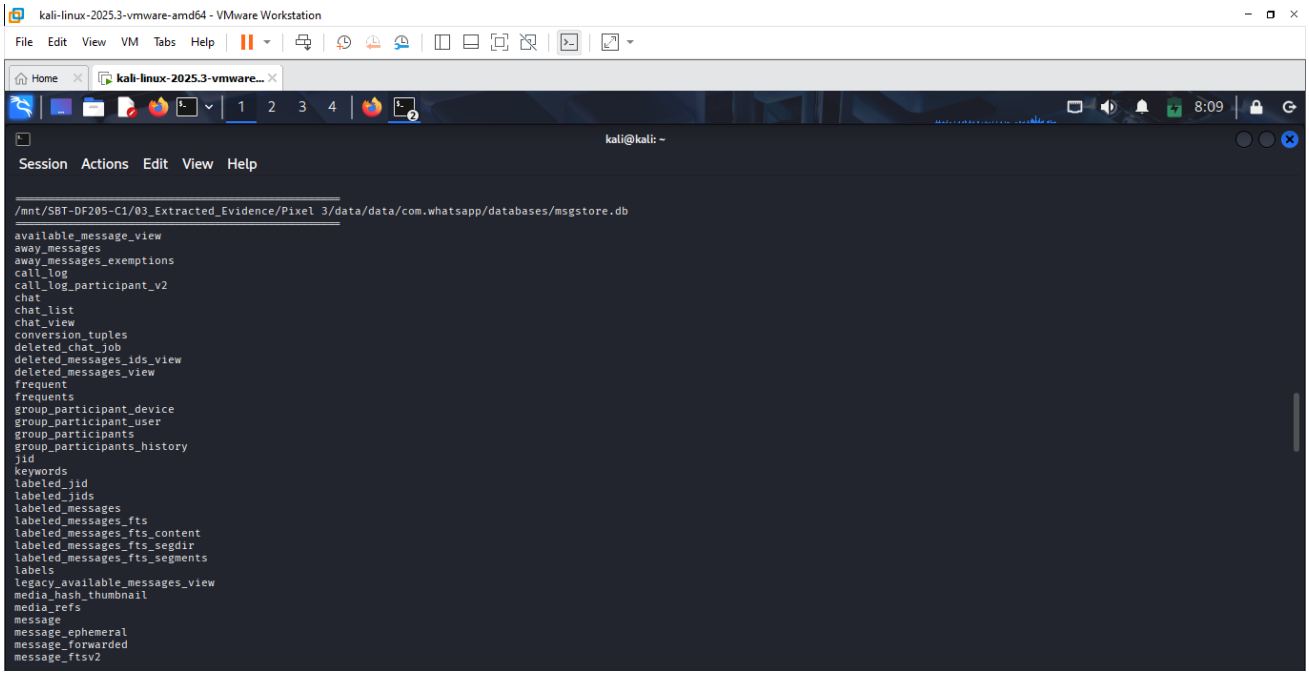


Fig 7.2 Inspecting all SQLite table names page 1.

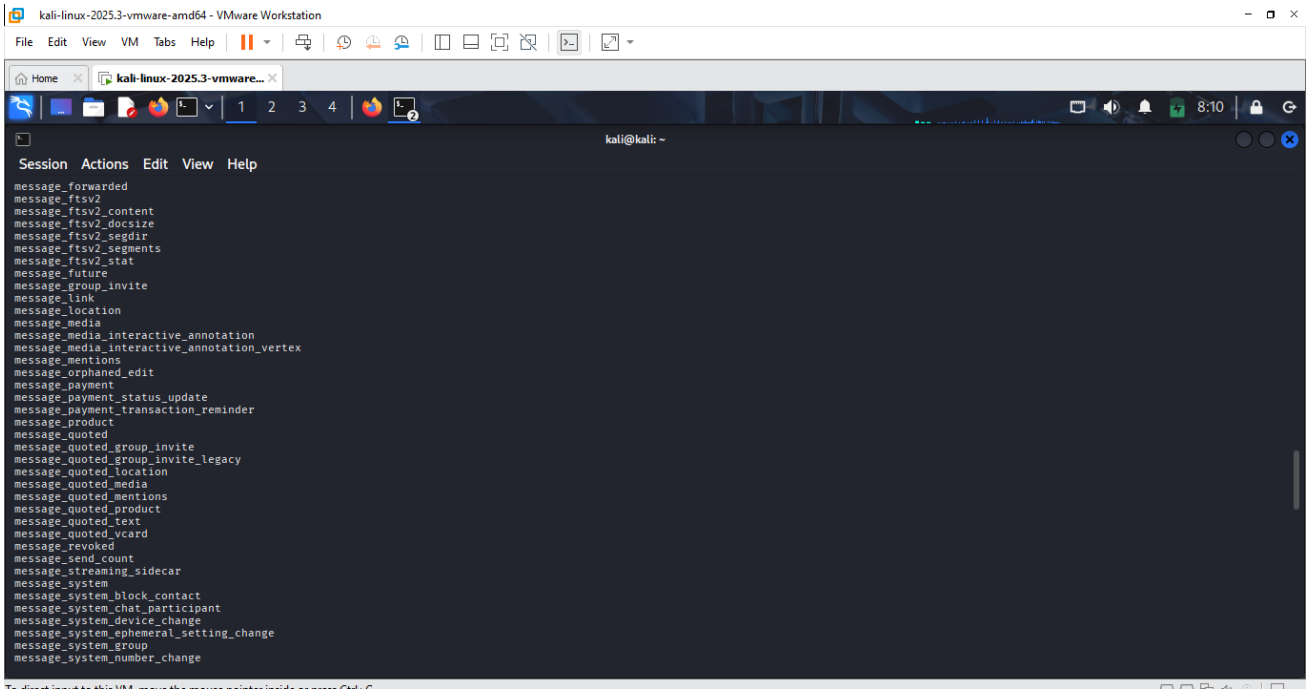


Fig 7.3 Inspecting all SQLite table names page 2.

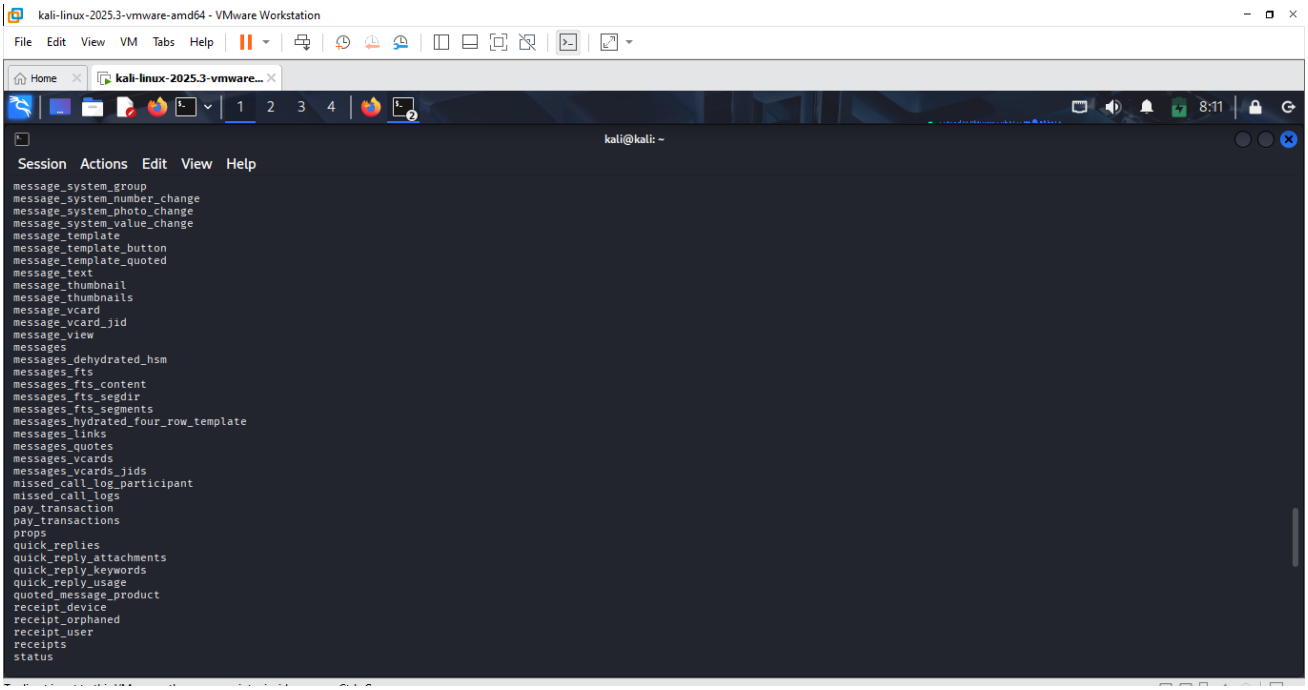


Fig 7.4 Inspecting all SQLite table names page 3.

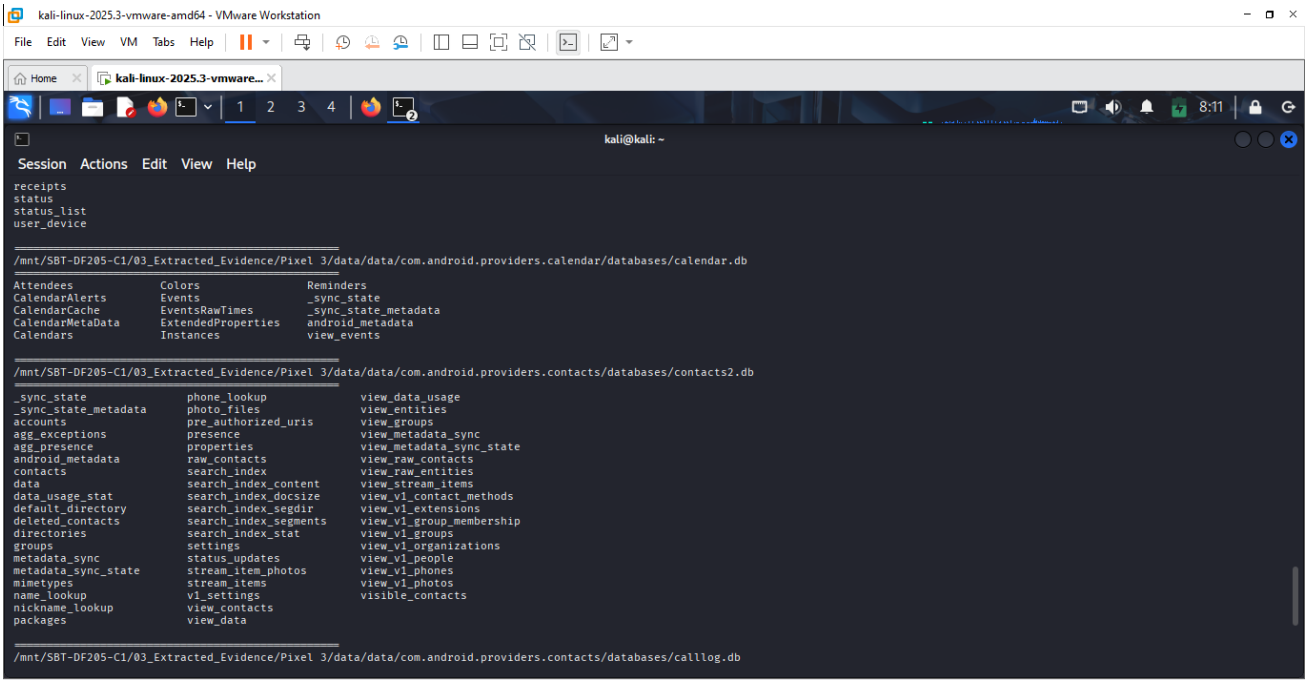


Fig 7.5 Inspecting all SQLite table names page 4.

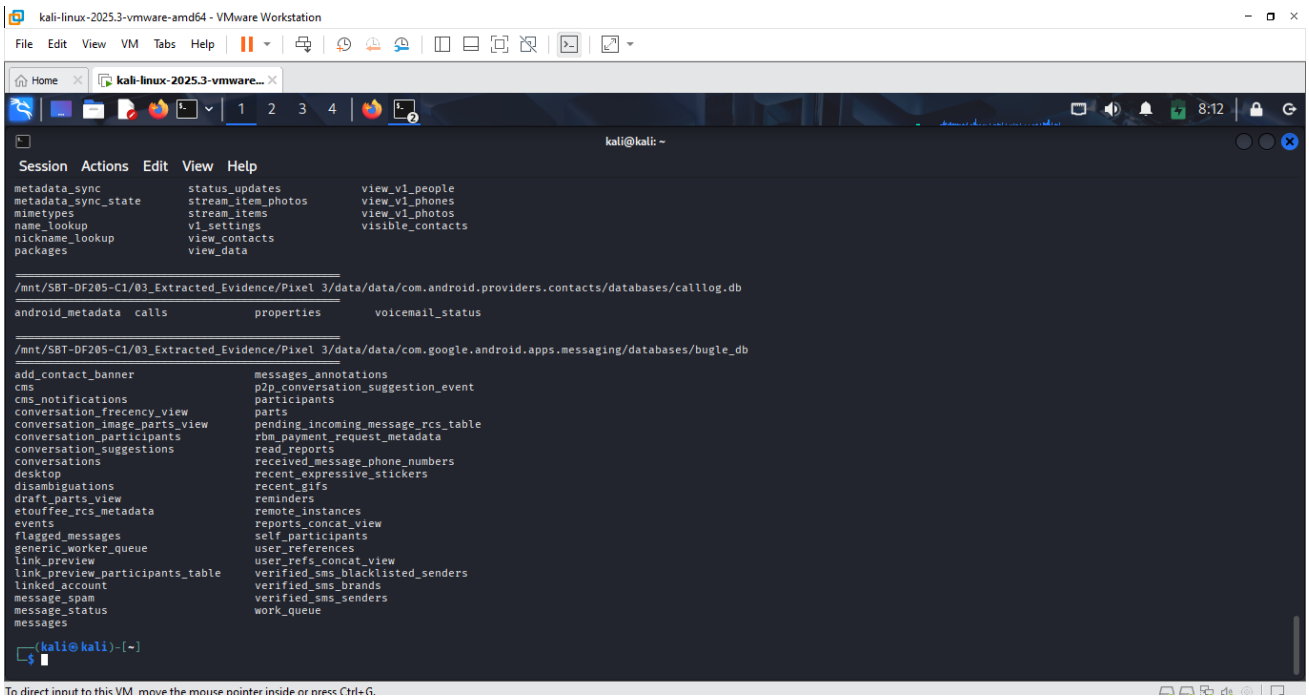


Fig 7.6 Inspecting all SQLite table names page 5.

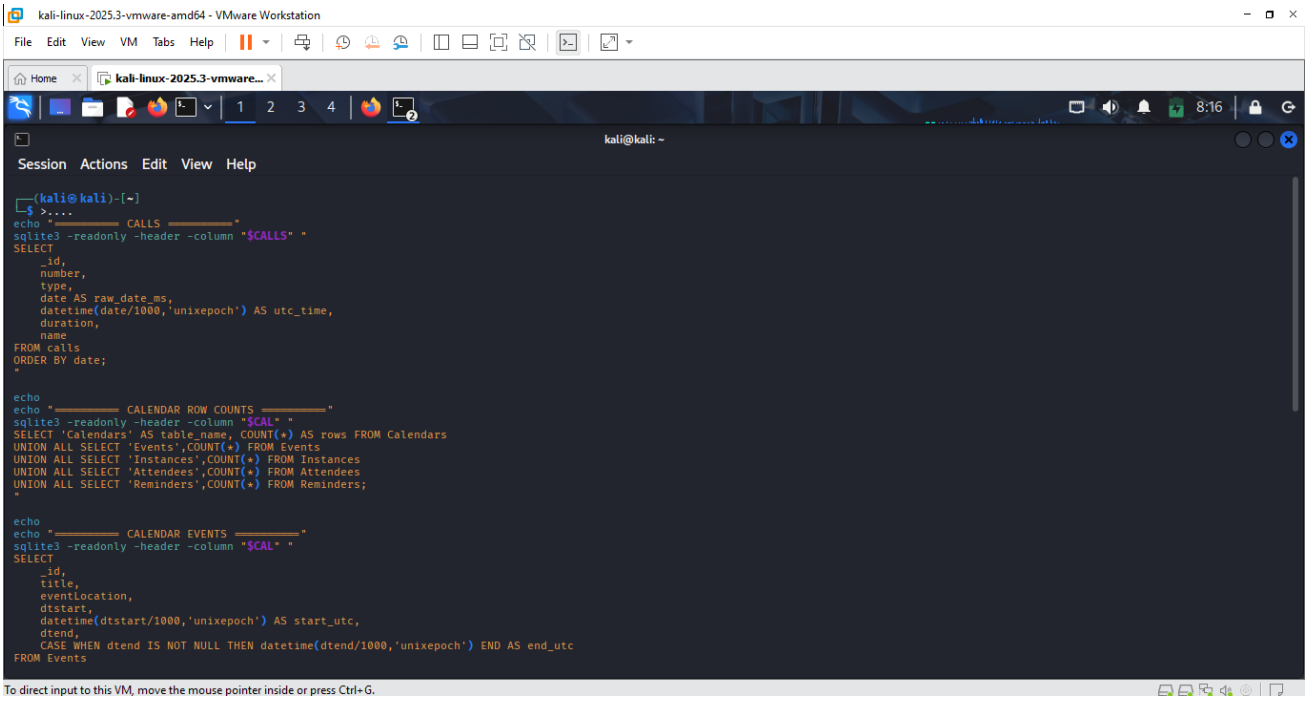


Fig 7.7 Contacts Calls and Calendar Forensic Findings.

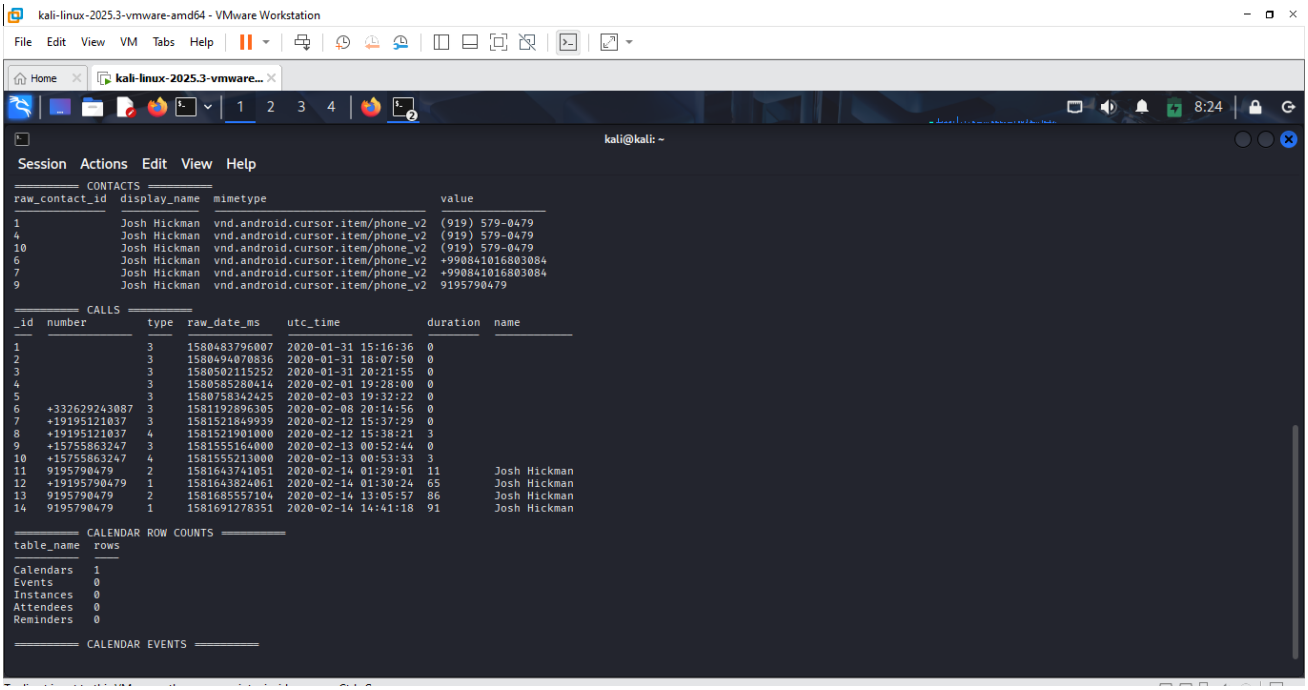


Fig 7.8 Contacts Calls and Calendar Forensic Findings page 1.

Explanation

The contact records linked the 9195790479 number and its variants to Josh Hickman, while the call log showed several calls involving the same contact. The Calendar Provider was also examined, but no event, instance, attendee or reminder records were recovered.

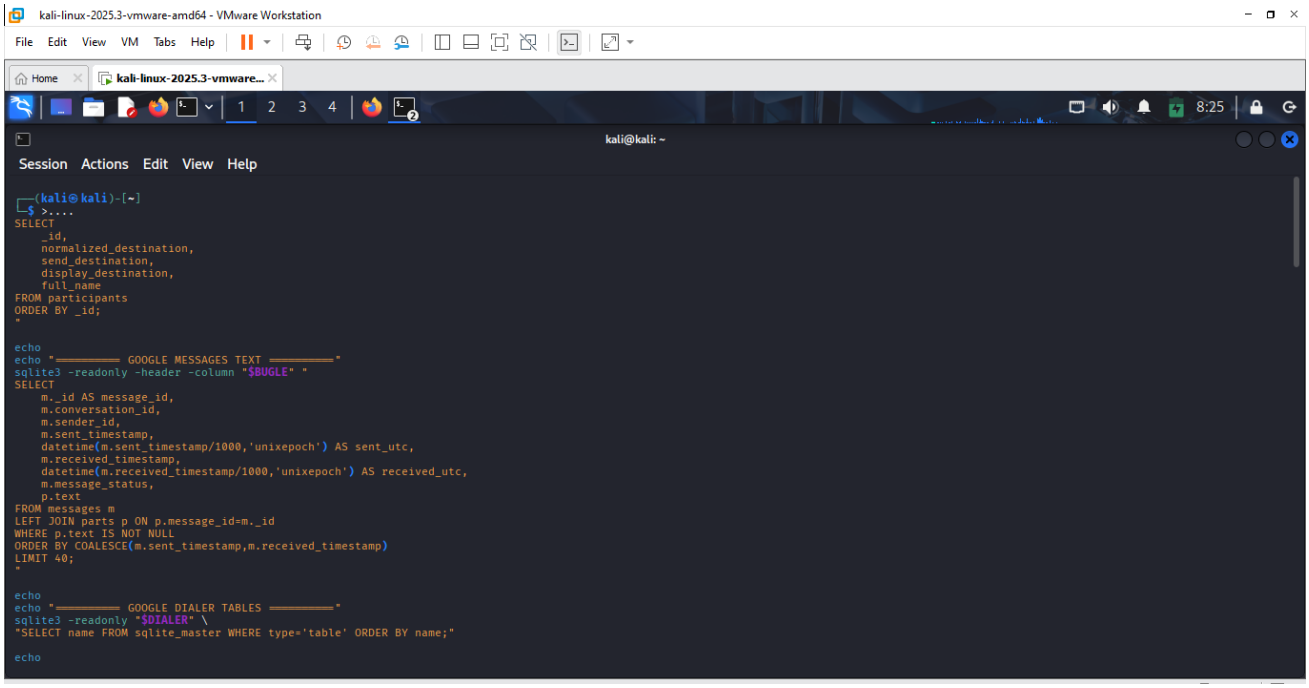


Fig 7.9 Google Messages and Dialer Forensic Records.

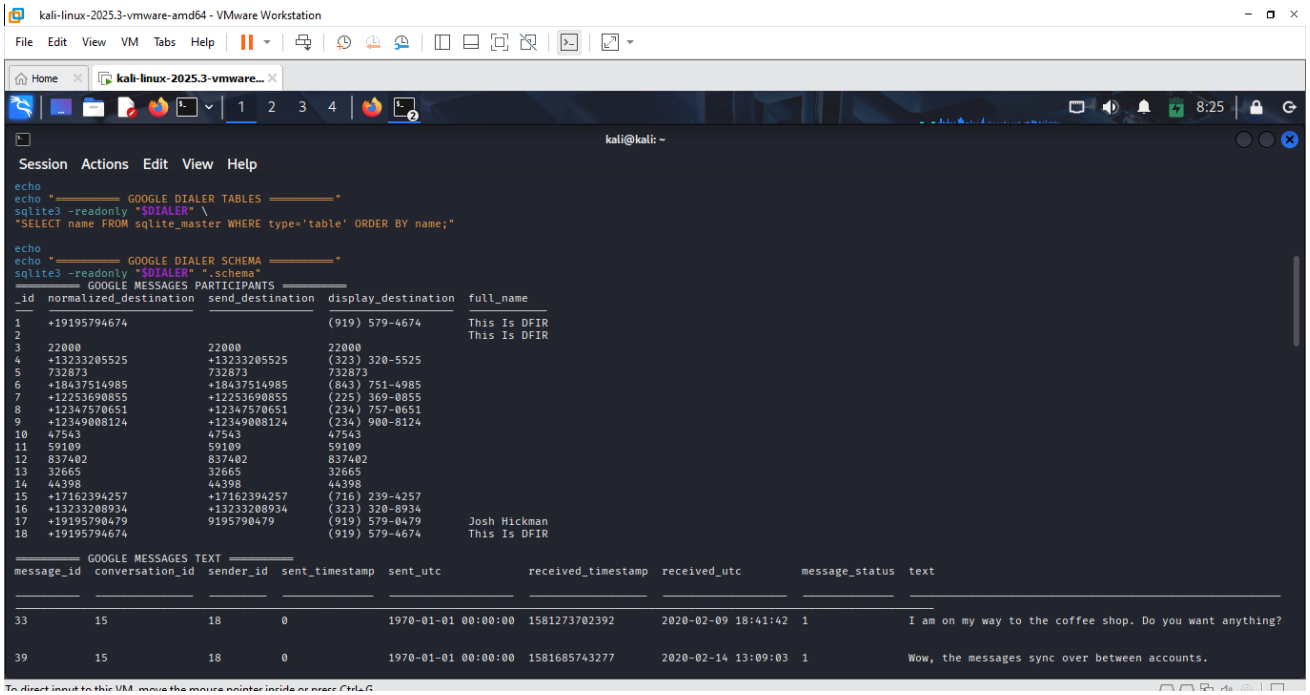


Fig 8.0 Google Messages and Dialer Forensic Records page 1.

Explanation

Google Messages contained message and participant information that could be compared with the AOSP Telephony Provider. Matching text and timestamps showed that several messages were duplicate representations of the same communication rather than separate events.

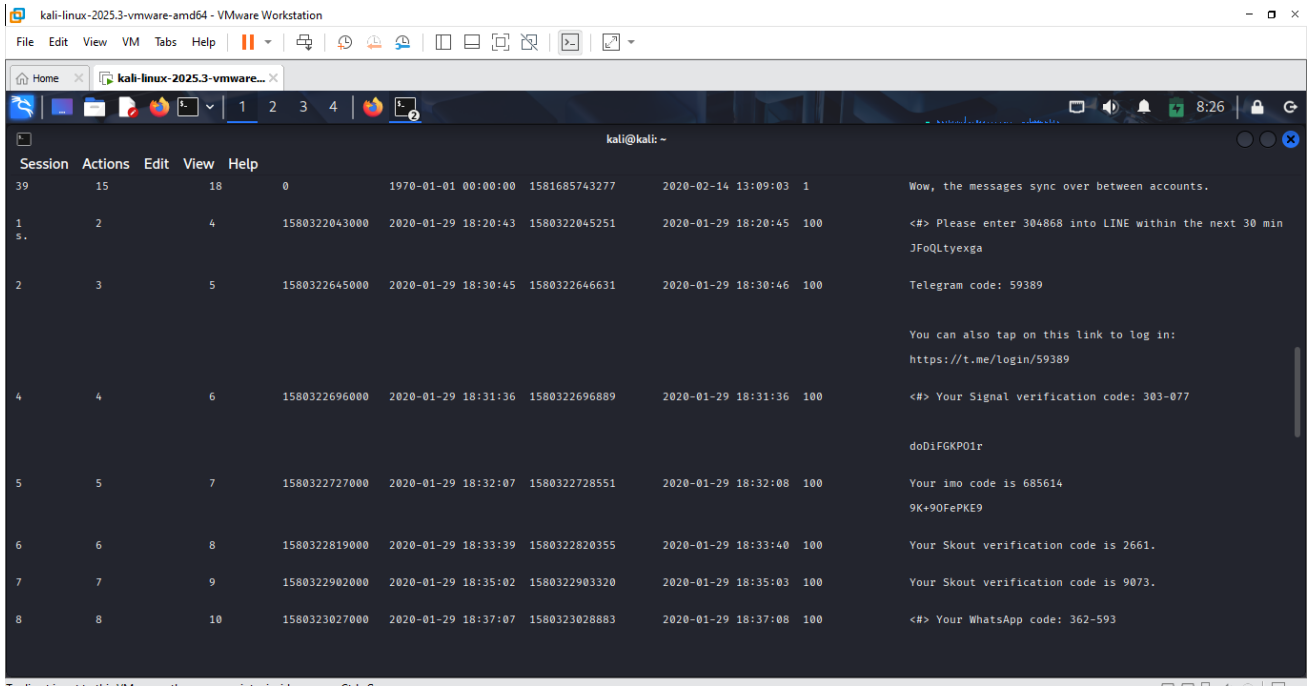


Fig 8.1 Google Messages and Dialer Forensic Records page 2.

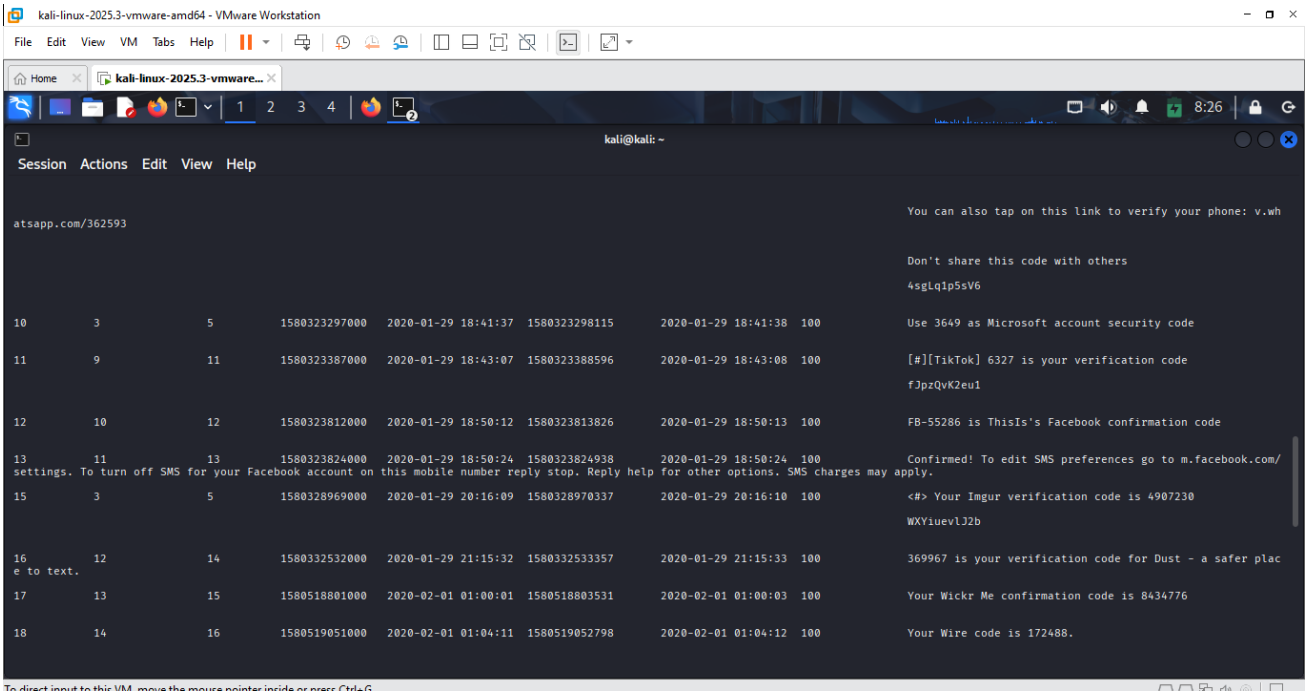


Fig 8.2 Google Messages and Dialer Forensic Records page 3.

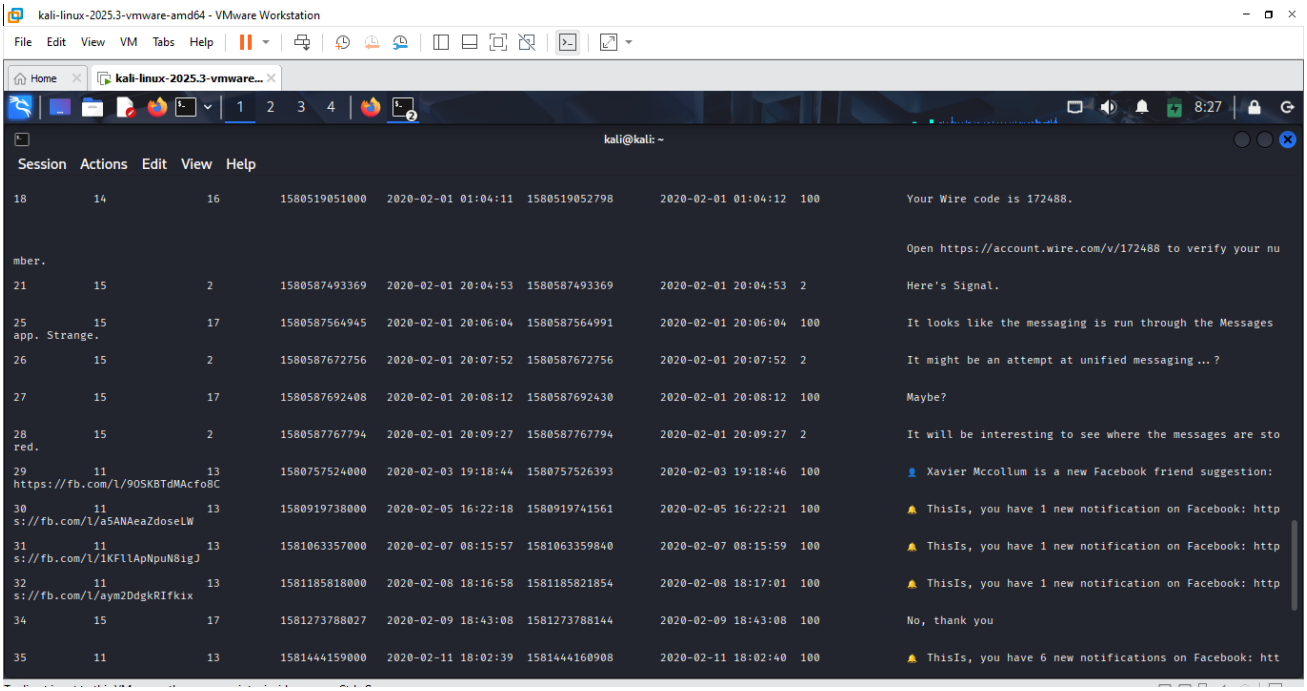


Fig 8.3 Google Messages and Dialer Forensic Records page 4.

Explanation

Google Messages contained message and participant information that could be compared with the AOSP Telephony Provider. Matching text and timestamps showed that several messages were duplicate represe

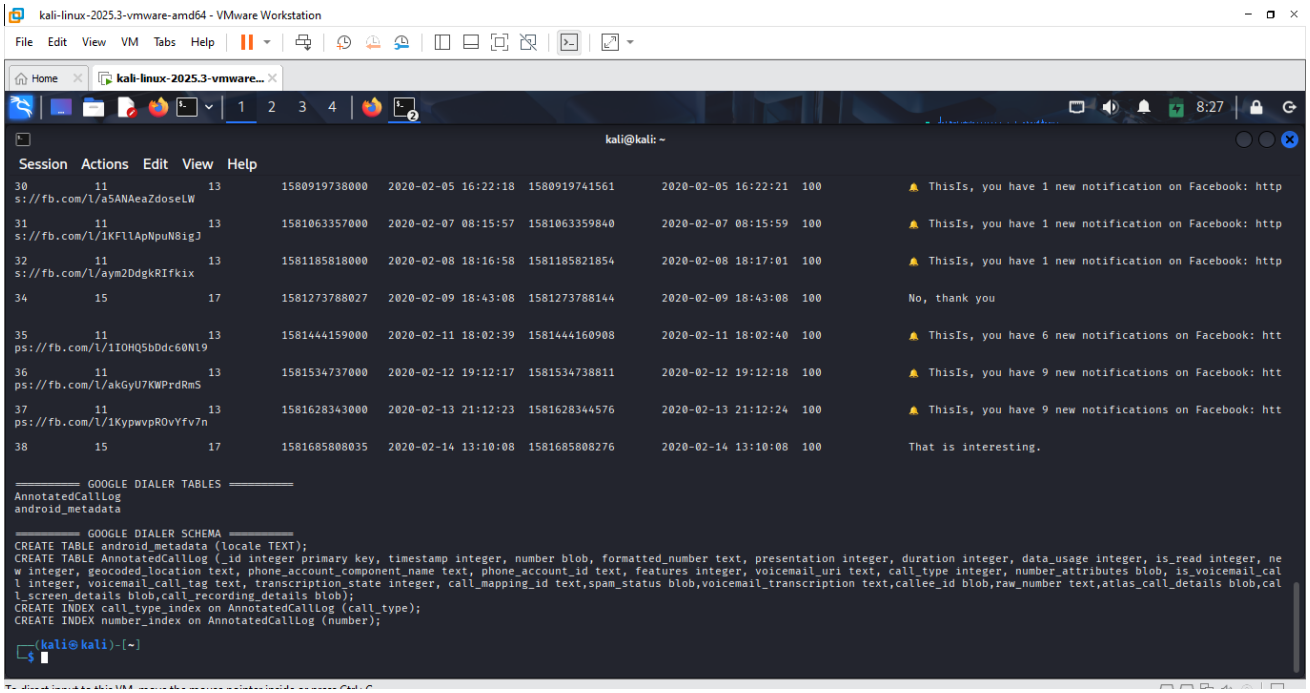


Fig 8.4 Google Messages and Dialer Forensic Records page 5.

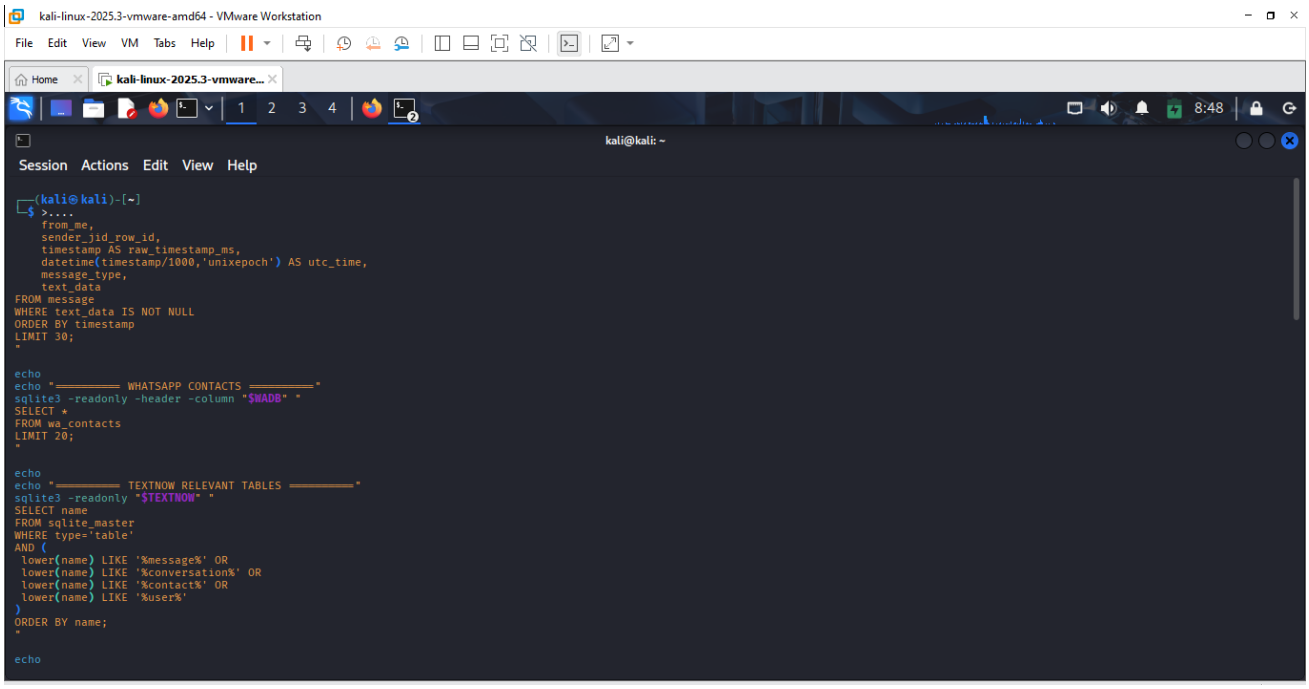


Fig 8.5 WhatsApp and TextNow Communication Evidence.

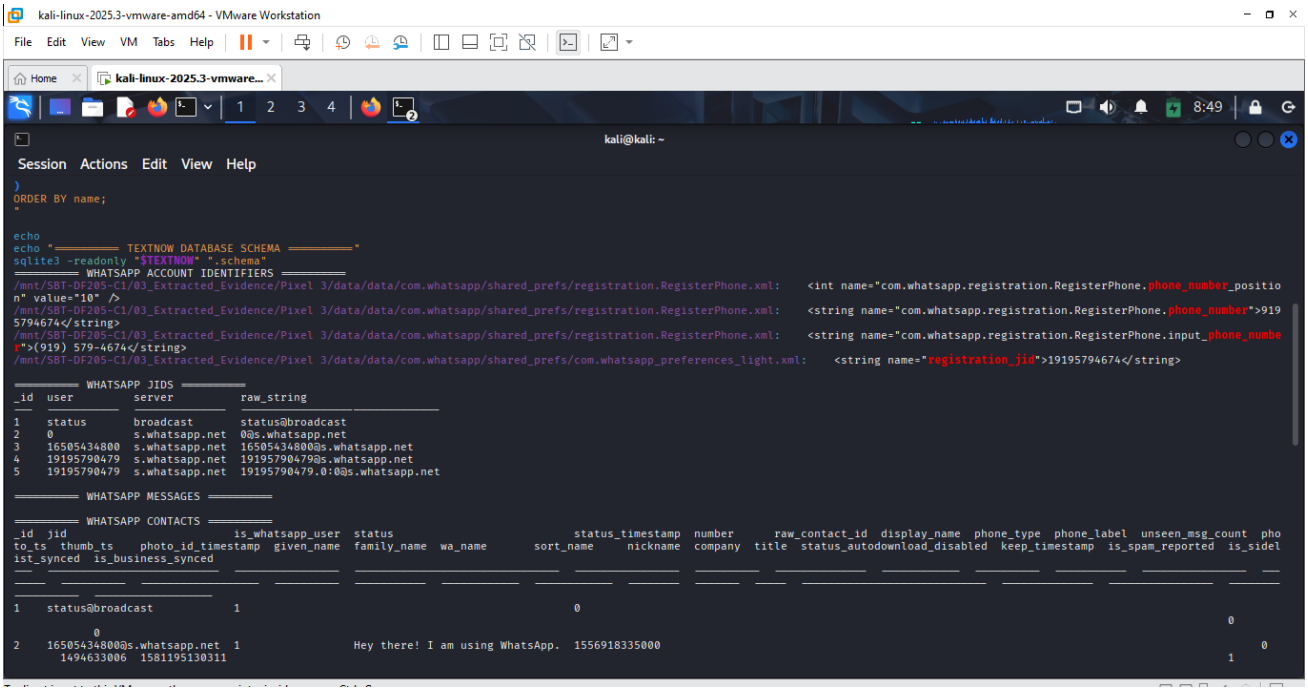


Fig 8.6 WhatsApp and TextNow Communication Evidence page 1.

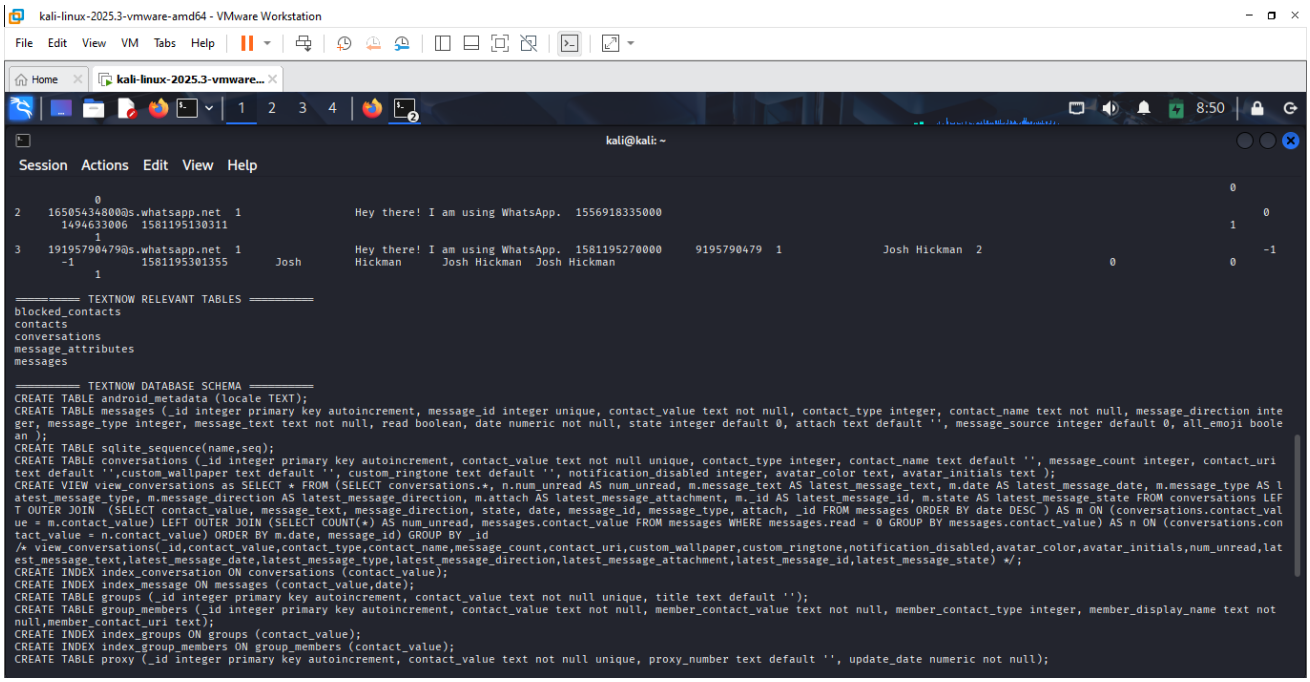


Fig 8.7 WhatsApp and TextNow Communication Evidence page 2.

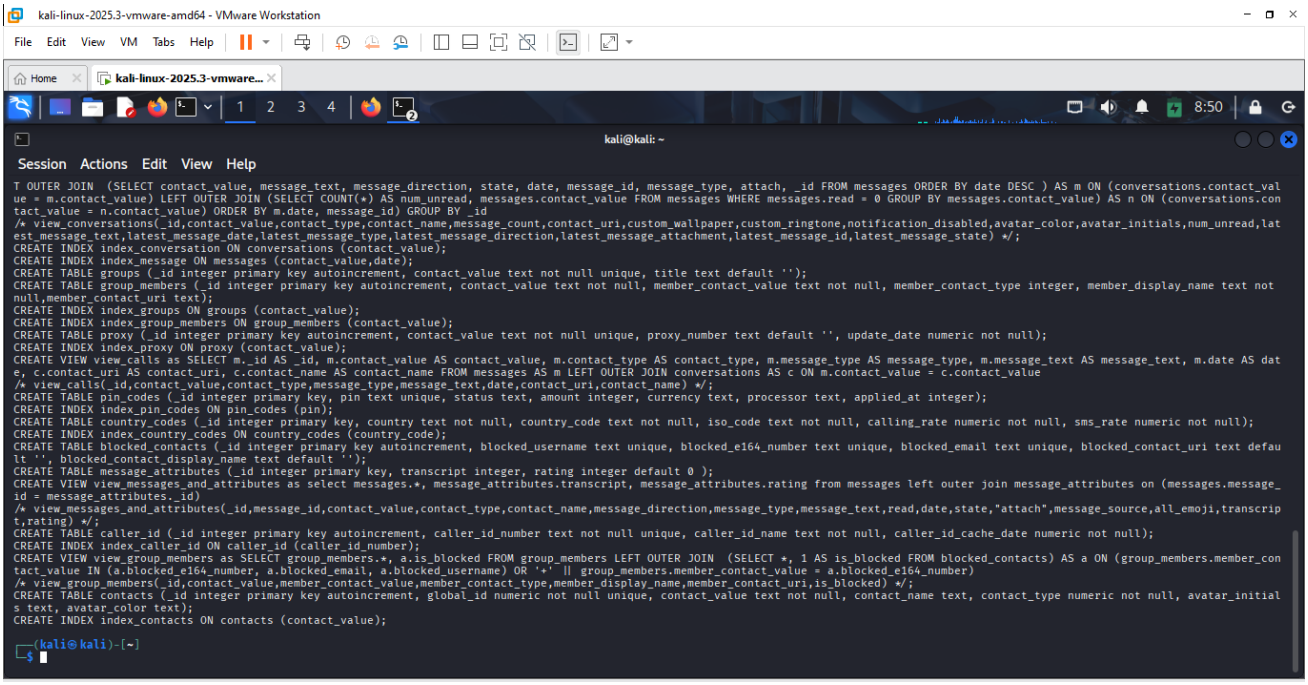


Fig 8.8 WhatsApp and TextNow Communication Evidence page 3.

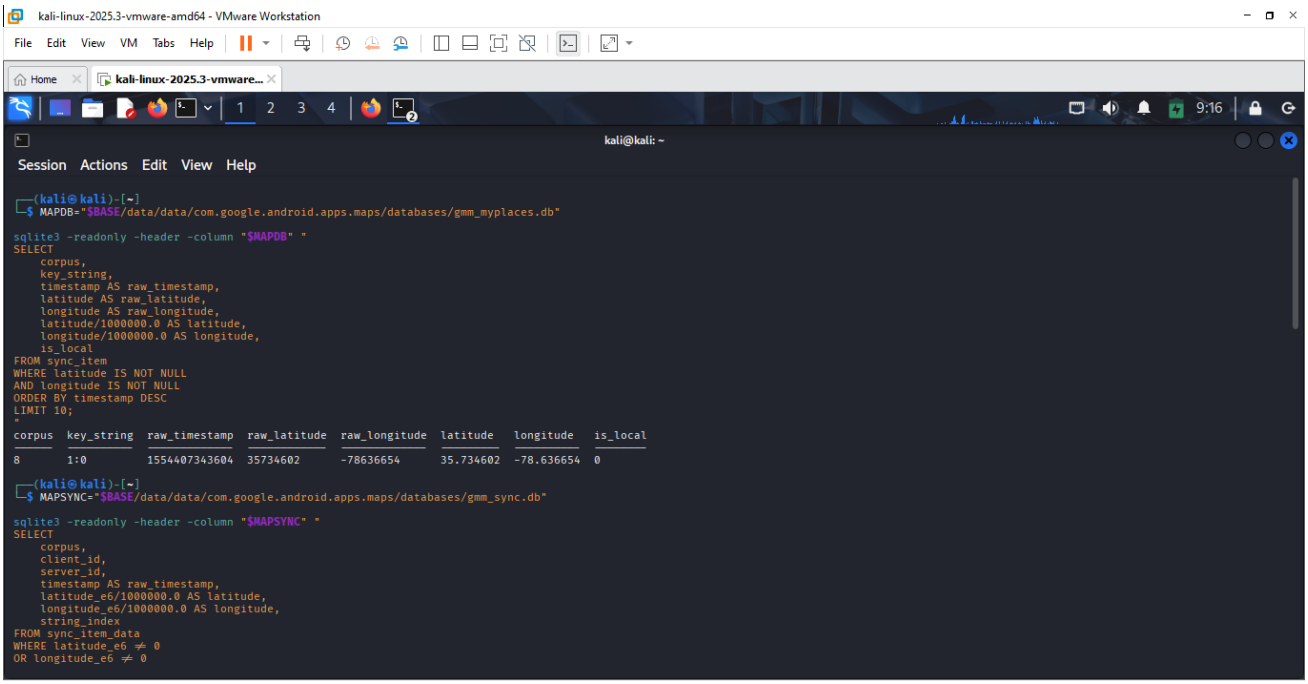


Fig 8.9 Google Maps Location Records.

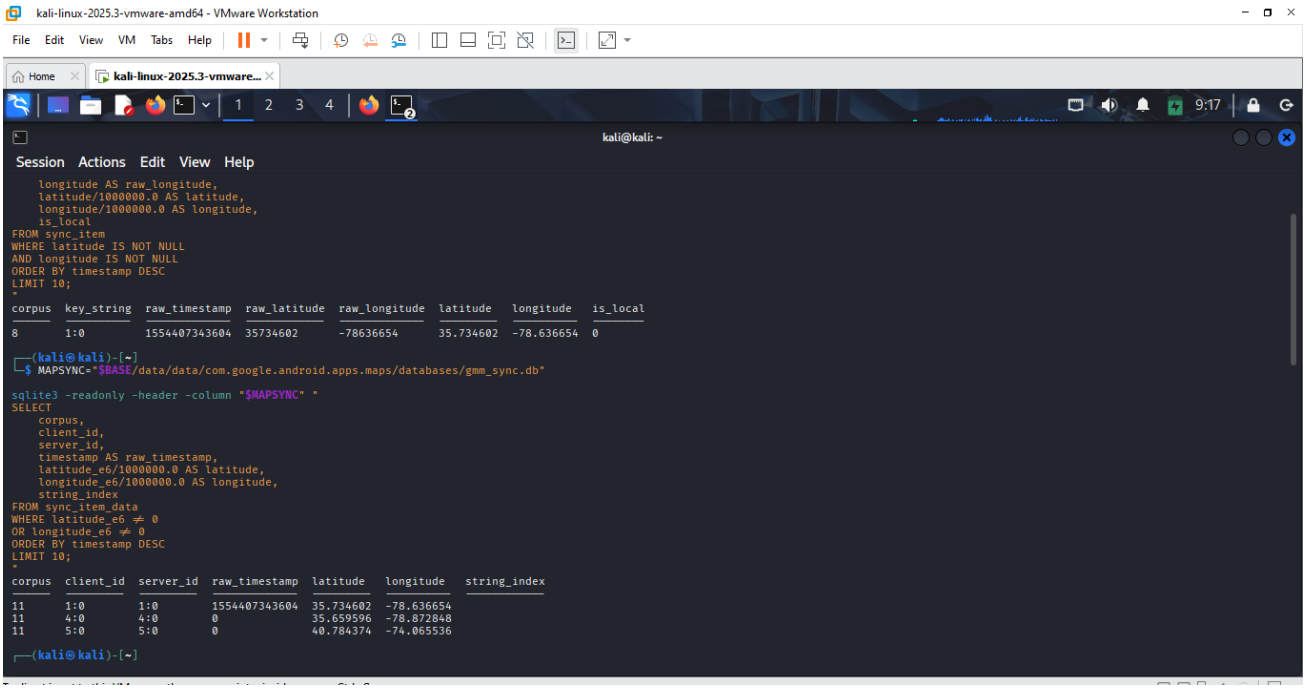


Fig 9.0 Google Maps Location Records page 1.

Explanation

The Maps databases contained stored coordinate records, including one entry with a usable Unix timestamp. I treated these as stored or synchronised Maps artefacts and did not assume that they proved the device user physically visited those locations.

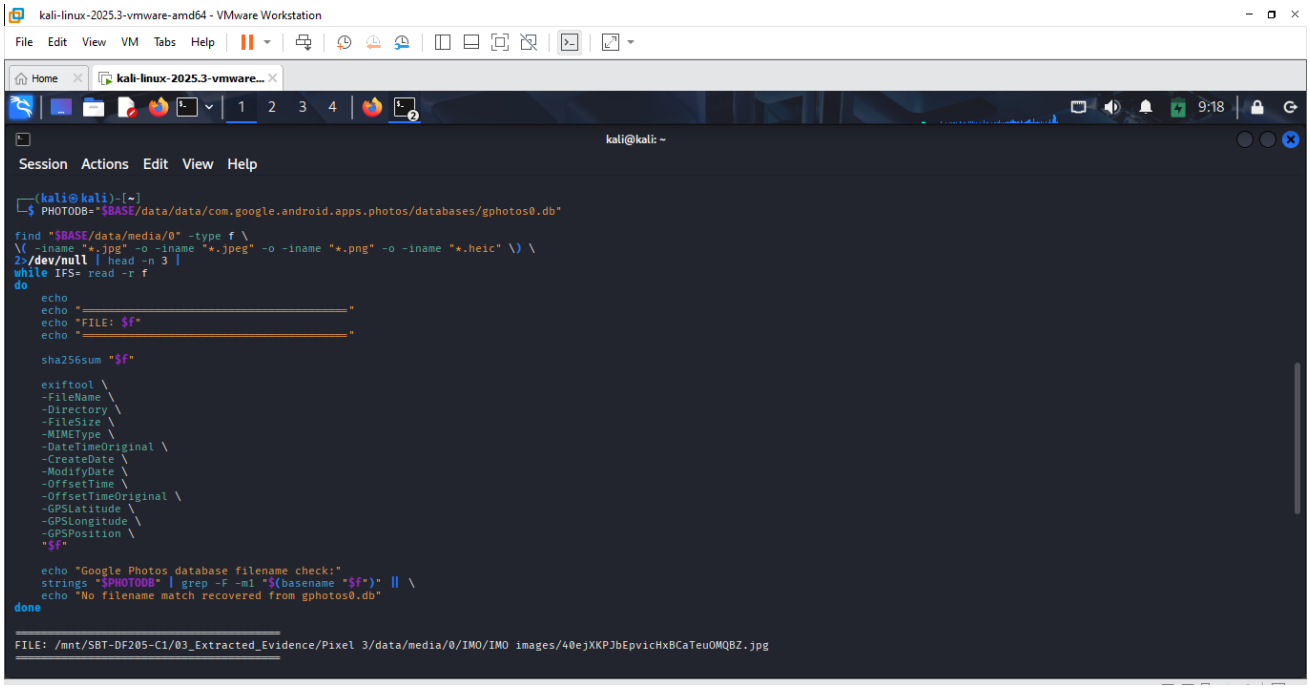


Fig 9.1 Photograph Metadata Hashes and GPS Evidence.

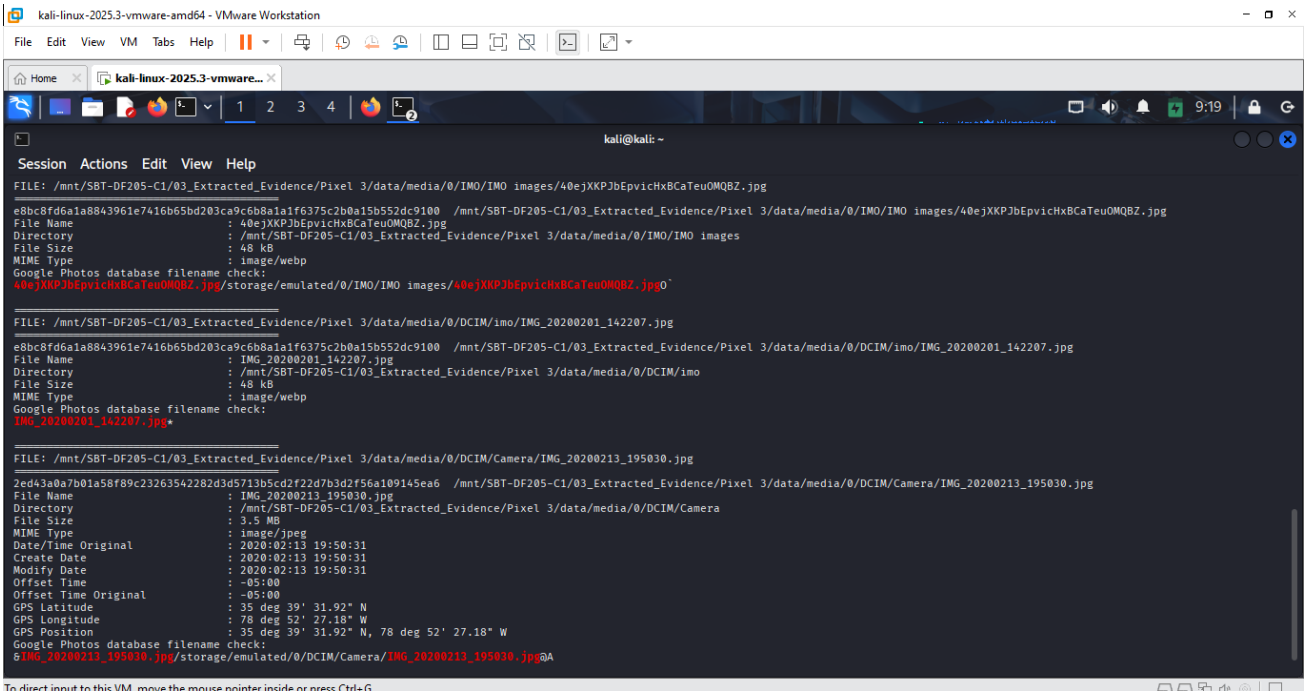


Fig 9.2 Photograph Metadata Hashes and GPS Evidence page 1.

Explanation

ExifTool identified the media type, file hashes and metadata of the examined photographs. The camera image contained an explicit -05:00 time-zone offset and GPS coordinates, allowing its capture time to be normalised to UTC and compared cautiously with the Maps evidence.

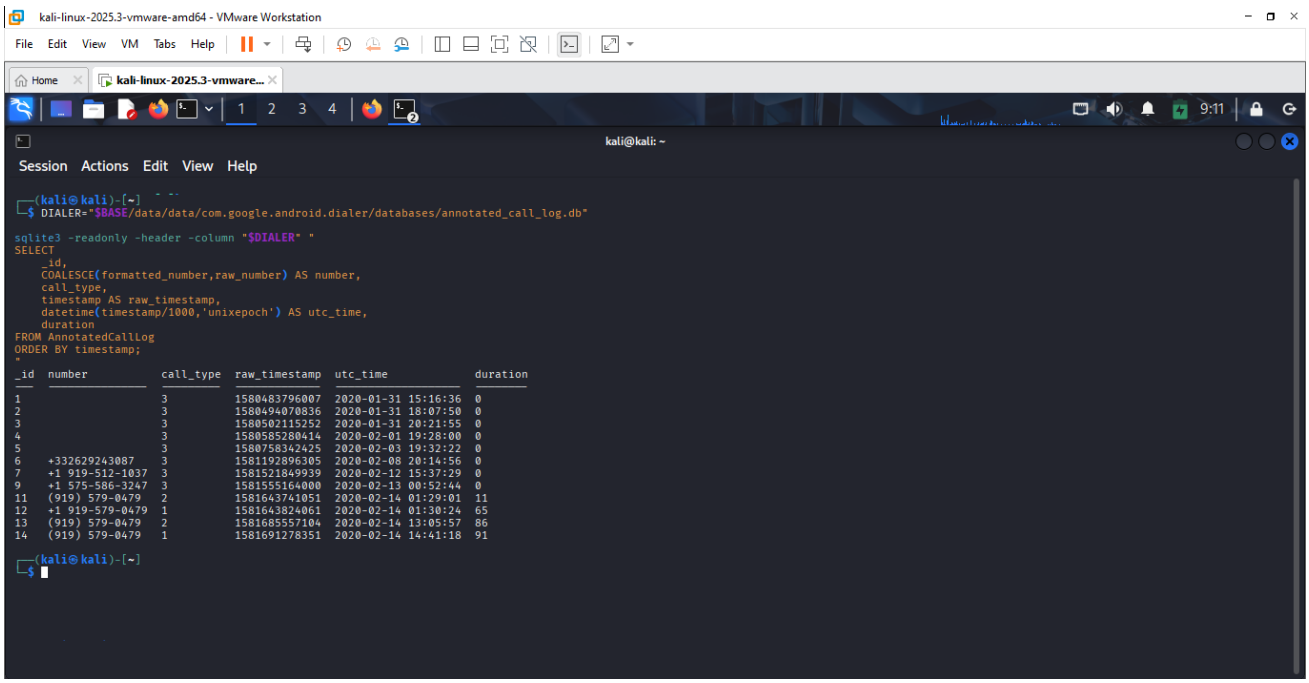


Fig 9.3 Google Dialer Call Records.

Explanation

The Google Dialer database contained calls with the same phone numbers, timestamps and durations found in the AOSP call log. This provided an independent application-level confirmation of selected call activity.

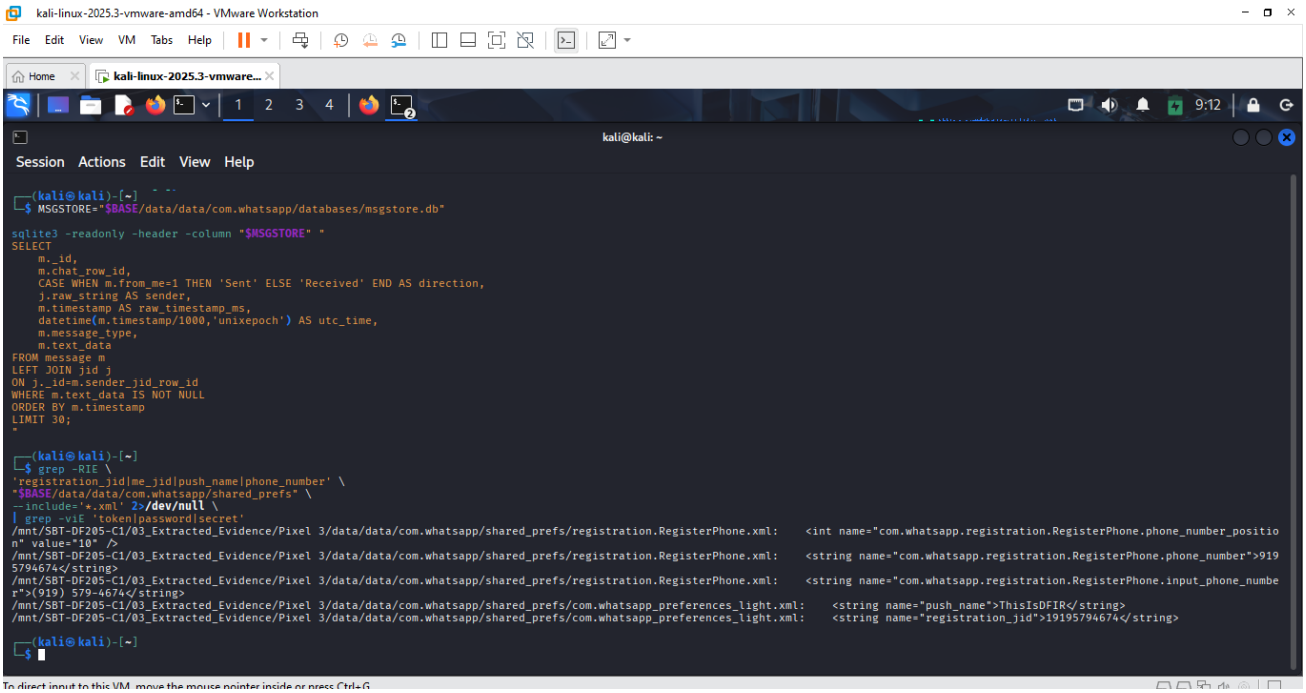


Fig 9.4 WhatsApp Communication Exchange.

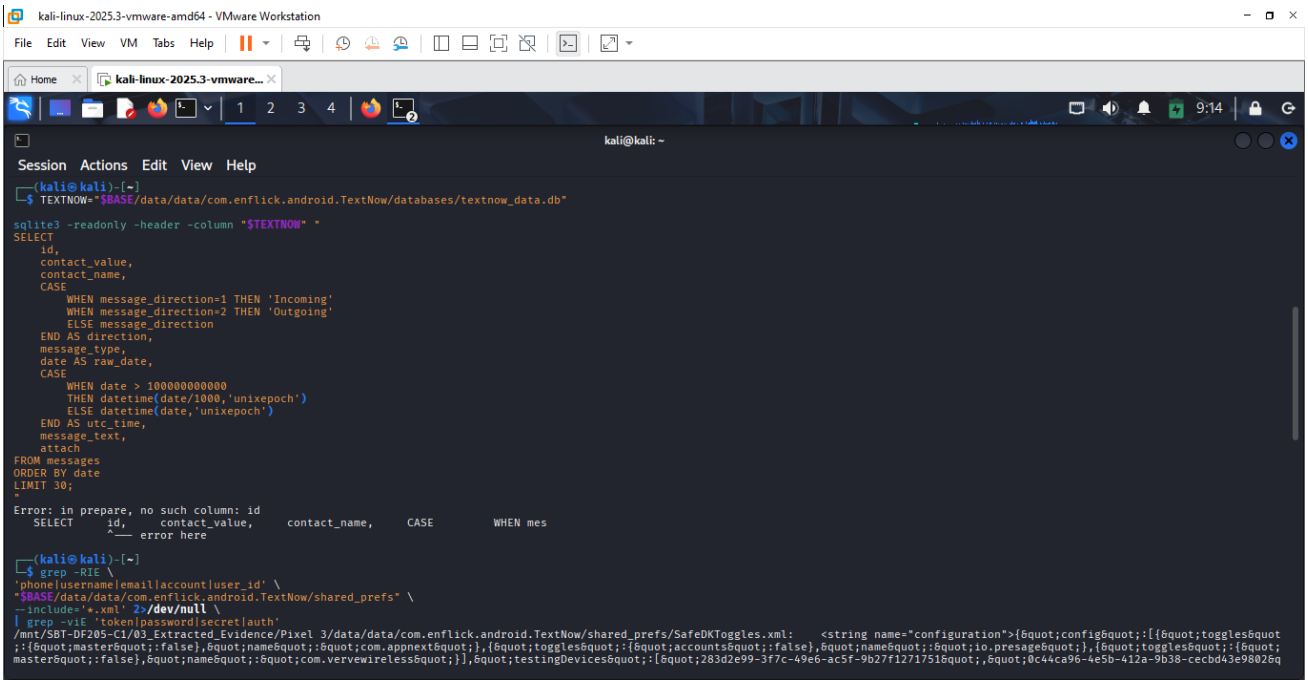


Fig 9.5 TextNow Communication Records.

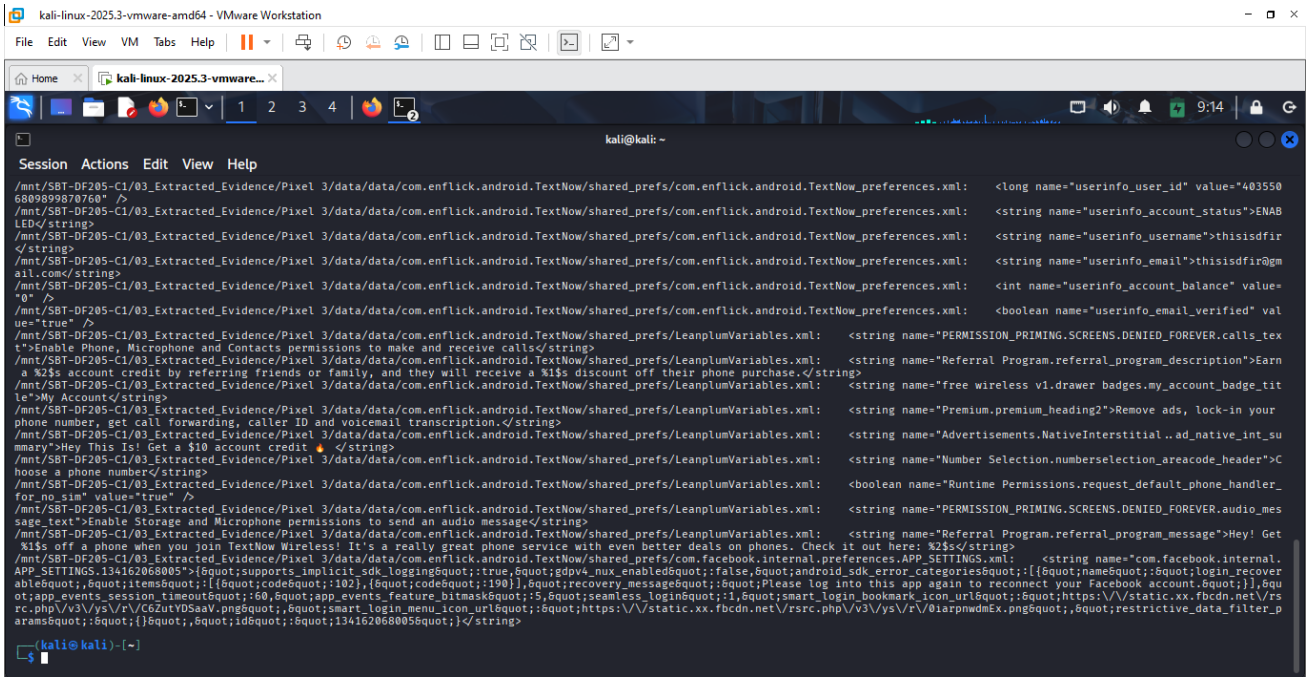


Fig 9.6 TextNow Communication Records page 1.

Task 1 – Evidence Handling, Integrity and Preparation

The supplied evidence consisted of the archive:

Pixel 3.zip

Before beginning the examination, I created a structured forensic case directory so that the original evidence, working copy, extracted files, reports, screenshots, queries and logs could be kept separate.

A dedicated 40 GB forensic evidence disk was added to the Kali Linux virtual machine. The disk was configured as an ext4 filesystem and mounted at:

/mnt/SBT-DF205-C1

The original archive was then moved into the preserved evidence directory:

/mnt/SBT-DF205-C1/01_Original_Evidence/Pixel 3.zip

The SHA-256 hash calculated for the original archive was:

ca6918ef8b20486b6a5ded15609ac51318f377829480f93be3ba15364a8aa00a

A separate working copy was created at:

/mnt/SBT-DF205-C1/02_Working_Copy/Pixel 3.zip

The working copy was hashed independently and produced the same SHA-256 value as the preserved original. This confirmed that the copy used for examination was identical to the supplied evidence and that no alteration had occurred during copying. The working-copy integrity check therefore passed.

Before extraction, the working archive was tested using unzip -t. Kali returned:

No errors detected in compressed data

The working copy was then extracted to:

/mnt/SBT-DF205-C1/03_Extracted_Evidence/

The resulting Android extraction occupied approximately 8.6 GB and contained the expected Pixel 3 file-system structure, including directories such as vendor, product, system, data, metadata and other Android system locations.

All subsequent examination was performed against this extracted working evidence rather than the preserved original archive.

This process maintained a clear separation between the source evidence and the material used for forensic analysis. The original archive remained preserved while hashes, extraction results and examination logs provided a reproducible record of how the evidence was handled.

Task 2 – Device Identity and User Profiles

Android build-property files were examined to establish the identity and software configuration of the recovered device.

The extracted properties identified the manufacturer as:

Google

The device model was:

Pixel 3

The Android operating-system version was:

Android 10

The build identifier was:

QQ1A.200105.003

The recorded Android security patch level was:

2020-01-01

These values were recovered from properties including:

ro.product.vendor.manufacturer=Google

ro.product.vendor.model=Pixel 3

ro.build.version.release=10

ro.build.id=QQ1A.200105.003

ro.build.version.security_patch=2020-01-01

Together, these records establish that the supplied extraction originated from a Google Pixel 3 running Android 10 with the January 2020 security patch.

The Android user configuration under data/system/users was then examined. The userlist.xml record identified two user profiles:

User 0

and

User 11

The detailed profile XML identified User 0 with the profile name:

This Is

User 0 had serial number 0. Its recorded created value was 0, so I did not interpret this as a genuine account creation timestamp.

The recorded last-login value for User 0 was:

1581695517124

which converts to:

2020-02-14 15:51:57.124 UTC

User 11 was identified with the profile name:

User 2

and serial number:

11

Its recorded creation timestamp was:

1581644358241

which converts to:

2020-02-14 01:39:18.241 UTC

Its last-login timestamp was:

1581684601922

which converts to:

2020-02-14 12:50:01.922 UTC

Both profiles also contained the Android build fingerprint:

google/blueline/blueline:10/QQ1A.200105.003/6079943:user/release-keys

This provides additional corroboration that both profiles belonged to the same Pixel 3 Android installation.

Account Manager databases were then examined separately for each profile.

User 0 was associated with the Google account:

thisisdfir@gmail.com

while User 11 was associated with:

thisisdfirtwo@gmail.com

Both records were stored as account type:

com.google

The two profile-to-account relationships documented in the evidence were therefore:

User 0 / “This Is” → thisisdfir@gmail.com

User 11 / “User 2” → thisisdfirtwo@gmail.com

These account findings are also recorded in the Task 2 table contained in the working report.

The account associations were not based on the Account Manager database alone. Android's sync configuration also contained both Google addresses, while Google Photos preferences contained:

thisisdfir@gmail.com

This provided independent corroboration of the User 0 Google account association across multiple artefact sources.

The records establish that these accounts were associated with their respective Android profiles. They do not, by themselves, prove which physical person was operating either profile at a particular time.

Task 3 – Android Settings, Wi-Fi and Bluetooth

The Android global and secure settings files for User 0 were examined to establish relevant device configuration.

The recovered global settings included:

adb_enabled = 1

development_settings_enabled = 1

wifi_on = 1

bluetooth_on = 0

airplane_mode_on = 0

device_name = Pixel 3

The secure settings included:

location_mode = 3

location_providers_allowed = gps,network

bluetooth_name = Pixel 3

mock_location = 0

These values show that Android Debug Bridge and developer settings were enabled in the stored configuration. Wi-Fi was stored as enabled, Bluetooth as disabled, airplane mode as disabled, and location services were configured to use GPS and network providers.

For the two main settings required for reporting, I identified:

adb_enabled = 1

and:

location_mode = 3

These values were taken directly from the stored Android settings rather than inferred from the behaviour of the device.

Wi-Fi Examination

Two principal Wi-Fi configuration files were identified:

data/misc_ce/0/wifi/WifiConfigStore.xml

and:

data/misc/wifi/WifiConfigStore.xml

The first file was approximately 966 bytes, while the second was approximately 626 bytes.

Both configuration files were examined for stored network information including SSIDs, BSSIDs, configuration keys and connection indicators.

The examination returned:

Network entries: 0

for both files.

No recoverable saved SSID, BSSID, ConfigKey or HasEverConnected record was identified in the examined Wi-Fi configuration files.

The presence of the configuration files and the global setting:

wifi_on = 1

shows that Wi-Fi functionality existed and was stored as enabled. However, the available artefacts do not establish that the handset was connected to a particular wireless network at a specific time.

Bluetooth Examination

Bluetooth-related evidence was recovered from several locations, including:

data/misc/bluedroid/bt_config.conf

and:

data/user_de/0/com.android.bluetooth/databases/bluetooth_db

The Bluetooth database was accompanied by its associated sidecar files:

bluetooth_db-wal

and:

bluetooth_db-shm

These files were retained together during examination.

The Bluetooth configuration contained a stored identifier:

b4:cc:02:73:ff:93

The Bluetooth metadata database contained two rows, including a remote-device-style address:

B4:EC:02:73:FF:93

The two addresses are similar but are not identical, because one contains CC while the other contains EC. I therefore did not treat them as corroborating records for the same Bluetooth device.

The stored Android setting:

bluetooth_on = 0

indicates that Bluetooth was recorded as disabled in the captured system settings. This does not mean that Bluetooth had never previously been enabled or used.

Overall, the Bluetooth artefacts demonstrate that Bluetooth configuration information existed within the extracted Android data, but the recovered evidence did not provide sufficient timestamp information to prove that a specific Bluetooth connection occurred at a particular time.

The Wi-Fi and Bluetooth findings were therefore treated as stored configuration evidence, rather than direct proof of timed network or device connections.

Task 4 – AOSP Communications, Contacts, Calls and Calendar

Two copies of mmssms.db were identified.

The main Telephony Provider database:

data/data/com.android.providers.telephony/databases/mmssms.db

had a size of 118,784 bytes and SHA-256:

cd5452f52933252f9dca9177491e0a316b542755ac7eb40e61b4b5e3c53f1342

It contained:

24 SMS records

7 PDU/MMS records

10 MMS part records

The separate user_de copy:

data/user_de/0/com.android.providers.telephony/databases/mmssms.db

was also 118,784 bytes, but had the different SHA-256:

d0ce9034defe1f2ba0409f1c86947447cd90f414a4fda5d847804d01ef34c827

Its SMS, PDU and PART tables contained zero records. The equal file size therefore did not mean that the two databases contained the same evidence.

The populated SMS database contained 22 received messages and 2 sent messages. A particularly useful conversation was thread 17. Two recovered sent messages were:

2020-02-09 18:41:42 UTC – “I am on my way to the coffee shop. Do you want anything?”

2020-02-14 13:09:03 UTC – “Wow, the messages sync over between accounts.”

The number associated with this thread was represented as both 9195790479 and 919-579-0479.

An MMS was reconstructed by correlating pdu, part and addr. PDU ID 1 belonged to thread 17 and was a sent MMS at 2020-02-01 20:04:53 UTC. The recovered text was:

“Here's Signal.”

The source and destination records identified +19195794674 and 9195790479.

Contact examination linked several representations of 9195790479 to Josh Hickman. The call log also contained multiple calls involving this contact. Examples included an 11-second call at 2020-02-14 01:29:01 UTC, a 65-second call at 01:30:24 UTC, an 86-second call at 13:05:57 UTC, and a 91-second call at 14:41:18 UTC.

The Calendar Provider contained one calendar record, but the examined Events, Instances, Attendees and Reminders tables contained zero records. No calendar activity was therefore inferred.

Task 5 – Google Messages and Google Dialer

The Google Messages bugle_db contained participant and message records that could be compared directly with the Telephony Provider.

The participant records linked the thread participant using the 919-579-0479 number format to Josh Hickman.

At least three message representations could be correlated with Telephony records by content and timestamp. Examples included:

the Facebook-related message represented in both databases;

“I am on my way to the coffee shop. Do you want anything?”; and

“Wow, the messages sync over between accounts.”

The same message appearing in both Google Messages and the Telephony Provider was treated as one communication event, not as two separate events.

The Google Dialer AnnotatedCallLog also reproduced calls found in the AOSP call log. Two particularly clear matches were:

919-579-0479, 2020-02-14 01:29:01 UTC, duration 11 seconds; and

+1 919-579-0479, 2020-02-14 01:30:24 UTC, duration 65 seconds.

The matching numbers, times and durations provide strong evidence that the AOSP and Google Dialer records describe the same calls.

Task 6 – WhatsApp and TextNow

WhatsApp preferences identified the registered phone number as (919) 579-4674 and the registration JID as 19195794674. The stored push name was ThisIsDFIR.

The WhatsApp contact database also contained JID/contact information, including an entry corresponding to Josh Hickman and the 9195790479 number.

However, the message query preserved in my working notes did not return a two-message WhatsApp text exchange. For that reason, I do not claim that the current evidence reconstructs the full WhatsApp exchange required by the task.

TextNow preference records identified:

username: thisisdfir
email: thisisdfir@gmail.com
account status: ENABLED
email verified: true

The TextNow database schema showed a messages table, but my message query used id while the actual column was _id. The query therefore failed before communication rows were extracted. The TextNow account association is supported, but two communication records were not recovered in the documented examination.

This is recorded as a limitation rather than filling the gap with assumed data.

Task 7 – Maps and Photographs

Google Maps artefacts were recovered from gmm_myplaces.db and gmm_sync.db.

One timestamped location-related record contained:

raw timestamp: 1554407343604

UTC: 2019-04-04 19:49:03.604

latitude: 35.734602

longitude: -78.636654

Additional synced coordinate records included:

35.659956, -78.872848

40.784374, -74.065536

The latter records contained a timestamp value of zero, so no reliable event time was assigned to them. I treated these records as stored/synchronised Maps location artefacts rather than proof that a specific person physically visited those coordinates.

Three media files were examined. Two files with .jpg extensions were identified by ExifTool as image/webp. They shared the same SHA-256 value, indicating duplicate underlying content stored at different paths.

The strongest photograph was:

IMG_20200213_195030.jpg

It was a 3.5 MB JPEG and contained:

Date/Time Original: 2020-02-13 19:50:31

Offset: -05:00

UTC equivalent: 2020-02-14 00:50:31 UTC

GPS latitude: 35° 0' 39.31" N

GPS longitude: 78° 52' 27.18" W

Its SHA-256 was:

2ed43a0a7b01a58f89c23263542282d3d5713b5cd2f22d7b3d2f56a109145ea6

The filename was also located in the Google Photos database, supporting the association between the local image and the Photos artefact.

The photograph's GPS location does not closely match the recovered timestamped Maps coordinate. Even the nearest of the other recovered Maps coordinates was approximately 72 km from the photograph location. The records therefore provide only a weak geographic comparison and do not prove that the photograph and Maps artefacts represent the same visit.

Task 8 – UTC Timeline

No.	Raw time / field	Format	UTC time	Event	Evidence
1	1554407343604 – Maps timestamp	Unix ms	2019-04-04 19:49:03.604	Maps sync/my-place coordinate stored at 35.734602, -78.636654	EV15
2	1580322646631 – SMS date	Unix ms	2020-01-29 18:30:46.631	Telegram verification SMS received; code omitted from report	EV07
3	1580322696889 – SMS date	Unix ms	2020-01-29 18:31:36.889	Signal verification SMS received; code omitted from report	EV07
4	1580587493 – PDU date	Unix seconds	2020-02-01 20:04:53	Sent MMS in thread 17: “Here's Signal.”	EV07
5	1580587564 – PDU date	Unix seconds	2020-02-01 20:06:04	MMS/thread text: “It looks like the messaging is run through the Messages app. Strange.”	EV07
6	1581273702392 – SMS date	Unix ms	2020-02-09 18:41:42.392	Sent to thread 17: “I am on my way to the coffee shop. Do you want anything?”	EV07
7	1581273788 – PDU date	Unix seconds	2020-02-09 18:43:08	Reply in thread 17: “No, thank you.”	EV07 / EV12
8	2020:02:13 19:50:31 -05:00 – DateTimeOriginal	EXIF local time with offset	2020-02-14 00:50:31	Camera photograph captured with GPS metadata	EV16
9	1581643741051 – call date	Unix ms	2020-02-14 01:29:01.051	11-second call involving Josh Hickman / 919-579-0479	EV10
10	1581643824061 – call date	Unix ms	2020-02-14 01:30:24.061	65-second call involving Josh Hickman / 919-579-0479	EV10
11	1581684601922 – lastLoggedIn	Unix ms	2020-02-14 12:50:01.922	Android User 11 last-login record	EV04
12	1581685743277 – SMS date	Unix ms	2020-02-14 13:09:03.277	Sent message: “Wow, the messages sync over between accounts.”	

Cross-Source Correlations

Correlation 1 – Telephony Provider and Google Messages

The Telephony Provider and Google Messages contained matching communications with the same content, participant/thread context and closely corresponding timestamps. The strongest examples were the coffee-shop message and the message about messages syncing between accounts.
This is a strong correlation because the records were recovered independently from mmssms.db and bugle_db.

Correlation 2 – Contacts, AOSP Call Log and Google Dialer

contacts2.db associated the number 9195790479 and its formatted variants with Josh Hickman. The same number appeared in calllog.db and the Google Dialer AnnotatedCallLog, where calls shared matching timestamps and durations.

This provides strong corroboration between identity, contact and call artefacts.

Correlation 3 – Android Account Manager, Sync and Application Records

Android User 0 contained the account thisisdfir@gmail.com. The same account was recovered from Android sync configuration and Google Photos preferences, and TextNow preferences used the same email and the username thisisdfir.

The repeated identifier across independent application and system artefacts provides good evidence of a common account association. It does not, however, prove who physically used the device at a particular time.

Limitations

The examination was restricted to the supplied Android 10 file-system extraction. I did not examine a live handset, access live cloud accounts, reset or root a device, or obtain provider-side records.

Some artefacts were incomplete or unavailable. The user_de Telephony database contained no SMS, PDU or PART rows even though the primary Telephony database contained communications. The Calendar Provider contained no event records. Wi-Fi configuration files contained no saved network entries, so they could not establish a specific network connection.

Bluetooth configuration was available, but the stored records did not provide sufficient time information to prove a connection event. The Bluetooth identifiers observed in different artefacts were not identical and were therefore not treated as a confirmed match.

The current WhatsApp analysis established the registration account and contacts but did not recover the required two-message exchange from the query preserved in the report. The TextNow message query also failed because the query referenced id instead of the actual _id column. I have therefore treated Task 6 communication reconstruction as incomplete rather than inferring records that were not displayed.

Some Maps records had a timestamp value of zero, limiting temporal interpretation. Maps coordinates and photograph GPS metadata demonstrate stored location information, but neither proves that a particular person visited the location.

Two .jpg media files were identified as WebP content, and both had the same SHA-256 hash. This suggests duplicated media content rather than two independent photographs.

Finally, account association, contact names and profile names support attribution only at the artefact level. They cannot independently establish which person physically operated the device.

Conclusion

The examination confirms that the Google Pixel 3 contained multiple user profiles and account associations, communication records across Android and Google applications, contact and call activity involving Josh Hickman, stored location artefacts, and media containing GPS metadata.

Several communications were strongly corroborated across independent sources. In particular, SMS records were reproduced in Google Messages, and calls associated with Josh Hickman's number were represented in both the AOSP call log and Google Dialer. The message stating that “the messages sync over between accounts” is relevant to the case because it demonstrates awareness of cross-account message behaviour.

However, the available evidence does not establish that the communications were unauthorised, nor does it demonstrate that confidential company information was actually transmitted to an unauthorised recipient.

For that reason, my final assessment is that the allegation remains UNRESOLVED.

One possible interpretation is that the multi-application communications and account syncing reflect deliberate sharing or movement of information between accounts. An alternative interpretation is that the activity was legitimate testing, normal personal communication, or authorised use of the company-issued device.

A stronger conclusion would require additional evidence such as company acceptable-use policies, records establishing who was authorised to use each account, complete message and attachment content from WhatsApp/TextNow or cloud sources, network or server logs, recipient-side evidence, and records showing whether any company information was actually transferred.

The recovered artefacts therefore demonstrate communication activity and cross-source consistency, but they do not provide sufficient evidence to classify that activity conclusively as unauthorised information sharing.

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ICDFA Materials.